

How to Classically Verify a Quantum Cat without Killing it

Yael Tauman Kalai* Dakshita Khurana† Justin Raizes‡

Abstract

Existing protocols for classical verification of quantum computation (CVQC) consume the prover’s witness state, requiring a new witness state for each invocation. Because QMA witnesses are not generally clonable, destroying the input witness means that amplifying soundness and completeness via repetition requires many copies of the witness. Building CVQC with low soundness error that uses only *one* copy of the witness has remained an open problem so far.

We resolve this problem by constructing a CVQC that uses a single copy of the QMA witness, has negligible completeness and soundness errors, and does *not* destroy its witness. The soundness of our CVQC is based on the post-quantum Learning With Errors (LWE) assumption.

To obtain this result, we define and construct two primitives (under the post-quantum LWE assumption) for non-destructively handling superpositions of classical data, which we believe are of independent interest:

1. A *state preserving* classical argument for NP.
2. Dual-mode trapdoor functions with *state recovery*.

*MIT. tauman@mit.edu

†UIUC and NTT Research. dakshita@illinois.edu

‡NTT Research. justin.raizes@ntt-research.com

Contents

1	Introduction	1
1.1	Results	2
1.2	Related Works	5
1.3	Open Problems	6
2	Technical Overview	7
2.1	Special Case: Almost-Perfect Completeness	8
2.2	ϵ -Repair with General Completeness.	11
2.3	Instantiating the Base CVQC	13
2.4	Amplifying Completeness and Soundness	14
3	Preliminaries	15
3.1	Quantum Computing	15
3.2	Tail Bounds	17
3.3	QMA	17
3.4	Jordan’s Lemma	18
3.5	State Repair	19
3.6	Classical Verification of Quantum Computation	20
4	Repairing High-Quality Witnesses	23
4.1	Estimating the Acceptance Probability of a High-Quality Witness	24
4.2	Result of Repairing a High-Quality Witness	27
5	Dual-Mode Trapdoor Functions with State Recovery.	28
5.1	Construction	31
6	State-Preserving Arguments for NP	35
6.1	Construction	36
7	Non-Destructive CVQC for QMA	39
7.1	Proof of Theorem 13 (ϵ -Non-Destructive CVQC)	45
7.2	Completeness and Soundness	48
7.3	Witness Recovery: High Completeness	50
7.4	Witness Recovery: Arbitrary Completeness	51
7.5	Proof of Theorem 14 (Sequential Repetition)	55
8	Acknowledgments	60
A	Proofs of No-Intrusion	64
A.1	Definition	64
A.2	Tools	66
A.3	Construction	68

1 Introduction

Imagine a quantum prover that would like to demonstrate possession of Schrödinger’s proverbial cat without killing the cat. Or picture a classical verifier seeking to assess whether a prover has sufficient quantum money to pay them. In both cases, the challenge is: how can one prove the existence of a valuable quantum state without destroying it in the process?

It is well-known that quantum states are precious, consumable resources. Some—like magic states—can be produced at a nontrivial but manageable cost, while others may be far harder to obtain. In particular, it is computationally infeasible to duplicate QMA witnesses, relative to an oracle [CKP25]. Since only one copy of a witness may be available, it is crucial to avoid destroying this witness while proving statements about it.

Non-Destructive Classical Verification of QMA. Mahadev’s breakthrough [Mah18] opened the door to classical verification of quantum computation (CVQC), enabling a quantum prover to prove QMA statements to an efficient classical verifier. This work spurred a number of exciting developments in CVQC [Mah18, ACGH20, CCY20, BKL⁺22, Zha22, KLVY23, MNZ24, GKNV25, BKM⁺25, BK25]. And yet, despite this progress, a fundamental limitation remains: all existing protocols for classically verifying QMA statements end up irreversibly destroying the prover’s precious witness in the process. In this work, we ask

*Can a quantum prover convince a classical verifier of a
QMA statement without sacrificing the witness?*

Using a Single Witness. In the setting where the verifier is *quantum*, non-destructive verification is exactly what enables deciding QMA with negligible error using only one copy of a witness. If a verifier can re-test the same witness repeatedly, completeness and soundness errors can be driven to $2^{-\lambda}$ without increasing the witness length. This idea is at the core of Marriott-Watrous’s fundamental result amplifying QMA with one copy of the witness [MW05]. Conversely, this near-perfect completeness also enables non-destructive QMA verification with a *quantum verifier*. Since the witness is accepted in [MW05] with overwhelming probability, the acceptance measurement disturbs it only negligibly.

In contrast, existing protocols for *classically verifying* QMA typically need sequential or parallel repetition to reduce completeness/soundness errors, but each repetition irreversibly damages the witness. As a result, these protocols require the prover to start with many copies of the QMA witness, a demand which is often unrealistic. We ask:

*Can a quantum prover convince a classical verifier of a
QMA statement using only one copy of the QMA witness?*

This Work. Our work answers *both questions* in the affirmative for CVQC, aligning the case of classical verification with the QMA picture. We note that unlike the QMA setting, even if near-perfect completeness were achieved in CVQC protocols, it is unclear whether the prover’s witness would be preserved in general.

In fact, our main result offers the best of both worlds: a CVQC for QMA with negligible error that uses a *single copy* of a Marriott-Watrous witness [MW05], and where the prover ends the protocol with a witness that has only negligible statistical distance from its original witness.

We achieve our main result via two generic transformations, both of which rely on the post-quantum hardness of LWE:

1. **Witness-Preservation for Near-Perfect Completeness:** We compile any non-adaptive¹ CVQC with $1 - \text{negl}(\lambda)$ completeness into a *witness-preserving* (non-adaptive) CVQC, where an honest prover ends with a state negligibly close to their original witness.

This transformation reduces our task to constructing a non-adaptive CVQC that has near-perfect completeness and uses a single witness (but may destroy the witness).

2. **Completeness and Soundness Amplification via Non-Destructive Verification:** We compile any non-adaptive CVQC for a language in $\text{QMA}_{1-2^{-\lambda}, 2^{-\lambda}}$ (where the honest prover uses a single witness that is accepted by the QMA verifier with overwhelming probability) into one with $1 - \text{negl}(\lambda)$ completeness and $\text{negl}(\lambda)$ soundness.

This compiler first converts any non-adaptive CVQC into one that only mildly destroys the prover’s witness, then it amplifies soundness and completeness by sequential repetition, using the resulting witness which is still “good enough.”

We obtain our main result by taking any non-adaptive CVQC that uses a single copy of the witness, and first applying the second compiler that amplifies soundness and completeness (while still using a single witness), and then applying the first compiler that makes the CVQC witness preserving.

1.1 Results.

Our main result is a CVQC which uses one copy of the witness and preserves it.

Theorem 1 (Informal). *There exists a CVQC protocol for any language $\mathcal{L} \in \text{QMA}_{1-2^{-\lambda}, 2^{-\lambda}}$ with the following properties:*

- **One-Copy Amplified Soundness and Completeness.** *It has $1 - \text{negl}(\lambda)$ statistical completeness and $\text{negl}(\lambda)$ computational soundness error, assuming the post-quantum hardness of LWE, using one copy of any witness $|w\rangle$ for $\mathcal{L} \in \text{QMA}_{1-2^{-\lambda}, 2^{-\lambda}}$. Furthermore, it is an argument of knowledge.*
- **Witness Preserving.** *At the end of the protocol, the prover is left with a state that is statistically close to $|w\rangle$.*

¹A non-adaptive CVQC is one where the verifier’s messages are independent of the prover’s. This is a broad class containing all CVQC protocols we are aware of and which naturally generalizes public-coin protocols.

Remark 1. *The prover may use one copy of any witness $|w\rangle$ which is accepted with probability $1 - 2^{-\lambda}$. Thus, if short, high-quality witnesses exist, the CVQC prover may use a short witness. In contrast, prior works require multiple copies of $|w\rangle$ regardless of how high-quality it is.*

Marriot and Watrous [MW05] show that every $\text{QMA}_{a,b}$ language, where $a-b = 1/\text{poly}(|x|)$, has short witnesses which are accepted with probability $1 - 2^{-\lambda}$. It should be noted that Marriot-Watrous witnesses have a special form and not every $\text{QMA}_{a,b}$ witness is a $\text{QMA}_{1-2^{-\lambda}, 2^{-\lambda}}$ witness. This limitation is inherent since thresholds below $1 - \text{negl}(\lambda)$ allow the possibility of a non-negligible amplitude on a “junk” state which cannot be amplified.

Being simultaneously an argument of knowledge and witness preserving may at first seem contradictory; if it is possible to extract the witness from the prover *and* have the prover keep their witness, have we not cloned the witness, which could in general be hard for QMA [CKP25]? Indeed, [VZ21] formalizes such an intuition for non-destructive *proofs* of knowledge (i.e., with statistical soundness).

Our approach dodges this issue by ensuring that extraction and preservation are both possible at the start of the protocol, but do not happen *simultaneously*. In a real execution, the prover’s witness is preserved. However, when extracting an adversarial prover’s witness, the extractor interacts differently with the prover in a mode which disables witness preservation. An adversarial prover cannot distinguish which mode is occurring, making our CVQC *both* witness preserving and an argument of knowledge. See Section 2.1 for more details.

One-Copy Amplification via Non-Destructive Verification. To obtain our main result, we build two compilers equipping any non-adaptive CVQC with the two desired properties. First, we show how to generically convert any non-adaptive CVQC with arbitrarily low completeness into one with near-perfect completeness and soundness, *without using extra copies of the witness*.

Theorem 2 (Informal). *Assuming the post-quantum hardness of LWE, there is an efficient compiler that converts any non-adaptive CVQC protocol with completeness c and soundness error s satisfying $c - s \geq 1/\text{poly}(\lambda)$ into a new CVQC protocol which has*

- **One-Copy Amplified Soundness and Completeness.** *It has $1 - \text{negl}(\lambda)$ statistical completeness and $\text{negl}(\lambda)$ computational soundness error using **one copy** of the witness $|w\rangle$ for the original CVQC.*

Furthermore, if the original CVQC was an argument of knowledge, so is the new one.

We prove this theorem by first converting the original CVQC into one which partially preserves the witness. Specifically, at the end of the protocol the prover has a new witness that is at most ϵ worse than its original witness, where $\epsilon = 1/\text{poly}(\lambda)$ is a tunable parameter. Then we sequentially amplify completeness and soundness using the leftover witness.

To use the amplification compiler (Theorem 2), we need to start with a non-adaptive CVQC that uses a single witness but may have poor completeness and soundness. The first

question the reader may ask is:

Do we even have a CVQC protocol that uses a single witness, even with poor completeness and soundness?

The answer to this question is a bit complicated, but for now, suppose the answer is yes (which is almost true, and the subtleties are deferred to the technical overview).²

Witness-Preservation via Near-Perfect Completeness. Second, we show how to compile any non-adaptive CVQC with near-perfect completeness into a witness preserving CVQC, again without using extra copies of the witness. We obtain our main result by applying this second compiler to our first one that amplifies completeness and soundness.

Theorem 3 (Informal). *Assuming the post-quantum hardness of LWE, there is an efficient compiler that converts any non-adaptive CVQC protocol with completeness $1 - \text{negl}(\lambda)$ and soundness error s into a new CVQC protocol with the same completeness and soundness which is*

- **Witness Preserving.** *At the end of the protocol, the prover is left with a state that is statistically close to its input witness $|w\rangle$.*

Furthermore, if the original CVQC was an argument of knowledge, so is the new one.

To construct our compilers, we develop two techniques that we believe are of independent interest.

Witness-preserving arguments for NP. We construct an interactive argument for NP with the guarantee that if the prover starts with a superposition over possible witnesses $\sum \alpha_w |w\rangle$ then this superposition is maintained at the end of the interactive argument. We refer to such an argument system as *witness preserving*.

Theorem 4 (Informal). *There exists a witness preserving interactive argument for NP assuming the post-quantum LWE assumption.*

We refer the reader to Section 6 for details. The construction uses the following primitive, which is also used by our compilers.

Dual-mode trapdoor function family with state recovery. We define and construct a family of randomized functions that can be sampled in two modes: **recovery** mode or **injective** mode. It has the guarantee that if a function is sampled in the **recovery** mode then it is *non-collapsing* in the sense that applying f to a state $\sum_w \alpha_w |w\rangle$ and measuring the output y , allows one to reconstruct the state $\sum_w \alpha_w |w\rangle$ given a trapdoor. On the other hand, if a function is sampled in the **injective** mode then it is injective and hence the state is collapsed to a single w , which can be computed given the trapdoor.

²Jumping ahead, our starting CVQC is a single repetition of Mahadev’s CVQC which uses only one witness (but ensures that $c - s \geq \frac{1}{\text{poly}}$ only conditioned on the test round passes with high probability).

Theorem 5 (Informal). *There exist dual-model trapdoor function family under the post-quantum **LWE** assumption.*

The construction is basically the same as the construction of dual-mode claw-free trapdoor functions from [BCM⁺18], with the addition of a new **Recover** algorithm. We refer the reader to Section 5 for details.

Witness Preservation Against Malicious Verifiers. Finally, we mention the implications of our results to a related interesting scenario where the verifier is malicious and attempts to destroy the prover’s witness. In other words,

*Can a quantum prover convince a verifier of a **QMA** statement without losing their witness **even if the verifier deviates from the protocol?***

It is not hard to see that any witness-preserving CVQC (not necessarily non-adaptive) can be generically converted into one with *malicious-verifier* witness preservation, assuming time-lock puzzles and zero-knowledge arguments for **NP**. The verifier sends the prover a time-lock puzzle, containing its randomness; then for each message it sends in the CVQC, it proves that this message is consistent with the time-locked randomness in zero-knowledge.³ If the verifier ever deviates from the protocol (e.g. by aborting), the prover can solve the time-lock puzzle and complete the protocol locally to recover its witness. On the other hand, the security of the time-lock puzzle and zero-knowledge argument ensure that a real prover has no information about the verifier’s randomness before the time-lock puzzle can be solved, so soundness is still guaranteed.

A conceptually even simpler approach uses classical witness encryption for **QMA** (for which there are no plain-model constructions, but there are constructions in the classical oracle model e.g. due to [BM22]). The verifier encrypts a random message m under the **QMA** statement to be proven, then the prover decrypts it and sends back m' , whereupon the verifier checks that $m' = m$. If the witness encryption is correctly generated, then decryption is almost deterministic and disturbs the prover’s witness negligibly. Honest encryption can be guaranteed using a zero-knowledge argument.

1.2 Related Works

Classical Proofs of Quantum Knowledge [VZ21]. Vidick and Zhang observed that proofs of quantum knowledge with classical communication are generally destructive unless the witness can be (inefficiently) cloned, and so an honest prover in a sequentially repeated protocol needs to use multiple copies of the witness in order to succeed. Their analysis applies to perfectly non-destructive, statistically sound proofs and they provide some discussion about extending the results to protocols which only damage the witness by a small amount.⁴ On the other hand, in this work we build arguments of knowledge that are non-destructive.

³For soundness, the zero-knowledge argument and CVQC must be together faster than the time to solve the time-lock puzzle.

⁴They mention that their no-go theorem does not apply to **QMA** since it is always possible to inefficiently find *some* witness. However, if the witness is not unique, it is still impossible to clone the *exact* witness used.

Proofs of No Intrusion [GR25]. Recently, Goyal and Raizes investigated a new primitive they called “proofs of no-intrusion” (PoNI). A PoNI for encryption is essentially a non-destructive classical argument that a ciphertext has not been “stolen”, in the sense that no external party can decrypt the ciphertext even given the key. Our results enable these by combining our state-preserving argument for NP with a public-key encryption scheme with publicly verifiable certified deletion (known from just public-key encryption [BKM⁺23, KNY23]). See Appendix A for more details.

However, their main new technical tool achieves something incomparable to our results. Goyal and Raizes introduce PoNIs for coset states, which is essentially a non-destructive classical argument of quantum knowledge specifically for coset states [CLLZ21]. Although conceptually related, PoNIs for coset states are incomparable to our results. Our results allow non-destructive arguments for the general class of QMA, where the statement being proven is public. On the other hand, PoNIs for coset states allow the prover to blindly prove a specific statement about coset states, where the prover *does not know the statement being proven*.

Disambiguation. [LMS22b] also used the term “state preserving argument” in the context of arguments of knowledge. There, “state preserving” refers to the ability to extract some knowledge from an adversarial prover without noticeably disturbing the prover’s internal state. This takes place in a sandbox where the communications with the adversarial prover can be purified or rewound. In contrast, our usage of state preserving refers to preserving an honest prover’s input even in a real execution.

1.3 Open Problems

We conclude this section with a set of open directions:

1. **Round Complexity.** Can we achieve single-copy CVQC for QMA with negligible errors in constant (*or even two*) rounds of interaction? Since high-error protocols can only be composed sequentially when there is a single copy of the witness, addressing this question likely requires additional techniques beyond those developed in this work.
2. **Succinctness.** Can we achieve single-copy CVQC for QMA with negligible errors with smaller communication than the instance size? Our compiler from $1 - \text{negl}(\lambda)$ completeness to witness preservation can make each message succinct using techniques from [GKNV25]. However, our compiler for amplifying completeness using one copy of the witness uses a number of sequential repetitions which grows with the instance size, preventing it from being succinct even with short messages.
3. **Relationship between Witness Preservation and Zero-Knowledge.** Witness preservation in CVQC seems to imply a form of witness hiding: intuitively, because the

Thus, their no-go theorem still applied to protocols where the original witness is preserved (as opposed to ensuring the prover retains an arbitrary valid witness after the protocol) and the extractor would output the original witness.

prover must keep the witness intact and quantum states cannot be cloned, an honest verifier cannot end the protocol holding the witness. Is there a more formal connection between witness preservation and zero-knowledge?

4. **Other Solutions for Malicious-Verifier Witness Preservation.** Can a quantum prover preserve its witness against a malicious verifier using weaker assumptions than time-lock puzzles and zero knowledge for NP or witness encryption for QMA? This question is even interesting using *quantum communication*, since the verifier is not guaranteed to cooperate with any request to uncompute the protocol.
5. **Further Applications.** Can witness-preserving CVQC serve as a foundation for new unclonable cryptographic primitives, such as classically verifiable quantum money, copy-protected software, or quantum credentials?

2 Technical Overview

In what follows, we describe our compiler that compiles any non-adaptive CVQC protocol, that uses a single copy of the QMA witness (but may have poor completeness and soundness parameters and may destroy the witness), into a CVQC protocol that uses a single copy of the QMA witness and is witness-preserving and has nearly perfect completeness and soundness.

Our compiler works in three stages.

- **Stage 1: ϵ -Non-Destruction and a Special Case.** First, we compile the CVQC protocol to an ϵ -non-destructive protocol with almost the same completeness $c - \text{negl}(\lambda)$ and soundness error $s + \text{negl}(\lambda)$. Roughly speaking, if the prover starts with a witness $|w\rangle$ which would be accepted by the QMA verifier with probability p (a “ p -good witness”), then at the end of the protocol they have a witness $|w'\rangle$ which is at least $(p - \epsilon)$ -good.⁵ The runtime scales with $1/\epsilon$, so we are limited to $\epsilon = 1/\text{poly}(\lambda)$.

As a special case of this compiler, if the underlying CVQC *starts* with $1 - \text{negl}(\lambda)$ completeness, then we achieve *witness preservation*: at the end of the protocol $|w'\rangle$ is negligibly far from $|w\rangle$. The runtime only blows up by a factor of the security parameter λ . This special case is used in Stage 3, below.

- **Stage 2: Sequential Amplification.** Next, we amplify completeness and soundness by repeating the protocol sequentially using the same witness. Since we can tune ϵ to any $1/\text{poly}(\lambda)$, if we start with a witness just slightly better than the QMA threshold, we can ensure that every intermediate witness still surpasses the QMA threshold. Thus, the CVQC verifier accepts with probability $c - \text{negl}(\lambda)$ in each execution.

⁵Our ϵ -non-destructive protocol actually uses a special set of “repairable” witnesses that include all Marriot-Watrous witnesses and more generally any $(1 - \text{negl}(\lambda))$ -good witness, but for now we make the simplifying assumption that the ϵ -non-destructive guarantee applies to any witness.

The parameters can also be tuned more aggressively to end up with a $(p - \epsilon')$ -good witness $|w'\rangle$ after all executions, but we are still limited to $\epsilon' = 1/\text{poly}(\lambda)$ by the runtime blowup from stage 1.

- **Stage 3: Witness Preservation.** Finally, we re-compile the amplified protocol *again*, using the special case of the stage 1 compiler. Since the amplified protocol has $1 - \text{negl}(\lambda)$ completeness, the final compiled protocol is witness preserving.

The bulk of the technical ideas appear in stage 1, which is the focus of this overview.

2.1 Special Case: Almost-Perfect Completeness

We begin with the special case of $c = 1 - \text{negl}(\lambda)$. The main body of the compiler is the same as the general completeness version, but the witness recovery step is significantly simpler and illustrates the core ideas well.

Classical Verification in Superposition. Ultimately, the goal in CVQC is to convert a quantum statement into a classical one so that the classical verifier can check it. The problem with existing approaches to CVQC is that measuring said transcript collapses the prover’s state. If the prover could keep the transcript “in superposition” throughout the protocol, then at the end of the protocol (after the verifier measures the verdict), the prover could uncompute the transcript and recover the witness. This yields something like the following very high-level approach.

1. Interactively, the prover *coherently* computes its answers a_i to each of the verifier’s ℓ queries q_i during the CVQC and does not measure them. At the end, the prover holds a superposition over answers $a_{[\ell]} = (a_1, \dots, a_\ell)$:

$$\sum_{a_{[\ell]}} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle$$

Since the verifier’s queries are non-adaptive, it doesn’t need to see the prover’s answers to send the next query.

2. The verifier sends its (classical) secret state from the CVQC to the prover⁶ and the prover uses a special argument for NP to prove that it holds an answer sequence $a_{[\ell]}$ which the verifier would accept, without collapsing its superposition over $a_{[\ell]}$.

Since the completeness is almost perfect, measuring whether $a_{[\ell]}$ is an accepting transcript disturbs the state negligibly.

3. Finally, the prover uncomputes its answers $\sum_{a_{[\ell]}} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle$ to recover the witness.

⁶Sending the verifier’s internal CVQC state is necessary so that the prover knows the NP statement it is proving. If the CVQC is publicly verifiable, this is not necessary, but publicly verifiable CVQC remains an open problem.

Of course, this approach is not sound as written because the prover can change its answers $a_{[\ell]}$ after seeing future queries or even the verifier’s internal state. So, the question is:

How do we bind the prover to $a_{[\ell]}$ without collapsing it or otherwise preventing the prover from uncomputing its answers and recovering the witness?

Dual-Mode Trapdoor Functions with State Recovery. Our solution to this conundrum is an observation of a novel property of [BCM⁺18, Mah18]’s trapdoor claw-free functions which we abstract out as a new primitive: dual-mode trapdoor functions with state recovery. This is a family of randomized functions and trapdoors $(f_{\text{pp}}, \text{td}_{\text{pp}})$. The tuples can be generated in two indistinguishable modes: **recovery** or **injective**.

In **injective** mode, two different inputs $x \neq x'$ cannot collide: $f_{\text{pp}}(x; r) \neq f_{\text{pp}}(x'; r')$ for all randomness r and r' . Given any $y = f_{\text{pp}}(x; r)$, the trapdoor td_{pp} can be used to find x . This mode will be important for proving soundness of our protocol.

In **recovery** mode, f_{pp} is 2^n -to-1, where n is the input size, and td_{pp} can be used to invert a preimage $(x; r_x)$ of $y = f_{\text{pp}}(x; r_x)$ for *any* $x \in \{0, 1\}^n$, where r_x is the randomness. Similarly to Mahadev’s measurement protocol, a quantum party can “commit” to a computational basis measurement of a quantum state $\sum_x \alpha_x |x\rangle$ by preparing a superposition over all randomness r then coherently evaluating $f_{\text{pp}}(x; r)$ and measuring the result y . This results in the state

$$\sum_x \alpha_x |x, r_x\rangle$$

where $f_{\text{pp}}(x, r_x) = y$ for all $x \in \{0, 1\}^n$. Here is where the recovery comes in: if the trapdoor td_{pp} were revealed, the quantum party can use it to coherently compute r_x controlled on x . By subtracting that from the randomness register, they disentangle their state and recover $\sum_x \alpha_x |x\rangle$.

Using the **recovery** mode, we can update the protocol above as follows.

1. The verifier generates a pair $(f_{\text{pp}}, \text{td}_{\text{pp}})$ in the **recovery** mode, and sends pp to the prover. Interactively, the prover “commits” to each answer a_i by evaluating f_{pp} as discussed previously. At the end, the verifier knows $y_{[\ell]} = (y_1, \dots, y_\ell)$ and the prover holds

$$\sum_{a_{[\ell]}} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle \otimes |r_{a_{[\ell]}}\rangle$$

where $y_i = f_{\text{pp}}(a_i; r_{a_i})$ for each $i \in [\ell]$.

2. The verifier reveals its secret state for the underlying CVQC and the prover uses the “state-preserving” argument for **NP** (which we elaborate on below) to prove that there exist preimages $(a_i; r_{a_i})$ of $y_{[\ell]}$ such that the verifier would accept $a_{[\ell]}$ in the CVQC.
3. The verifier sends td_{pp} to the prover, who uses it to uncompute each $|r_{a_{[\ell]}}\rangle$. Since the superposition over answers $a_{[\ell]}$ is no longer entangled with any external state, the prover can uncompute its answers to recover the witness.

If the state-preserving argument for NP does not disturb the superposition $\sum_{a_{[\ell]}} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle \otimes |r_{a_{[\ell]}}\rangle$, then the **recovery** mode trapdoor allows recovering the prover's original witness up to negligible disturbance.

State-Preserving Arguments for NP. Roughly, we need the prover to be able to use a superposition of NP witnesses $\sum_i \alpha_i |w_{\text{NP},i}\rangle$ to prove the NP statement, and at the end of the argument they should have a state close to their original witness superposition.

A very simple construction is possible using witness encryption for NP. The verifier encrypts a random message m under the statement to be proven. Then, the prover coherently decrypts m' using their witness superposition. The verifier accepts if $m = m'$. Since decryption is almost deterministic using a valid witness, measuring m' disturbs the prover's superposition negligibly. Unfortunately, witness encryption is a relatively strong assumption, which we would like to avoid (and do avoid).

The next idea is to use a *statistically* witness-indistinguishable (WI) argument. Intuitively, since the verifier's view is statistically independent of which witness was used, the prover avoids measuring their state. However, some care is needed here. During the argument, the prover may also entangle their witness with the randomness used. To recover the witness superposition, it is crucial that the prover be able to unentangle their state later.⁷

We show that the well-known 3-coloring WI argument, when instantiated using dual-mode trapdoor functions with state recovery, is a state-preserving argument for NP. The analysis requires a careful accounting of the prover's randomness – both for the trapdoor function and for randomness inherent to the 3-coloring protocol – to show that it can be safely uncomputed at the end of the protocol.

Soundness. The proof of soundness reduces to the soundness of the underlying CVQC by extracting an accepting transcript $a_{[\ell]}$ from the prover. As a result, the new CVQC is a proof of knowledge if the underlying CVQC is. If $(f_{\text{pp}}, \text{td}_{\text{pp}})$ is generated in **injective** mode, then there is only one preimage a_i of each y_i in step 1. The reduction can use td_{pp} to extract each answer a_i from y_i as the prover sends it, then forward a_i to the CVQC verifier.

The subtle part of the proof is in showing that the extracted $a_{[\ell]}$ will be accepted by the verifier. Intuitively, the indistinguishability of **injective** and **recovery** mode ensures that the adversarial prover cannot *knowingly* change its behavior between the extractor and a real execution. However, the adversarial prover does not know td_{pp} , so anything extracted using it might change when we switch modes.⁸

To prevent this possibility, we break the proof into two parts. First, we observe that in **injective** mode, the *existence* of a preimage $a_{[\ell]}$ of $y_{[\ell]}$ which the CVQC verifier would accept,

⁷It is not hard to come up with examples where unentangling is computationally hard, for example using a claw-free lossy function *without* a trapdoor.

⁸As a concrete example, suppose that the function was specified by an Fully Homomorphic Encryption (FHE) ciphertext $\text{ct} = \text{Enc}(b)$. If the mode is **recovery**, $b = 0$, and if the mode is **injective**, $b = 1$. An evaluation of x is just a homomorphic multiplication of $x \cdot b$. Although an adversarial evaluator might not be able to detect a switch from $b = 0$ to $b = 1$, homomorphic evaluation allows them to encrypt different messages in the two cases.

is sufficient to extract an accepting $a_{[\ell]}$, since said preimage is unique.

Then, we use the soundness of the **NP** argument to establish that this holds with almost the same probability that the adversarial prover would convince the non-destructive verifier. If the **NP** argument verifier would accept (in **recovery** mode), there exists such a preimage, although it is not necessarily the unique preimage. Since the **NP** argument can be checked *without* td_{pp} , we can rely on the indistinguishability of **recovery** and **injective** modes to show that the probability of the **NP** argument verifier accepting is almost the same in **injective** mode as in **recovery** mode.

Dodging the Cloning Bullet. As mentioned previously, at first it may seem that the ability to both extract and retain the prover’s witness implies the ability to clone it, which would violate the no-cloning theorem. Now that we have explained the core of the protocol, it becomes clearer why our approach does not imply cloning.

During our proof for (knowledge) soundness, we reduce to the knowledge soundness of the original CVQC by using an **injective** mode f_{pp} to extract the prover’s answers for the original CVQC. However, **injective** mode does not have the nice recovery property we used to preserve the witness. In fact, measuring an evaluation of an injective mode f_{pp} *permanently* collapses the witness. The only way to extract a witness is to take it away from the prover!

We also mention how our protocol avoids Vidick and Zhang’s more formal cloning implication for *proofs* of knowledge [VZ21]. Their technique relies on the non-destructivity to query the prover on every possible verifier message and learn its classical response. The resulting table of responses acts as an inefficient clone of the prover. At that point, the extractor can be applied to each copy of the prover, extracting the prover’s witness twice. In our case, the extraction and witness recovery modes have two disjoint, but indistinguishable sets of verifier messages. The prover’s behavior in the witness recovery mode could indeed be cloned, but it would not be possible to extract a witness from the cloned prover. In fact, the cloned prover would be able to distinguish between an **injective** and **recovery** mode f_{pp} simply because it does not know how to respond when the verifier sends f_{pp} in **injective** mode.

2.2 ϵ -Repair with General Completeness.

In the more general setting of an arbitrary c , the prover’s measurement in step 2 of whether they hold an accepting CVQC transcript $a_{[\ell]}$ is no longer gentle. Since that measurement can noticeably disturb the state, uncomputing the trapdoor function randomness $r_{a_{[\ell]}}$ in step 3 does not allow recovering the witness.

Still, we have made some progress. In the original CVQC, the prover needed to measure the entire transcript, which can have exponentially many outcomes. In the new CVQC, the prover only makes a measurement with two outcomes, limiting the possible damage. Limiting to two possible outcomes enables a repair technique from [CMSZ22].

CMSZ State Repair. Suppose one had a state $|\psi\rangle$ which passed some verification procedure V with probability p , but then measured it with an N -outcome projective measurement.

The measurement damages the state, resulting in $|\psi'\rangle$. Is it possible to repair $|\psi'\rangle$ so that it is accepted by V with probability close to p again?

[CMSZ22] consider exactly this scenario and give an algorithm to repair $|\psi'\rangle$ with overwhelming probability. For the purposes of non-destructive CVQC, we set V to be the QMA verifier and the damaging measurement to be coherently generating a CVQC transcript and measuring whether it is accepting. If the algorithm is successful in restoring $|\psi'\rangle$'s original success probability, then the prover still has a valid witness, albeit potentially a different one.

Runtime Considerations. CMSZ's Repair algorithm is extremely useful, but has a few important limitations. First, the expected runtime scales with N , the number of outcomes for the damaging measurement. By substituting a measurement with exponentially many outcomes for a measurement with $N = 2$ outcomes using our approach so far, we keep the expected runtime polynomial.

Second, Repair only repairs the success probability to *approximately* p , plus or minus ϵ . Although ϵ can be tuned arbitrarily, the runtime *also* grows with $1/\epsilon$, so we are limited to $\epsilon = 1/\text{poly}(\lambda)$ if we want Repair to run in expected polynomial time. Fortunately, as discussed at the start of the overview, an $\epsilon = 1/\text{poly}(\lambda)$ decay suffices for our purposes.

What Can CMSZ Repair? The third and final limitation of Repair is that it can only make strong repair guarantees for a particular set of “repairable” states. If the original state $|\psi\rangle$ was a superposition of two repairable states which were accepted with probability p_1 and p_2 , respectively, then the repair procedure will not repair to the average of p_1 and p_2 ; instead, it will essentially select one at random and repair to that one. As an example, imagine that $|\psi\rangle$ was a superposition over a “good” QMA witness which is accepted with probability 1 and some “junk” which is accepted with probability 0. If both the “good” witness and “junk” were individually repairable, then there is a possibility that Repair picks the junk state and repairs the success probability to close to 0, ruining the witness.

We show a more precise picture of which states CMSZ's Repair almost certainly will repair to an acceptance probability $\geq p^*$, which we call “ p^* -repairable”. The overall effect of repairing a state using CMSZ's techniques can be thought of in three steps:

1. Estimate the probability p that $|\psi\rangle$ passes the verification procedure V . This potentially disturbs the state.
2. Damage the state by measuring it.
3. Repair the state back to acceptance probability p , plus or minus a small error ϵ .

The primary factor determining the acceptance probability of the repaired state is the estimate in step 1. CMSZ requires a special estimation procedure that differs significantly from Marriot-Watrous's QMA verification (which can be viewed as a probability estimation

algorithm) to make it compatible with **Repair** (see Section 4.1 for a more in-depth discussion of this point). By closely inspecting their estimation algorithm, we observe that it approximately measures $|\psi\rangle$ in the eigenbasis of the operator⁹

$$\frac{1}{2}V + \frac{1}{4}I$$

and outputs the corresponding eigenvalue, rescaled. This operator has the same eigenstates as V with rescaled eigenvalues. As such, the set of witnesses which can be repaired almost certainly to better than p^* acceptance probability is the span of eigenstates of V with eigenvalues $\geq p^* + \epsilon$.

Letting a be the QMA threshold, the set of $(a - \epsilon)$ -repairable witnesses includes all Marriot-Watrous witnesses. More generally, it can be seen that any witness which is accepted by V with overwhelming probability is overwhelmingly close to being $(1 - \text{negl}(\lambda))$ -repairable.

2.3 Instantiating the Base CVQC

To obtain our main result (and to instantiate our compilers, described below), we need a “base” CVQC protocol where the honest prover uses a single copy of the QMA witness, and has arbitrary completeness c and soundness s , as long as $c - s \geq 1/\text{poly}(|x|, \lambda)$.

One way to obtain such a CVQC protocol is to use Mahadev’s protocol [Mah18] as the underlying CVQC protocol. This requires some care since Mahadev’s CVQC consists of many repetitions of an underlying CVQC (which [Mah18] constructs). As a result, the prover needs many copies of the witness, one copy for each repetition.

It is tempting to use only one copy of Mahadev’s protocol, however, a single copy of this protocol has worse completeness than soundness! To demystify this, we note that what makes this protocol useful is that it has a special structure: The verifier V is defined via two verification algorithms (V_T, V_C) ; it implements V_T , which is a test phase, with probability $1/2$, and implements V_C , which checks the validity of the witness, with probability $1/2$. Importantly, V_T has completeness 1 while V_C has (low) completeness c .¹⁰ The guarantee is that for every $x \in \mathcal{L}_{\text{no}}$ and every cheating prover P^* that convinces V_T to accept with probability close to 1, can convince V_C to accept with probability at most s which is smaller than c . We denote this notion of soundness by *testable soundness*.

We ensure that both our compilers in Theorems 2 and 3 also compile CVQC protocols with testable soundness s . We refer the reader to Section 7 (and in particular to Theorems 13 and 14 for the formal theorems).

We note that an alternative route is to use the [KLVY23] compiler to obtain an underlying CVQC, as follows:

1. Take any any MIP^* protocol for $\mathcal{L}$, with arbitrary completeness c and soundness s , such that $c - s \geq 1/\text{poly}(|x|, \lambda)$, that uses a *single* copy of a (repairable) witness.

⁹We abuse notation here by using V to mean the “accept” operator of the POVM induced by V .

¹⁰This follows from the fact that the QMA witness is converted into a Morimae-Fitzimons [MF16] QMA witness, which has low completeness.

For example, one can take the MIP^* from [Ji15].

2. Apply the compiler from [KLVY23] to convert this MIP^* protocol into a CVQC.

The soundness of the [KLVY23] compiler is still under investigation. It was proven to be equivalent to the quantum value (assuming the existence of a secure quantum FHE) for any game where the optimal strategy is a finite one [NZ23, KMP⁺25, BLJŠ25, BKL⁺25]. Unfortunately, it is not clear whether for every QMA language $\mathcal{L} = (\mathcal{L}_{\text{yes}}, \mathcal{L}_{\text{no}})$ and every MIP^* for $\mathcal{L}$, it holds that for every $x \in \mathcal{L}_{\text{no}}$ the optimal strategy for proving that $x \in \mathcal{L}_{\text{yes}}$ is finite.

2.4 Amplifying Completeness and Soundness

Let us now focus on achieving negligible errors with a single copy of the witness (while potentially allowing the witness to eventually degrade by the end of the protocol). To do this, we will have the prover and verifier sequentially repeat an ϵ -non-destructive CVQC, with completeness c and soundness s where $c - s \geq 1/\text{poly}(|x|, \lambda)$, N times, where N is a large enough polynomial in $(\lambda, |x|, (c - s)^{-1})$. Such an ϵ -non-destructive CVQC can be obtained by applying the ϵ -repair technique described in Section 2.2 to the base CVQC described in Section 2.3.

In our sequentially repeated protocol, the verifier will accept if and only if all the test rounds accept, and at least $\left(\frac{c+s}{2}\right) \cdot N$ of the check rounds accept. If we start with an ϵ “repairable” witness that is accepted with probability p , then after N executions, with overwhelming probability we are left with a witness that is accepted with probability $p - N\epsilon$, where we are able to set $N\epsilon$ to be an arbitrarily small inverse polynomial value. The degradation is small enough that an honest prover is still able to cause the verifier to accept in $\left(\frac{c+s}{2}\right) \cdot N$ of the check rounds, except with negligible probability. This ensures negligible completeness error.

To argue that the protocol is a proof of knowledge, we will build an extractor that outputs a QMA witness for x with oracle access to any prover that generates accepting transcripts for x with noticeable probability. Our extractor will simply pick a random execution $j \leftarrow [N]$ and run the extractor of the underlying CVQC on this execution: we will prove that this extractor succeeds with noticeable probability in finding a good QMA witness for x .

To prove correctness of this extractor, we will rely on the extractability of the underlying CVQC: namely, given any prover P^* that convinces V_T to accept with probability close to 1, and convinces V_C to accept with probability noticeably larger than s , the underlying extractor outputs a QMA witness with noticeable probability.

Then our goal is to simply prove that with noticeable probability, the session j picked by the outer extractor satisfies both the constraints above. We prove this in two parts: first, we prove that for accepting transcripts, nearly all prefixes of the transcript satisfy the following: the prover, conditioned on this prefix, will pass V_T with high probability in the upcoming round. We call such a transcript a **Good** transcript.

Next, we prove that for nearly all **Good** transcripts τ , there is an i such that in session i , the prover convinces V_T to accept with probability close to 1 as before, but additionally,

passes V_C with probability greater than s . This is proved by contradiction: suppose this were not the case, then we show that the number of check rounds that would accept will be smaller than $(c + s)/2$, and the transcript would be rejected by the verifier. This requires careful tail bounds for non-independent random variables, and in particular we are able to use Azuma's inequality to obtain these bounds.

The proof of soundness proceeds similarly, by noting that for any prover that outputs accepting proofs with noticeable probability, there must exist a round where the prover passes the test round with probability close to 1, and passes the check round with probability greater than s , which gives us a contradiction to the soundness of the base CVQC.

This gives us a protocol that uses a single copy of the witness, and achieves negligible soundness and completeness errors, albeit at the cost of slightly degrading the witness. We can then apply the compiler described in Section 2.1 *again* to the resulting protocol to make it witness preserving (i.e. with negligible disturbance to the witness state).

3 Preliminaries

3.1 Quantum Computing

Let $\mathcal{H}$ and $\mathcal{A}$ be Hilbert spaces. A quantum algorithm V is a unitary circuit U_V together with a (without loss of generality, computational basis) measurement $\Pi_V = \{\Pi_V^x\}_{x \in X}$. To evaluate it on a state $|\psi\rangle \in \mathcal{H}$, first append $|0\rangle \in \mathcal{A}$ to it, then compute $U_V(|\psi\rangle \otimes |0\rangle)$ and finally measure the resulting state with respect to Π . The ancilla registers are discarded afterwards. The probability of outcome x is

$$\|\Pi_V^x U_V(|\psi\rangle \otimes |0\rangle)\|^2$$

In the case where V outputs a bit $x \in \{\text{Accept}, \text{Reject}\}$, we overload V to also mean the operator representing an accepting outcome:

$$V := (I \otimes \langle 0|) U_V^\dagger \Pi_V^{\text{Accept}} U_V (I \otimes |0\rangle)$$

Then the probability of accepting a state $|\psi\rangle$ is

$$\text{Tr}[V |\psi\rangle] = \langle \psi | V | \psi \rangle.$$

Any eigenvector of V with eigenvalue p is accepted by V with probability p . A general quantum channel may not be described in terms of a unitary, in which case we refer to a quantum algorithm that implements the channel, i.e. a unitary circuit U_Φ and a measurement Π_Φ , as a unitary dilation of the channel.

Definition 1. Given an algorithm V , we define the Hilbert space $\mathcal{ES}_{\geq a}(V)$ to be the span of eigenvectors of V with eigenvalue $\geq a$. $\mathcal{ES}_{\leq b}(V)$ is defined similarly.

The trace distance between two quantum states ρ and σ is $\frac{1}{2}\|\rho - \sigma\|_1$, where $\|\cdot\|_1$ denotes the trace norm. The distance between two quantum channels is measured by how far apart they map the same state. In other words,

Definition 2 (Diamond Distance). *The diamond distance, which is induced by the completely bounded trace norm, between two channels Φ_1 and Φ_2 on n qubits is given by*

$$\frac{1}{2}\|\Phi_1 - \Phi_2\|_\diamond = \max_{\rho} \frac{1}{2}\|(\Phi_1 \otimes I_n)\rho - (\Phi_2 \otimes I_n)\rho\|_1$$

where $\|\cdot\|_1$ denotes the trace norm.

Lemma 1 (Mixed to Pure). *Let V be a binary-outcome measurement (not necessarily projective). Then any ρ such that*

$$\Pr[\text{Accept} \leftarrow V(\rho)] \geq 1 - \gamma$$

can be decomposed as $\rho = q_{\text{good}}\rho_{\text{good}} + q_{\text{bad}}\rho_{\text{bad}}$, a mixture over two orthogonal mixed states where

$$q_{\text{bad}} \leq \sqrt{\gamma}$$

and ρ_{good} is supported completely on pure states $|\psi\rangle$ such that

$$\Pr[\text{Accept} \leftarrow V(|\psi\rangle)] \geq 1 - \sqrt{\gamma}$$

Proof. Without loss of generality, we may write

$$\rho = \sum_i q_i |\psi_i\rangle\langle\psi_i|$$

for some orthogonal basis $\{|\psi_i\rangle\}_i$. Let

$$\mathcal{I}_{\text{good}} = \{i : \Pr[\text{Accept} \leftarrow V(|\psi_i\rangle)] \geq 1 - \sqrt{\gamma}\}$$

and define $\mathcal{I}_{\text{bad}}$ to be its complement. Define

$$q_{\text{good}} = \sum_{i \in \mathcal{I}_{\text{good}}} q_i \quad \text{and} \quad \rho_{\text{good}} = q_{\text{good}}^{-1} \sum_{i \in \mathcal{I}_{\text{good}}} q_i |\psi_i\rangle\langle\psi_i|.$$

Define q_{bad} and ρ_{bad} analogously. By definition, ρ_{good} satisfies the second half of the claim.

Thus,

$$\begin{aligned} 1 - \gamma &\leq \\ \Pr[\text{Accept} \leftarrow V(\rho)] &= \\ \sum_i q_i \Pr[\text{Accept} \leftarrow V(|\psi_i\rangle)] &= \\ \sum_{i \in \mathcal{I}_{\text{good}}} q_i \Pr[\text{Accept} \leftarrow V(|\psi_i\rangle)] + \sum_{i \in \mathcal{I}_{\text{bad}}} q_i \Pr[\text{Accept} \leftarrow V(|\psi_i\rangle)] &\leq \\ 1 - \sqrt{\gamma} \sum_{i \in \mathcal{I}_{\text{bad}}} q_i & \end{aligned}$$

Therefore,

$$q_{\text{bad}} = \sum_{i \in \mathcal{I}_{\text{bad}}} q_i \leq \gamma / \sqrt{\gamma} = \sqrt{\gamma},$$

as desired. □

3.2 Tail Bounds

Our analysis of sequential repetition will use tail bounds for certain non-independent processes called Martingales. A martingale is a stochastic process in which the expected value of the next observation, given all prior observations, is equal to the most recent value. We formally define such a process and tail bounds for this process below (focusing on the simplified finite case, which suffices for our setting).

Definition 3 (Martingales). *A sequence of random variables $(S_m)_{m \geq 0}$ from a finite universe is called a **martingale** if $\mathbb{E}[S_{m+1} \mid S_1, \dots, S_m] = S_m$ for all $m \geq 0$.*

Theorem 6 (Azuma–Hoeffding inequality for Martingales with bounded differences). *Let $(S_m)_{m \geq 0}$ be a martingale and let $c_1, c_2, \dots, c_m$ be constants such that for all $1 \leq i \leq m$, $|S_i - S_{i-1}| \leq c_i$. Then for any $t > 0$,*

$$\Pr[S_m - S_0 \geq t] \leq \exp\left(\frac{-2t^2}{\sum_{k=1}^m c_k^2}\right).$$

3.3 QMA

Definition 4. *Let $a, b : \mathbb{N} \rightarrow [0, 1]$. there exists a polynomial p such that for every instance size n there exists a family $\{V_x\}_{x \in \mathcal{L}_{\text{yes}} \cup \mathcal{L}_{\text{no}}}$ of quantum polynomial-time algorithms which take as input a quantum state on $p(n)$ qubits and output a decision bit such that the following properties hold.*

- **Efficiency.** *There exists a quantum polynomial-time algorithm that takes as input a classical string $x \in \{0, 1\}^*$ and outputs a description of V_x .*
- **Completeness.** *For every $x \in \mathcal{L}_{\text{yes}}$, there exists a quantum state $|w\rangle$*

$$\Pr[\text{Accept} \leftarrow V_x(|w\rangle)] \geq a(n)$$

Such states $|w\rangle$ are called witnesses.

- **Soundness.** *For every $x \in \mathcal{L}_{\text{no}}$ and every state $|\psi\rangle$,*

$$\Pr[\text{Accept} \leftarrow V_x(|\psi\rangle)] < b(n)$$

In general, there may be many verifiers $V = \{V_x\}_{x \in \mathcal{L}_{\text{yes}} \cup \mathcal{L}_{\text{no}}}$ satisfying these conditions. We say $\{V_x\}_{x \in \mathcal{L}_{\text{yes}} \cup \mathcal{L}_{\text{no}}}$ decides $\mathcal{L} \in \text{QMA}_{a,b}$ if it satisfies the above conditions. We say $|w\rangle$ is a witness for x with respect to V if it satisfies the completeness condition using V_x .

In general, the set of witnesses is not a subspace because a witness may be the superposition over a state which is accepted with probability 1 and a state which is accepted with probability 0. Marriot and Watrous [MW05] showed that eigenstates of V_x with eigenvalues $\geq a$ can be amplified to exponentially small completeness error, and moreover that their procedure rejects eigenstates with eigenvalues $\leq b$ with exponentially small soundness error.

Theorem 7 ([MW05]). *Let $a, b : \mathbb{N} \rightarrow [0, 1]$ such that $a(n) - b(n) = 1/\text{poly}(n)$ for all $n \in \mathbb{N}$. There exists a quantum polynomial-time algorithm $\text{MW}_{V_x}(1^\lambda, \cdot)$ such that for every promise language $(\mathcal{L}_{\text{yes}}, \mathcal{L}_{\text{no}})$ in $\text{QMA}_{a,b}$, every $x \in \mathcal{L}_{\text{yes}} \cup \mathcal{L}_{\text{no}}$ with instance size $n \in \mathbb{N}$, and every $\lambda \in \mathbb{N}$, the following properties hold.*

- **Completeness.** *For all $|w\rangle \in \mathcal{ES}_{\geq a}(V_x)$,*

$$\Pr[\text{Accept} \leftarrow \text{MW}_{V_x}(1^\lambda, |w\rangle)] \geq 1 - 2^{-\lambda}$$

- **Soundness.** *For all $|\psi\rangle \in \mathcal{ES}_{\leq b}(V_x)$,¹¹*

$$\Pr[\text{Accept} \leftarrow \text{MW}_{V_x}(1^\lambda, |\psi\rangle)] \leq 2^{-\lambda}$$

3.4 Jordan's Lemma

We provide here a few useful facts for reasoning about the interaction of two projectors.

Lemma 2 (Jordan's Lemma). *For any two Hermitian projectors Π_A and Π_B on a Hilbert space $\mathcal{H}$, there exists an orthogonal decomposition of $\mathcal{H} = \bigoplus_j S_j$ into one-dimensional and two-dimensional subspaces $\{\mathcal{S}_j\}_j$ (referred to as the Jordan subspaces), where each $\mathcal{S}_j$ is invariant under both Π_A and Π_B . Moreover:*

- *in each one-dimensional subspace, Π_A and Π_B act as identity or rank-zero projectors*
- *and in each two-dimensional subspace S_j , Π_A and Π_B are rank-one projectors. In particular, there exist distinct orthogonal bases $\{|\text{jor}_{j,1}^A\rangle, |\text{jor}_{j,0}^A\rangle\}$ and $\{|\text{jor}_{j,1}^B\rangle, |\text{jor}_{j,0}^B\rangle\}$ such that Π_A projects onto $|\text{jor}_{j,1}^A\rangle$ and Π_B projects onto $|\text{jor}_{j,1}^B\rangle$.*

Lemma 3. *Let Π_A and Π_B be Hermitian projectors on a Hilbert space $\mathcal{H}$ and let $\mathcal{H} = \bigoplus_j S_j$ be the Jordan decomposition of the space corresponding to Π_A and Π_B . Then $\Pi_B \Pi_A \Pi_B$ can be eigen-decomposed as*

$$\Pi_B \Pi_A \Pi_B = \sum_j p_j |\text{jor}_{j,1}^B\rangle \langle \text{jor}_{j,1}^B|$$

where $p_j = |\langle \text{jor}_{j,1}^B | \text{jor}_{j,1}^A \rangle|^2$.

Proof. For each $|\text{jor}_{j,1}^B\rangle$, we have

$$\begin{aligned} \Pi_B \Pi_A \Pi_B |\text{jor}_{j,1}^B\rangle &= \Pi_B (|\text{jor}_{j,1}^A\rangle \langle \text{jor}_{j,1}^A| |\text{jor}_{j,1}^B\rangle) \\ &= |\langle \text{jor}_{j,1}^B | \text{jor}_{j,1}^A \rangle|^2 |\text{jor}_{j,1}^B\rangle \end{aligned}$$

Furthermore, $\text{Tr}[\Pi_B \Pi_A \Pi_B |\text{jor}_{j,0}^B\rangle] = 0$ and $\mathcal{H} = \bigoplus_j \mathcal{S}_j$, so this characterizes all eigenvectors of $\Pi_B \Pi_A \Pi_B$ with nonzero eigenvalue. $\square$

¹¹Note that if $x \in \mathcal{L}_{\text{no}}$, then $\mathcal{ES}_{\leq b}(V_x)$ is the entire Hilbert space on $p(n)$ qubits.

3.5 State Repair

Definition 5 (Almost Projective Measurement). *A real-valued measurement $\mathbf{M}$ on $\mathcal{H}$ is (ϵ, δ) -almost-projective if applying $\mathbf{M}$ twice in a row to any state $\rho \in S(\mathcal{H})$ produces measurement outcomes p, p' where*

$$\Pr[|p - p'| \leq \epsilon] \geq 1 - \delta$$

We borrow the following lemmas from [CMSZ22] (using the formalism from [LMS22a]).

Lemma 4 (Value Estimation, [CMSZ22, LMS22a]). *Let $\mathcal{H}$ be a Hilbert space. There exists a quantum algorithm ValEst that satisfies the following guarantees:*

1. $(\rho^*, p^*) \leftarrow \text{ValEst}_V(\rho, \epsilon, \delta)$ is given black-box access to the unitary dilation of a quantum verifier V which takes as input the state ρ and outputs a bit $\{0, 1\}$.¹² On input a quantum state $\rho \in S(\mathcal{H})$ and accuracy parameters $\epsilon, \delta \in (0, 1]$, ValEst_V outputs a quantum state $\rho^* \in S(\mathcal{H})$ and value $p^* \in X \subseteq [-\frac{1}{2}, \frac{3}{2}]$ for some discrete set X where $N_{\epsilon, \delta} := |X| = O(\frac{1}{\epsilon} \log \frac{1}{\delta})$.¹³
2. ValEst is an oracle circuit with $O(\frac{1}{\epsilon} \log \frac{1}{\delta})$ gates.
3. $\mathbb{E}[p^* \mid (\rho^*, p^*) \leftarrow \text{ValEst}_V(\rho, \epsilon, \delta)] = \Pr[V(\rho) = 1]$.
4. For every $p^* \in \mathbb{R}$, if

$$\Pr_{(p, \rho') \leftarrow \text{ValEst}_V(\rho, \epsilon, \delta)}[p \geq p^*] \geq 1 - \gamma, \tag{1}$$

then

$$\Pr[\text{Accept} \leftarrow V(\rho')] \geq p^* - \gamma - \epsilon - \delta.$$

5. $\text{ValEst}_{\epsilon, \delta}^V := \text{ValEst}_V(\cdot, \epsilon, \delta)$ is an **Almost Projective Family**:

$$\Pr \left[|p^* - p^{**}| \geq \max\{\epsilon, \epsilon'\} \mid \begin{array}{l} (\rho^*, p^*) \leftarrow \text{ValEst}_V(\rho, \epsilon, \delta) \\ (\rho^{**}, p^{**}) \leftarrow \text{ValEst}_V(\rho^*, \epsilon', \delta') \end{array} \right] \leq \max\{\delta, \delta'\}.$$

Lemma 5 (Repair, [CMSZ22, LMS22a]). *Let $\mathcal{H}$ be a Hilbert space. There exists a quantum algorithm Repair that satisfies the following guarantees:*

1. $\sigma^* \leftarrow \text{Repair}_{\mathbf{M}, \Pi}(\sigma, y, p, T)$ is given black-box access to an (ϵ, δ) -almost projective measurement $\mathbf{M}$ and a projective measurement $\Pi = (\Pi_y)_{y \in Y}$ on $\mathcal{H}$. On input a quantum state $\sigma \in S(\mathcal{H})$, an outcome $y \in Y$, a probability $p \in [0, 1]$, and a maximum runtime $T \in \mathbb{N}$, it outputs a quantum state $\sigma^* \in S(\mathcal{H})$.

¹² V is implemented by a unitary U acting on registers $\mathcal{X} \otimes \mathcal{A}$. It takes as input a state on register $\mathcal{X}$, then initializes $\mathcal{A}$ to $|0^n\rangle$ and runs U . By black-box access to the unitary dilation, we mean that ValEst_V has black-box access to the unitaries U and $U^\dagger$, along with direct access to $\mathcal{A}$.

¹³The reader should think of p^* as a noisy estimate of the success probability, whose expectation is in $[0, 1]$.

2. For any (ϵ, δ) -almost projective measurement $\mathbf{M}$ on $\mathcal{H}$, any projective measurement Π , and any mixed state ρ ,

$$\Pr \left[|p^* - p^{**}| \geq 2\epsilon \mid \begin{array}{l} (\rho_{p^*}, p^*) \leftarrow \mathbf{M}(\rho) \\ (\sigma, y) \leftarrow \Pi(\rho_{p^*}) \\ \sigma^* \leftarrow \text{Repair}_{\mathbf{M}, \Pi}(\sigma, y, p^*, T) \\ (\rho^{**}, p^{**}) \leftarrow \mathbf{M}(\sigma^*) \end{array} \right] \leq N \cdot (\delta + 1/T) + 4\sqrt{\delta},$$

where $N = |Y|$ is the number of outcomes the projective measurement Π can obtain.

3. **Repair** is a variable-runtime oracle algorithm making $N + 4T\sqrt{\delta} + 1$ oracle queries in expectation.

Setting $T = 1/\sqrt{\delta}$, the expected number of oracle calls is $N + 5$ and the probability that $|p^* - p^{**}| \geq 2\epsilon$ is $O(N\sqrt{\delta})$.

3.6 Classical Verification of Quantum Computation

Definition 6. A *Classical Verification of Quantum Computation (CVQC)* protocol for a $\text{QMA}_{a,b}$ language $\mathcal{L} = (\mathcal{L}_{\text{yes}}, \mathcal{L}_{\text{no}})$ associated with a verifier $V_{\text{QMA}} = \{V_{\text{QMA},x}\}_{x \in \mathcal{L}_{\text{yes}} \cup \mathcal{L}_{\text{no}}}$ deciding $\mathcal{L} \in \text{QMA}_{a,b}$, is an interactive protocol between a quantum prover P and a classical verifier V .

It satisfies the following for some parameters $c = c(\lambda, |x|)$ and $s = s(\lambda, |x|)$.

- **Completeness c .** For every $\lambda \in \mathbb{N}$, every $x \in \mathcal{L}_{\text{yes}}$ of size at most 2^λ , and every witness $|w\rangle$ such that

$$\Pr[V_{\text{QMA},x}(|w\rangle) = 1] \geq a,$$

it holds that

$$\Pr[(P(|w\rangle), V)(1^\lambda, x) = 1] \geq c(\lambda, |x|).$$

Remark 2. Many of the CVQC protocols in the literature (such as [Mah18]) need many copies of the witness to obtain completeness. The focus of this work is on constructing CVQC protocols where the prover uses only a single copy of the witness, and hence we define completeness in this restrictive manner.

- **Computational soundness s .** For every QPT cheating prover P^* , every $\lambda \in \mathbb{N}$ and every $x \in \mathcal{L}_{\text{no}}$ it holds that for every state $|\psi\rangle$,

$$\Pr[P^*(|\psi\rangle), V)(1^\lambda, x) = 1] \leq s(\lambda, |x|).$$

- **Efficiency.** For every λ and every $x \in \mathcal{L}_{\text{yes}}$ with a corresponding witness $|w\rangle$, the honest prover $P(1^\lambda, x, |w\rangle)$ is a (uniform) quantum circuit of size $\text{poly}(\lambda, |x|)$, and the verifier $V(1^\lambda, x)$ is a classical probabilistic polynomial time circuit of size $\text{poly}(\lambda, |x|)$.

A CVQC protocol can additionally be a *classical proof of quantum knowledge* [VZ21], which is a generalization of the classical concept of a proof of knowledge to the quantum setting.

Definition 7 (η -Argument of Knowledge). *Let $\mathcal{R}$ be a QPT algorithm outputting a decision bit, which we call the “QMA relation”. An interactive protocol between a classical verifier and a quantum prover is an η -argument of (quantum) knowledge for $\mathcal{R}$ if there exists a QPT extractor $\mathcal{E}$ such that for every statement $x \in \mathcal{L}_{\text{yes}} \cup \mathcal{L}_{\text{no}}$ and every dishonest QPT prover P^* which convinces the verifier with probability $s(\lambda, |x|)$, the extractor outputs a quantum state satisfying*

$$\Pr[\text{Accept} \leftarrow \mathcal{R}(1^\lambda, x, \rho) : \rho \leftarrow \mathcal{E}(1^\lambda, P^*, x)] \geq \eta(\lambda, |x|)s(\lambda, |x|).$$

This definition is weaker than the one in the classical setting, which guarantees that if P^* convinces the verifier w.p. $\frac{1}{p(\lambda)}$ then the extractor extracts a valid witness with overwhelming probability (after running in time $\text{poly}(\lambda, p(\lambda))$). This difference is inherent since if P^* has a state $\alpha|w\rangle + \beta|\text{junk}\rangle$ then we cannot hope to extract $|w\rangle$ with probability greater than $|\alpha|^2$.

Testable Soundness and Proof of Knowledge. For use in our technical sections, we now define *testable* soundness and proof of knowledge. Intuitively, we amplify by sequentially repeating a base CVQC protocol with high completeness and soundness errors, and indeed where the completeness error is higher than the soundness error. Fortunately, the usefulness of this protocol comes from a special structure: the verifier V is specified by a pair of verification algorithms $(V_{\text{T}}, V_{\text{C}})$. On any execution, the verifier chooses between them uniformly at random: it runs a test phase V_{T} with probability $1/2$ and a check phase V_{C} with probability $1/2$. Crucially, V_{T} has *completeness* 1, whereas V_{C} has a *lower completeness* c ¹⁴.

The protocol’s security guarantee is as follows: for every $x \in \mathcal{L}_{\text{no}}$ and any cheating prover P^* that succeeds at convincing V_{T} with probability close to 1, the probability of convincing V_{C} is bounded above by $s < c$. We define *testable soundness* below, parameterized by s —the upper bound on the probability that V_{C} accepts $x \in \mathcal{L}_{\text{no}}$.

Definition 8 (s -Testable Soundness). *A CVQC protocol for language $\mathcal{L}$ has testable soundness $s = s(\lambda, |x|)$ if there is a constant $\zeta > 0$ such that the following holds. The verifier uniformly randomly picks one out of two verification algorithms: a “test” algorithm V_{T} and a “check” algorithm V_{C} , where V_{T} has perfect completeness, and for every QPT cheating prover P^* , every $\lambda \in \mathbb{N}$ and every $x \in \mathcal{L}_{\text{no}}$ it holds that for every (mixed) state ρ consisting of at most $\text{poly}(\lambda, |x|)$ qubits, if*

$$\Pr[P^*(\rho), V_{\text{T}}(1^\lambda, x) = 1] \geq \zeta \tag{2}$$

then

$$\Pr[P^*(\rho), V_{\text{C}}(1^\lambda, x) = 1] \leq s(\lambda, |x|) \tag{3}$$

¹⁴This arises because the original QMA witness is transformed into a Morimae–Fitzsimons [MF16] QMA witness, which has low completeness.

The following definition generalizes the definition of testable soundness to an argument-of-knowledge setting, where one can extract a witness from a prover that succeeds in both phases. The key property is as follows: whenever a prover succeeds in both (i) convincing V_T with high probability and (ii) convincing V_C with at least some threshold probability s , there exists a *quantum polynomial-time extractor* $\mathcal{E}$ that can recover a valid witness σ for the underlying QMA relation $\mathcal{R}$ with probability at least δ .

Definition 9 ($((d, s, \delta)$ -Testable Argument of Knowledge). *Let $\mathcal{R}$ be a QPT algorithm outputting a decision bit, which we call the “QMA relation”. A CVQC protocol for language $\mathcal{L}$ is a $((d, s, \delta)$ testable argument of knowledge for $\mathcal{R}$ if the following holds, where d is a constant, $s = s(\lambda, |x|)$ is a polynomial, and δ is a function of $(\lambda, |x|)$ that takes values between 0 and 1. The verifier uniformly randomly picks one out of two verification algorithms: a “test” algorithm V_T and a “check” algorithm V_C , where V_T has perfect completeness, and there is a QPT extractor $\mathcal{E}$ such that for every QPT prover P^* , every $\lambda \in \mathbb{N}$ and every $x \in \mathcal{L}_{\text{yes}} \cup \mathcal{L}_{\text{no}}$ it holds that for every state ρ consisting of at most $\text{poly}(\lambda, |x|)$ qubits, if*

$$\Pr[P^*(\rho), V_T(1^\lambda, x) = 1] \geq 1 - (1/|x|^{d\lambda^d}) \quad (4)$$

and

$$\Pr[P^*(\rho), V_C(1^\lambda, x) = 1] \geq s(\lambda, |x|) \quad (5)$$

then

$$\Pr[\text{Accept} \leftarrow \mathcal{R}(1^\lambda, x, \sigma) : \sigma \leftarrow \mathcal{E}^{P^*}(\rho, 1^\lambda, x)] \geq \delta(\lambda, |x|)$$

We note that a single repetition of Mahadev’s CVQC protocol [Mah18] has completeness c w.r.t. V_C and satisfies s -testable soundness with $(c - s) > \frac{1}{\text{poly}(\lambda, |x|)}$. It is also a $((d, s, \delta)$ -testable argument of knowledge for a large enough constant d , with $(c - s) > \frac{1}{\text{poly}(\lambda, |x|)}$ and $\delta(\lambda) > 1 - 1/\lambda$. The testable argument of knowledge property is proven implicitly in Theorem 6.5 of [VZ21].

Non-adaptive Verifiers. Finally, we will only compile protocols that admit non-adaptive verifiers, which we define below. First, we provide some notation.

If the CVQC protocol consists of ℓ back-and-forth rounds, we denote by $(q_1, \dots, q_\ell)$ the verifier’s messages, and by $(a_1, \dots, a_\ell)$ the prover’s messages, where in round i the verifier sends query q_i and the verifier responds with answer a_i .

Definition 10. *An ℓ -round CVQC protocol is said to have a non-adaptive verifier if the verifier’s queries do not depend on the provers answers; namely, a non-adaptive verifier can be partitioned into two PPT algorithm $V = (V_1, V_2)$, where V_1 is the query sampler*

$$(\text{st}, q_1, \dots, q_\ell) \leftarrow V_1(x, 1^\lambda)$$

and V_2 is the verdict function

$$V_2(\text{st}, a_1, \dots, a_\ell) \in \{0, 1\}.$$

We note that all CQVC protocols that we are aware of have a non-adaptive verifier [Mah18, ACGH20, CCY20, BKL⁺22, Zha22, KLVY23, MNZ24, GKNV25, BKM⁺25, BK25]¹⁵ (though they don't all satisfy our desired completeness guarantee that completeness holds with a single witness).

4 Repairing High-Quality Witnesses

In this section, we prove that [CMSZ22]'s state repair procedure can repair high quality QMA witnesses – those that are accepted with overwhelming probability – with overwhelming probability.

In particular, [CMSZ22]'s state repair procedure uses a special probability estimation algorithm **ValEst**.¹⁶ Their repair procedure can be thought of as using **ValEst** to estimate the probability p of $V(|\psi\rangle)$ accepting, then repairing a damaged $|\psi\rangle$ so that the repaired state is accepted with probability $p - \epsilon$. However, **ValEst** is only guaranteed to be accurate *on average*. For example, if $|\psi\rangle$ has success probability p , and is a superposition of $|\psi_1\rangle$ and $|\psi_2\rangle$, where $|\psi_1\rangle$ is accepted with probability 1 and $|\psi_2\rangle$ is accepted with probability 0, the **ValEst** procedure may collapse $|\psi\rangle$ into a mixture of $|\psi_1\rangle$ and $|\psi_2\rangle$. If the witness is collapsed to $|\psi_2\rangle$, then the repair procedure will never return the success probability to $\approx p$. Although intuitively an overwhelmingly-good witness cannot collapse like this, that fact surprisingly does not follow from the black-box properties of **ValEst** presented in [CMSZ22].¹⁷

To ensure that we can repair a QMA witness $|\psi\rangle$ for QMA, we need to ensure that **ValEst**($|\psi\rangle$) gives an estimate close to the probability that $|\psi\rangle$ passes the QMA verification procedure (e.g. $2/3$) *with overwhelming probability*. This will rule out the possibility of a problematic collapse, except with negligible probability.

We show in this section that if $|\psi\rangle$ is an eigenvector of the verification procedure V with eigenvalue p , then **ValEst**($|\psi\rangle$) returns an estimate $p \pm \epsilon$ with overwhelming probability. Such states are exactly Marriot-Watrous witnesses when p is greater than the QMA threshold (e.g. $2/3$). As a corollary, any damaged witness which was originally accepted by V with overwhelming probability can be repaired to a state which is accepted with $1 - 3\epsilon - \text{negl}(\lambda)$ probability.¹⁸

We prove a concentration inequality for **ValEst** on high-quality witnesses in Section 4.1 (Lemma 6 and Corollary 1). Then, we use this to show in Section 4.2 that repairing a

¹⁵[BK25]'s verifier messages technically depend on the prover's messages. However, that part of the protocol is independent of the statement being proven, so we can view it as a non-adaptive CVQC in the (classical) preprocessing model, which is also sufficient for our compilers.

¹⁶Technically their state repair procedure can be used with any almost-projective measurement, but a probability estimation algorithm is required to repair the probability of acceptance of a QMA witness.

¹⁷In more detail, they only give the implication that $p = \Pr[\text{Accept} \leftarrow V(\rho)]$ implies that **ValEst**(ρ) outputs p on expectation (property 3 of Lemma 4). Unfortunately, even if p is close to 1 this is not enough to argue that **ValEst** outputs a value close to 1 with high probability because **ValEst** can return an estimate which is larger than 1, since its output is in $[-\frac{1}{2}, \frac{3}{2}]$.

¹⁸Running **ValEst** may reduce the accepting probability by an additive factor of ϵ , and running **Repair** may reduce the accepting probability by another additive factor of 2ϵ (see Lemma 5).

damaged high-quality witness results in a state ρ which is overwhelmingly supported on witnesses of $1 - 3\epsilon$ quality.

A Note on Other Probability Estimation Algorithms. [MW05]’s original probability estimation algorithm is unfortunately *not* compatible with the repair technique, because it is not approximately projective. [Zha20] refines [MW05]’s algorithm so that it becomes approximately projective, although the resulting algorithm runs in *expected* polynomial time as opposed to strict polynomial time. Unfortunately, [CMSZ22]’s repair procedure requires the (ϵ, δ) -approximately-projective measurement to run in *strict* polynomial time.¹⁹ [CMSZ22] modifies Zhandry’s algorithm so that it can be truncated while still being approximately projective.

4.1 Estimating the Acceptance Probability of a High-Quality Witness

Lemma 6. *Let V be a binary-outcome quantum algorithm and let ValEst be [CMSZ22]’s probability estimation algorithm. For every eigenstate $|\psi\rangle$ of V with eigenvalue p^* ,*

$$\Pr[|p - p^*| > \epsilon : (p, \rho) \leftarrow \text{ValEst}_V(|\psi\rangle, \epsilon, \delta)] \leq \delta.$$

Furthermore, if $|\phi\rangle = \sum_j \alpha_j |\psi_j\rangle$ for eigenstates $|\psi_j\rangle$ of V with eigenvalues p_j and there is a range $[p_1^, p_2^*]$ such that*

$$\sum_{j: p_j \notin [p_1^*, p_2^*]} |\alpha_j|^2 \leq \eta,$$

then

$$\Pr[p \notin [p_1^* - \epsilon, p_2^* + \epsilon] : (p, \rho) \leftarrow \text{ValEst}_V(|\phi\rangle, \epsilon, \delta)] \leq \delta + \eta$$

Using this lemma, we can bound the probability that $\text{ValEst}(|\psi\rangle)$ gives an estimate close to 1 when $|\psi\rangle$ is accepted by V with overwhelming probability by reasoning about its representation in the eigenbasis of V .

Corollary 1. *For any $|\phi\rangle$ such that $\Pr[\text{Accept} \leftarrow V(|\phi\rangle)] \geq 1 - \eta$, where $\sqrt{\eta} = o\left(\frac{1}{\epsilon \log(\frac{1}{\delta})}\right)$,*

$$\Pr[|1 - p| \geq \epsilon : (p, \rho) \leftarrow \text{ValEst}_V(|\phi\rangle, \epsilon, \delta)] \leq \delta + \sqrt{\eta}.$$

¹⁹For example, it is tempting to simply truncate the runtime in Zhandry’s algorithm. However the extra runtime is necessary to return the state to its “original form” so that the procedure is approximately projective. Specifically, if the true value being estimated is p , it takes $\approx 1/p$ time to return the state to its “original form.”

Proof. Decompose $|\phi\rangle = \sum_j \alpha_j |\psi_j\rangle$ in terms of eigenstates $|\psi_j\rangle$ of V with eigenvalues p_j . The probability of acceptance is

$$\begin{aligned}
\langle \phi | V | \phi \rangle &= \sum_j |\alpha_j|^2 p_j \\
&= \sum_{j:p_j \geq 1-\sqrt{\eta}} |\alpha_j|^2 p_j + \sum_{j:p_j < 1-\sqrt{\eta}} |\alpha_j|^2 p_j \\
&\leq \sum_{j:p_j \geq 1-\sqrt{\eta}} |\alpha_j|^2 + \sum_{j:p_j < 1-\sqrt{\eta}} |\alpha_j|^2 (1-\sqrt{\eta}) \\
&= 1 - \sqrt{\eta} \sum_{j:p_j < 1-\sqrt{\eta}} |\alpha_j|^2
\end{aligned}$$

Comparing to the bound $\Pr[\text{Accept} \leftarrow V(|\phi\rangle)] \geq 1 - \eta$ and rearranging,

$$\sum_{j:p_j < 1-\sqrt{\eta}} |\alpha_j|^2 \leq \sqrt{\eta}.$$

Setting $p_1^* = 1 - \sqrt{\eta}$ and noting that the maximum eigenvalue of V is 1, the second half of Lemma 6 implies that $p \in [1 - \sqrt{\eta} - \epsilon, 1 + \epsilon]$ except with probability $\delta + \sqrt{\eta}$. Finally, $\text{ValEst}_V(\cdot, \epsilon, \delta)$ has precision $1/\Theta(\frac{1}{\epsilon} \log(\frac{1}{\delta}))$. We may round up the lower bound to the nearest multiple of the precision, which is $1 - \epsilon$ since $\sqrt{\eta}$ is asymptotically smaller than the precision.²⁰ $\square$

Proof of Lemma 6. We recall **ValEst** for completeness. Let the unitary U_V and the projective measurement Π_V denote the implementation of V . Let $\mathcal{H}$ be the input register to V . $t = \Theta(\frac{1}{\epsilon} \log(\frac{1}{\delta}))$ is a parameter controlling the runtime – see [CMSZ22] for the exact value. $\text{NReps}(\mathbf{b})$ is the number of consecutive repeated bits in $\mathbf{b}$, divided by $|\mathbf{b}| - 1$; for example, $\text{NReps}(0011110) = 3/5$.

1. Initialize register $\mathcal{R}$ to $|0\rangle$ and initialize register $\mathcal{T}$ to $|_{+\tau, \perp}\rangle := \frac{1}{\sqrt{2}}(|0\rangle + |\top/\perp\rangle)$, where $|_{\top/\perp}\rangle := \frac{1}{\sqrt{2}}(|\top\rangle + |\perp\rangle)$. Here, $|\top\rangle$ and $|\perp\rangle$ are special basis elements representing “automatically” winning or losing the game, respectively.
2. Define measurement $M_G = (\Pi_G, I - \Pi_G)$ where

$$\Pi_G := \left(U_V^\dagger \Pi_V^{\text{Accept}} U_V \right) \otimes |0\rangle\langle 0|_{\mathcal{T}} + I_{\mathcal{H}, \mathcal{R}} \otimes |\top\rangle\langle \top|_{\mathcal{T}}$$

In words, Π_G evaluates V when $\mathcal{T}$ contains $|0\rangle$ and accepts any input when $\mathcal{T}$ contains $|\top\rangle$. It rejects any input when $\mathcal{T}$ contains $|\perp\rangle$.

Define the measurement $M_{\text{reset}} = (|0, +\tau, \perp\rangle\langle 0, +\tau, \perp|_{\mathcal{R}, \mathcal{T}}, I - |0, +\tau, \perp\rangle\langle 0, +\tau, \perp|_{\mathcal{R}, \mathcal{T}})$.

3. For $i = 1$ to t :

²⁰We assume that $1/\epsilon$ is an integer, since taking the floor of $1/\epsilon$ does not change the algorithm.

- (a) Apply M_G , obtaining outcome $L_{2i-1} \in \{0, 1\}$.
- (b) Apply M_{reset} to registers $(\mathcal{R}, \mathcal{T})$, obtaining outcome $L_{2i} \in \{0, 1\}$.
4. If $L_{2t} = 1$, skip to the next step. Otherwise, apply M_G and M_{reset} in an alternating fashion as above until M_{reset} outputs 1 or a further $2t$ measurements have been applied.
5. Discard registers $\mathcal{R}$, $\mathcal{R}'$, and $\mathcal{T}$, then output $p := 2\text{NReps}(1, L_1, \dots, L_{2t})$.

This is equivalent to [CMSZ22]’s algorithm, but the description varies slightly. In [CMSZ22], they consider V to take in ancillas of the form $\frac{1}{\sqrt{|R|}} \sum_{r \in R} |r, r\rangle$, rather than $|0\rangle$. These are easily accounted for by having U_V first map $|0\rangle$ to $\frac{1}{\sqrt{|R|}} \sum_{r \in R} |r, r\rangle$. The other difference is that we separate the “automatic” win/loss symbols into their own register, rather than squeezing them into $\mathcal{R}$ as well. Since $|\top\rangle$ and $|\perp\rangle$ are already orthogonal to the other computational basis vectors in [CMSZ22], this has the same effect but makes the definition of Π_G easier to read.

Applying Jordan’s lemma (Lemma 2) to Π_G and $|_{+\top, \perp}\rangle\langle_{+\top, \perp}|_{\mathcal{R}, \mathcal{R}', \mathcal{T}}$ allows the orthogonal decomposition $\mathcal{H} \otimes \mathcal{R} \otimes \mathcal{T} = \bigoplus_j \mathcal{S}_j$ into one- and two-dimensional subspaces $\mathcal{S}_j$, which we call *Jordan subspaces*. Each $\mathcal{S}_j$ is invariant under both Π_G and $|0, +\top, \perp\rangle\langle 0, +\top, \perp|_{\mathcal{R}, \mathcal{T}}$. The projection of $\mathcal{S}_j$ onto Π_G is a state $|\text{jor}_{j,1}^G\rangle$. Similarly, the projection of $\mathcal{S}_j$ onto $|0, +\top, \perp\rangle\langle 0, +\top, \perp|_{\mathcal{R}, \mathcal{T}}$ is a state $|\text{jor}_{j,1}^{\text{reset}}\rangle$. Define

$$p_j := |\langle \text{jor}_{j,1}^G | \text{jor}_{j,1}^{\text{reset}} \rangle|^2$$

Claim 1 ([CMSZ22], Lemma 4.5 + Proposition 4.7 + Chernoff Bound). *If*

$$|\psi\rangle \otimes |0, +\top, \perp\rangle = \sum_j \alpha_j |\text{jor}_{j,1}^{\text{reset}}\rangle$$

then $\text{ValEst}_V(|\psi\rangle, \epsilon, \delta)$ is distributed as

1. Sample j with probability $|\alpha_j|^2$.
2. Sample $p' \sim \text{Binomial}(2t, p_j)$ and output $2p' - 1/2$.

In particular, conditioned on choosing j , $\Pr[|p - p_j| \geq \epsilon] < \delta$.

Thus, it suffices to show that if $|\psi\rangle$ is an eigenstate of

$$V = (I \otimes \langle 0|) U_V^\dagger \Pi_V^{\text{Accept}} U_V (I \otimes |0\rangle)$$

with eigenvalue p^* , then $|\psi\rangle \otimes |0, +\top, \perp\rangle = |\text{jor}_{j,1}^{\text{reset}}\rangle$ with corresponding Jordan value $p_j = p^*/2 + 1/4$. If this is the case, the first statement of the lemma holds immediately. The second statement, for $|\phi\rangle = \sum_j \alpha_j |\psi_j\rangle$, follows from the observation that a j such that $p_j \in [p_1^*, p_2^*]$ will be sampled with probability $\geq 1 - \eta$.

By Lemma 3, it in turn suffices to show that $|\psi\rangle \otimes |0, +\top, \perp\rangle$ is an eigenstate of

$$\left(I_{\mathcal{H}} \otimes |0, +\top, \perp\rangle\langle 0, +\top, \perp| \right) \Pi_G \left(I_{\mathcal{H}} \otimes |0, +\top, \perp\rangle\langle 0, +\top, \perp| \right)$$

with eigenvalue $(p^*/2 + 1/4)$. The eigenstates of this operator are the same as the eigenstates of

$$\left(I \otimes \langle 0, +_{\top, \perp} | \right) \Pi_G \left(I_{\mathcal{H}} \otimes |0, +_{\top, \perp} \rangle \right)$$

with the same eigenvalues, when the latter are appended with $|0, +_{\top, \perp}\rangle$. Expanding this operator,

$$\begin{aligned} \left(I_{\mathcal{H}} \otimes \langle 0, +_{\top, \perp} | \right) \Pi_G \left(I_{\mathcal{H}} \otimes |0, +_{\top, \perp} \rangle \right) &= \frac{1}{2} \left(I_{\mathcal{H}} \otimes \langle 0, 0 | \right) U_V^\dagger \Pi_V^{\text{Accept}} U_V \left(I_{\mathcal{H}} \otimes |0, 0 \rangle \right) \\ &\quad + \frac{1}{2} \left(I_{\mathcal{H}} \otimes \langle 0, 0 | \right) \Pi_G \left(I_{\mathcal{H}} \otimes |0, \top \rangle \right) \\ &\quad + \frac{1}{2} \left(I_{\mathcal{H}} \otimes \langle 0, \top | \right) \Pi_G \left(I_{\mathcal{H}} \otimes |0, 0 \rangle \right) \\ &\quad + \frac{1}{4} \left(I_{\mathcal{H}} \otimes \langle 0, \top | \right) I \left(I_{\mathcal{H}} \otimes |0, \top \rangle \right) \end{aligned}$$

Since Π_G is diagonal on $|0\rangle_{\mathcal{T}}$, $|\top\rangle_{\mathcal{T}}$, and $|\perp\rangle_{\mathcal{T}}$ the cross terms cancel and we are left with

$$\frac{1}{2}V + \frac{1}{4}I_{\mathcal{H}}$$

Finally, any eigenstate of V with eigenvalue p^* is also an eigenstate of $\frac{1}{2}V + \frac{1}{4}I_{\mathcal{H}}$ with eigenvalue $\frac{p^*}{2} + \frac{1}{4}$. □

4.2 Result of Repairing a High-Quality Witness

We prove here that repairing a damaged high-quality witness will, with overwhelming probability, produce a witness which is accepted with probability $1 - 3\epsilon - \text{negl}(\lambda)$. Using corollary 1, estimating the value of a high quality witness $|w\rangle$ will return $\geq 1 - \epsilon$ with overwhelming probability. Lemma 5 guarantees that the repaired mixed state ρ then returns an estimate within 2ϵ of this with overwhelming probability, i.e. the final estimate is $\geq 1 - 3\epsilon$. Finally, we show in the following lemma that this implies that ρ is overwhelmingly supported on *pure state* witnesses which are accepted with probability $\geq 1 - 3\epsilon - \text{negl}(\lambda)$.

Lemma 7. *Let ρ be a state such that*

$$\Pr[p \geq p^* : (p, \rho') \leftarrow \text{ValEst}_V(\rho, \epsilon, \delta)] \geq 1 - \gamma$$

for some p^ . Then $\rho = q_{\text{good}}\rho_{\text{good}} + q_{\text{bad}}\rho_{\text{bad}}$ can be decomposed as a mixture over two orthogonal mixed states where*

$$q_{\text{bad}} \leq \sqrt{\gamma}$$

and ρ_{good} is supported completely on pure states $|\psi\rangle$ such that

$$\Pr[\text{Accept} \leftarrow V(|\psi\rangle)] \geq p^* - \sqrt{\gamma} - \epsilon - \delta$$

Proof. Consider the binary-outcome measurement where one (coherently) estimates $p \leftarrow \text{ValEst}_V(|\psi\rangle, \epsilon, \delta)$ and measures whether $p \geq p^*$ or $p < p^*$. By Lemma 1, we can decompose $\rho = q_{\text{good}}\rho_{\text{good}} + q_{\text{bad}}\rho_{\text{bad}}$ where

$$q_{\text{bad}} \leq \sqrt{\gamma}$$

and ρ_{good} is supported completely on pure states $|\psi\rangle$ such that

$$\Pr[p \geq p^* \leftarrow \text{ValEst}_V(|\psi\rangle, \epsilon, \delta)] \geq 1 - \sqrt{\gamma}$$

By property 4 of Lemma 4, each such $|\psi\rangle$ also satisfies

$$\Pr[\text{Accept} \leftarrow V(|\psi\rangle)] \geq p^* - \sqrt{\gamma} - \epsilon - \delta$$

□

5 Dual-Mode Trapdoor Functions with State Recovery.

In this section we define the notion of a *dual-mode trapdoor function family with state recovery*. Intuitively, this is a family of (randomized) functions that have two modes: an *injective mode* and a *recovery mode*, and it is hard to distinguish between the two. Functions in the injective mode are injective and are associated with a trapdoor which can be used to invert the function efficiently. Functions in the recovery mode are also associated with a trapdoor, where this trapdoor is used to recover the state (this roughly corresponds to the lossy/two-to-one mode in the existing literature on dual-mode TCFs). More specifically, functions in recovery mode have the property that if we compute the function coherently on a quantum state and measure the output value then this does not disturb the state, and we can use the trapdoor to recover the quantum state.

Definition 11 (Dual-Mode Trapdoor Function Family with State Recovery.). *A **dual-mode trapdoor function family with state recovery** is a family of (potentially randomized) functions*

$$\mathcal{F} = \{f_{\text{pp}} : \mathcal{X} \times \mathcal{R} \rightarrow \mathcal{Y}\}_{\text{pp}}$$

*satisfying the following properties.*²¹

- **Efficient Generation.** *There exists a probabilistic poly-time algorithm **Setup** distribution, which takes as input security parameter 1^λ along with a flag $\text{mode} \in \{\text{recovery}, \text{injective}\}$, and outputs*

$$(\text{pp}, \text{td}) \leftarrow \text{Setup}(1^\lambda, \text{mode}).$$

²¹Formally, $\mathcal{X}, \mathcal{R}, \mathcal{Y}$ all depend on the security parameter with respect to which pp was generated. This is omitted from the notation for the sake of simplicity.

- **Dual-Mode Indistinguishability.**

$$\{\text{pp} : (\text{pp}, \text{td}) \leftarrow \text{Setup}(1^\lambda, \text{injective})\} \approx_c \{\text{pp} : (\text{pp}, \text{td}) \leftarrow \text{Setup}(1^\lambda, \text{recovery})\}$$

where by $\approx_c$ we mean computational indistinguishability against QPT algorithms.

- **Efficient Evaluation and Range Superposition.** There is a poly-time computable evaluation function that given any pp , in the image of Setup and given any pair $(x, r) \in (\mathcal{X} \times \mathcal{R})$ outputs $f_{\text{pp}}(x, r)$.

Furthermore, f_{pp} is associated with a distribution over randomness $\mathcal{R}$ with probability mass function p such that the state $|\text{rand}_{\text{pp}}\rangle = \sum_{r \in \mathcal{R}} \sqrt{p(r)} |r\rangle$ is efficiently preparable. This state is a superposition over randomness used to evaluate f_{pp} .

- **Injective Mode: Trapdoor.** There exists a poly-time algorithm Ext such that for all $(\text{pp}, \text{td}) \in \text{SUPP}(\text{Setup}(1^\lambda, \text{injective}))$ and all input/randomness pairs $(x, r) \in \mathcal{X} \times \mathcal{R}$,

$$\Pr[x \leftarrow \text{Ext}(\text{td}, f_{\text{pp}}(x; r))] = 1$$

- **Recovery Mode: Trapdoor.** Consider the following experiment $\text{Exp}(\text{pp}, |\psi\rangle)$, parameterized by a public parameter pp and a quantum state $|\psi\rangle = \sum_x \alpha_x |\phi_x\rangle_A \otimes |x\rangle_B$ in registers A and B .

1. Prepare $|\text{rand}_{\text{pp}}\rangle$ in register C and evaluate f_{pp} coherently on register B to obtain $\sum_x \alpha_x |\phi_x, x\rangle_{A,B} \otimes \sum_r \sqrt{p(r)} |r, f_{\text{pp}}(x; r)\rangle$.
2. Measure the register containing $f_{\text{pp}}(x; r)$ in the computational basis to obtain y . Let $|\varphi_y\rangle$ be the residual state.

There exists a QPT algorithm Recover acting on registers B and C such that for all $(\text{pp}, \text{td}) \in \text{SUPP}(\text{Setup}(1^\lambda, \text{recovery}))$ and all quantum states $|\psi\rangle = \sum_x \alpha_x |x\rangle$,

$$\Pr \left[\frac{1}{2} \| |\psi\rangle\langle\psi| - \rho \|_1 \leq \text{negl}(\lambda) : \begin{array}{l} (|\varphi_y\rangle, y) \leftarrow \text{Exp}(\text{pp}, |\psi\rangle) \\ \rho \leftarrow \text{Recover}(\text{td}, |\varphi_y\rangle, y) \end{array} \right] = 1 - \text{negl}(\lambda)$$

In other words, the channel $\text{Recover}_{\text{td}} \circ \text{Exp}_{\text{pp}}$ is close in diamond distance to the identity channel, where

$$\text{Recover}_{\text{td}} \triangleq \text{Recover}(\text{td}, \cdot, \cdot) \quad \text{and} \quad \text{Exp}_{\text{pp}} \triangleq \text{Exp}(\text{pp}, \cdot).$$

As a more granular definition, if the recovered state ρ has trace distance δ to the original state $|\psi\rangle$, we refer to the function family as having $(1 - \delta)$ -state recovery.

Because these function families are evaluated in the computational basis, they support coherently implemented classical computation on the encoded data followed by recovery. More concretely, imagine that one had already evaluated f_{pp} on a state $\sum_x \alpha_x |x\rangle$, but wanted to measure some classical predicate of x . Computing on x and measuring the result, and then performing Recover produces a state which is close to the state obtained if the computation had been performed directly on $\sum_x \alpha_x |x\rangle$. This is formalized in the following lemma.

Lemma 8. *Let $U = \sum_{x \in \mathcal{X}} U_x \otimes |x\rangle\langle x|$ be a unitary acting on registers $\mathcal{A}$ and $\mathcal{B}$ for some set of unitaries $\{U_x\}_{x \in \mathcal{X}}$. Then the following two channels are negligibly far in diamond distance for all $(\text{pp}, \text{td}) \in \text{SUPP}(\text{Setup}(1^\lambda, \text{recovery}))$:*

$$\left| (I_{\mathcal{A}} \otimes \text{Recover}_{\text{td}}) \circ (U \otimes I_{\mathcal{C}}) \circ (I_{\mathcal{A}} \otimes \text{Exp}_{\text{pp}}) - (U \otimes I_{\mathcal{C}}) \right|_{\diamond} = \text{negl}(\lambda)$$

where $\text{Recover}_{\text{td}}$ and Exp_{pp} act only on register $\mathcal{B}$ and an ancilla register $\mathcal{C}$ (which is used to contain the randomness r).

If $\mathcal{F}$ has $(1 - \delta)$ -state recovery, then the distance is $\leq \delta$.

Proof. First, observe that $U \otimes I_{\mathcal{C}}$ commutes with $I_{\mathcal{A}} \otimes \text{Exp}_{\text{pp}}$ because both U and Exp_{pp} are diagonal in the computational basis on register B and otherwise operate on disjoint registers. Then, using submultiplicativity of the diamond norm to factor out U and the recovery mode trapdoor property imply the claim. $\square$

Theorem 8. *For any $n \in \mathbb{N}$, there exists a dual-mode trapdoor function family with $(1 - 2^{-\lambda})$ -state recovery for $\mathcal{X} = \{0, 1\}^n$ assuming the post-quantum hardness of LWE .*

The construction is very similar to the one from [BCM⁺18, Mah18], except for the new Recover algorithm. At a high level, the result of evaluating [BCM⁺18, Mah18]’s trapdoor claw-free function in superposition and measuring the result y is close to

$$\sum_x \alpha_x |x, r_x\rangle$$

where $f_{\text{pp}}(x; r_x) = y$ for all x . By using the trapdoor to invert r_x from y (controlled on x), the second register can be returned to 0, recovering the original state.

Other Constructions. In addition to the LWE -based construction presented here, we sketch a few other constructions.

- **Classical Commitments to Quantum States.** Any classical commitment to a quantum state [GKNV25] satisfies the recovery property. If the receiver sends the committer their secret state, the prover can open and decode a measurement of their state in either the computational or Hadamard basis. This permits the committer to implement the extractor from [VZ21, GKNV25], which succeeds with overwhelming probability since the committer can generate accepting openings with overwhelming probability.

To create a dual-mode trapdoor function family with state recovery, the commitment scheme only needs to additionally have an indistinguishable injective mode. For example, [GKNV25]’s work satisfies this since they build on [BCM⁺18, Mah18]’s dual-mode claw-free trapdoor functions.

- **Group Actions.** [GV24] showed how to build dual-mode claw-free trapdoor functions from group actions by building on [AMR22]. It is likely that this construction

also satisfies the recovery property using a similar approach of using the trapdoor to coherently uncompute the entangled randomness. The main caveat is that their basic construction does not have $1 - \text{negl}(\lambda)$ overlap in recovery mode and they rely on an XOR amplification lemma (Lemma 5 in [GV24]) to achieve $1 - \text{negl}(\lambda)$ overlap. The amplification step allows there to be many r such that $f(x; r) = y$ for each y . One would need to verify that the amplified variant still allows uncomputation of a superposition over such r .²²

5.1 Construction

We give a version with domain $\mathcal{X} = \{0, 1\}$ and note that the domain can be extended by appending evaluations of the same function, at the cost of an additive overhead in recovery distance (e.g. for $\mathcal{X} = \{0, 1\}^n$, the concatenated family has $(1 - n2^{-\lambda})$ -state-recovery). The recovery parameter can be adjusted by modifying λ .

The construction is lattice-based, and makes use of the following theorem from [MP11].

Theorem 9 (Theorem 5.1 in [MP11]). *Let $n, m \geq 1$ and $q \geq 2$ be such that $m = \Omega(n \log q)$. There is an efficient randomized algorithm $\text{TrapGen}_{\text{MP}}(1^n, 1^m, q)$ that returns a matrix $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$ together with a trapdoor $\mathbf{t}_{\mathbf{A}}$ such that the distribution of $\mathbf{A}$ is negligibly (in n) close to the uniform distribution. Moreover, there is an efficient algorithm $\text{Invert}_{\text{MP}}$ such that with overwhelming probability over $(\mathbf{A}, \mathbf{t}_{\mathbf{A}}) \leftarrow \text{TrapGen}_{\text{MP}}(1^n, 1^m, q)$, the following holds: for every $\mathbf{s} \in \mathbb{Z}_q^n$ and every $\mathbf{e} \in \mathbb{Z}_q^m$ s.t. $\|\mathbf{e}\| \leq \frac{q}{C\sqrt{n \log q}}$ (where C is a universal constant),*

$$\text{Invert}_{\text{MP}}(\mathbf{A}, \mathbf{t}_{\mathbf{A}}, \mathbf{A} \cdot \mathbf{s} + \mathbf{e}) = (\mathbf{s}, \mathbf{e})$$

Remark 3. *We mention that the guarantee in the theorem above actually holds for every $(\mathbf{A}, \mathbf{t}_{\mathbf{A}})$ generated by $\text{TrapGen}_{\text{MP}}(1^n, 1^m, q)$. The reason is that $\text{TrapGen}_{\text{MP}}(1^n, 1^m, q)$ works as follows:*

1. Set $k := \lceil \log q \rceil$.
2. Let $\bar{m} = m - nk$.
3. Sample $\bar{\mathbf{A}} \leftarrow \mathbb{Z}_q^{\bar{m} \times n}$.
4. Let $\mathbf{G} \in \mathbb{Z}_q^{nk \times n}$ be a gadget matrix such that there exists an efficient algorithm that for every $\mathbf{x}$ and every $\mathbf{e}$ of bounded norm, given $\mathbf{G}\mathbf{x} + \mathbf{e}$ outputs $\mathbf{x}$. Such a gadget matrix exists by Lemma 4.1 in [MP11].
5. Let $\mathbf{A}' = \begin{pmatrix} \bar{\mathbf{A}} \\ \mathbf{G} \end{pmatrix} \in \mathbb{Z}_q^{m \times n}$.
6. Sample $\mathbf{R} \leftarrow \{0, 1\}^{nk \times nk}$.

²²Thanks to [Gup] for discussing this point with us.

7. Let $\mathbf{T} = \begin{pmatrix} \mathbf{I} & -\mathbf{R} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \in \mathbb{Z}_q^{m \times m}$. where $\mathbf{I}$ is the $\bar{m}$ -by- $\bar{m}$ identity matrix.

8. Let $\mathbf{A} = \mathbf{T}\mathbf{A}'$.

9. Let $\mathbf{T}^{-1} = \begin{pmatrix} \mathbf{I} & \mathbf{R} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$.

10. Output $(\mathbf{A}, \mathbf{t}_\mathbf{A})$ where $\mathbf{t}_\mathbf{A} = \mathbf{T}^{-1}$.

Given $\mathbf{A}\mathbf{x} + \mathbf{e}$, multiply this vector with $\mathbf{t}_\mathbf{A}$ to obtain $\mathbf{A}'\mathbf{x} + \mathbf{T}^{-1}\mathbf{e} = \mathbf{A}'\mathbf{x} + \mathbf{e}'$. From this one can efficiently compute $\mathbf{G}\mathbf{x} + \mathbf{e}''$ and thus invert $\mathbf{x}$.

Construction 1 (Dual-Mode Trapdoor Functions with State Recovery). The construction for input domain $\mathcal{X} = \{0, 1\}$ is as follows.

- **Setup:** This algorithm takes as input the security parameter 1^λ along with a flag $\text{mode} \in \{\text{recovery}, \text{injective}\}$, and outputs a public parameter $\mathbf{pp}$ along with a trapdoor $\mathbf{td}$. It is associated with the following parameters:
 - A prime $q \approx 2^{2\lambda}/\text{poly}(\lambda)$.
 - Parameters $n = n(\lambda)$ and $m = m(\lambda)$, both polynomially bounded functions of λ , such that $m \geq n \log(q)$ and $n \geq \ell \log(q) + \lambda$.
 - Two error distributions $\chi_B, \chi_{B'}$ over $\mathbb{Z}_q$. These are discrete Gaussians truncated to $B \in \mathbb{N}$ and $B' \in \mathbb{N}$, respectively. The truncation bounds satisfy:
 1. $\frac{B}{B'} = \frac{1}{2^{2\lambda}(2\pi m)}$.
 2. $B' \leq \frac{q}{2C\sqrt{n \cdot m \cdot \log q}}$, where C is the universal constant from Theorem 9.
 3. $\Pr_{\mathbf{e} \leftarrow \chi^m} [\|\mathbf{e}\| > B] = \text{negl}(\lambda)$.
 4. $\Pr_{\mathbf{e}' \leftarrow (\chi')^m} [\|\mathbf{e}'\| > B'] = \text{negl}(\lambda)$.

It does the following:

1. Generate $(\mathbf{A}, \mathbf{t}_\mathbf{A}) \leftarrow \text{TrapGen}_{\text{MP}}(1^n, 1^m, q)$.
2. If $\text{mode} = \text{recovery}$ then sample a uniform random string $\mathbf{s} \leftarrow \mathbb{Z}_q^n$ and a random error vector $\mathbf{e} \leftarrow \chi_B^m$, and set $\mathbf{u} = \mathbf{A} \cdot \mathbf{s} + \mathbf{e}$.
3. If $\text{mode} = \text{injective}$ then sample a uniform $\mathbf{u} \leftarrow \mathbb{Z}_q^m$.
 If we want the injective mode inversion algorithm to succeed in inverting with probability 1 (as opposed to $1 - \text{negl}(\lambda)$) then we should use $\mathbf{t}_\mathbf{A}$ to check if that $\mathbf{u}$ is not of the form $\mathbf{A}\mathbf{s} + \mathbf{e}$ for some low-norm $\mathbf{e}$ (a condition that is satisfied with overwhelming probability). If it is of this form, then we can add the vector $(\lfloor \frac{q}{2} \rfloor, \dots, \lfloor \frac{q}{2} \rfloor)$ to $\mathbf{u}$.
4. Output $\mathbf{pp} = (\mathbf{A}, \mathbf{u})$ and $\mathbf{td} = (\mathbf{A}, \mathbf{u}, \mathbf{t}_\mathbf{A})$.

- **Function f_{pp} :** The domain of f_{pp} is $\mathcal{X} \times \mathcal{R}$, where $\mathcal{X} = \{0, 1\}$. Its range is $\mathcal{R} = \mathbb{Z}_q \times [-B', B']^m$, and its range is $\mathbb{Z}_q^m$. f_{pp} is defined as follows:

$$f_{\mathbf{A}, \mathbf{u}}(b, (\mathbf{x}, \mathbf{e}')) = \mathbf{A}\mathbf{x} + b\mathbf{u} + \mathbf{e}'$$

- **Injective mode Ext algorithm.** The poly-time algorithm Ext takes as input $(\mathbf{td}, \mathbf{y})$ and does the following:

1. Parse $\mathbf{td} = (\mathbf{A}, \mathbf{u}, \mathbf{t}_{\mathbf{A}})$.
2. Compute $(\mathbf{x}', \mathbf{e}) = \text{Invert}_{\text{MP}}(\mathbf{A}, \mathbf{t}, \mathbf{y})$.
3. If $\mathbf{y} = \mathbf{A}\mathbf{x}' + \mathbf{e}$ then output $b = 0$.
4. Otherwise, output $b = 1$.

- **Recover algorithm:** Recover is a QPT algorithm that takes as input $\mathbf{td}, \mathbf{y}$, and a state

$$|\varphi_{\mathbf{y}}\rangle = \sum_{b \in \{0, 1\}} \alpha_b |b\rangle \sum_{(\mathbf{x}, \mathbf{e}'): f_{\text{pp}}(b, \mathbf{x}, \mathbf{e}') = \mathbf{y}} |\mathbf{x}, \mathbf{e}', \mathbf{y}\rangle = \sum_{b \in \{0, 1\}} \alpha_b |b, \mathbf{x}_b, \mathbf{e}'_b\rangle$$

where for every $b \in \{0, 1\}$,

$$f_{\text{pp}}(b, \mathbf{x}_b, \mathbf{e}'_b) = \mathbf{y}.$$

It computes $\sum_{b \in \{0, 1\}} \alpha_b |b\rangle$, as follows:²³

1. Parse $\mathbf{td} = (\mathbf{A}, \mathbf{u}, \mathbf{t}_{\mathbf{A}})$.
2. Compute $(\mathbf{s}, \mathbf{e}_{\mathbf{s}}) = \text{Invert}_{\text{MP}}(\mathbf{A}, \mathbf{t}_{\mathbf{A}}, \mathbf{u})$.
3. Compute $(\mathbf{x}, \mathbf{e}_{\mathbf{x}}) = \text{Invert}_{\text{MP}}(\mathbf{A}, \mathbf{t}_{\mathbf{A}}, \mathbf{y})$.
4. Let $\mathbf{x}_0 = \mathbf{x}$ and let $\mathbf{x}_1 = \mathbf{x} - \mathbf{s}$.
5. Let $\mathbf{e}'_0 = \mathbf{e}_{\mathbf{x}}$ and let $\mathbf{e}_1 = \mathbf{e}_{\mathbf{x}} - \mathbf{e}_{\mathbf{s}}$.
6. Apply the unitary mapping

$$|b, (\mathbf{r}_{\mathbf{x}}, \mathbf{r}_{\mathbf{e}})\rangle \mapsto |b, (\mathbf{r}_{\mathbf{x}} - \mathbf{x}_b, \mathbf{r}_{\mathbf{e}} - \mathbf{e}'_b)\rangle$$

to $|\varphi_{\mathbf{y}}\rangle$.

7. Output the resulting state.

The dual-mode indistinguishability follows immediately from the post-quantum hardness of LWE. The inversion algorithm Ext works due to Theorem 9.

Claim 2. *The above construction has $(1 - 2^{-\lambda})$ -state recovery.*

²³This entire computation can be done coherently controlled on b , so that in an n -time parallel repetition the Recover algorithm can compute the 2^n preimages in superposition, instead of computing them individually and blowing up the runtime.

Proof. Without loss of generality, we may delay the measurement of y in Exp by coherently copying it to another register which is measured later. Let Exp' be the experiment with the delayed measurement, before the extra register is measured. We first compute Exp' for $b = 0$ and $b = 1$, which we call $|\text{Eval}(0)\rangle$ and $|\text{Eval}(1)\rangle$, respectively. We have

$$\begin{aligned} |\text{Eval}(0)\rangle &\propto |0\rangle \otimes \sum_{(\mathbf{x}, \mathbf{e}')} \sqrt{p(\mathbf{e}')} |\langle \mathbf{x}, \mathbf{e}' \rangle\rangle \otimes |\mathbf{Ax} + \mathbf{e}'\rangle^{\otimes 2} \\ |\text{Eval}(1)\rangle &\propto |1\rangle \otimes \sum_{(\mathbf{x}, \mathbf{e}')} \sqrt{p(\mathbf{e}')} |\langle \mathbf{x}, \mathbf{e}' \rangle\rangle \otimes |\mathbf{Ax} + (\mathbf{As} + \mathbf{e}) + \mathbf{e}'\rangle^{\otimes 2} \end{aligned}$$

where $(\mathbf{As} + \mathbf{e}) = \mathbf{u}$, where p is the probability density function of $\chi_{B'}$, and where the normalization constant over $\mathbf{x}$ is suppressed since $\mathbf{x}$ is uniform and independent of $\mathbf{e}'$. Following [BCM⁺18], the distribution defined by sampling $y = \mathbf{Ax} + (\mathbf{As} + \mathbf{e}) + \mathbf{e}'$ with probability $p(\mathbf{e}')$ has Hellinger distance

$$1 - e^{-\frac{2\pi m B}{B'}}$$

to the distribution defined by sampling $y = \mathbf{Ax} + \mathbf{As} + (\mathbf{e} + \mathbf{e}')$ with probability $p(\mathbf{e}' + \mathbf{e})$. Since we set $\frac{B}{B'} = \frac{1}{2^\lambda(2\pi m)}$, the distance is $\leq 2^{-\lambda}$. Since the trace distance of two superpositions over samples from distributions g_1, g_2 is related to the Hellinger distance $H^2(g_1, g_2)$ by $\sqrt{1 - (1 - H^2(g_1, g_2))^2}$, the state $|\text{Eval}(1)\rangle$ has trace distance

$$\leq \sqrt{1 - (1 - 2^{-2\lambda})^2} \leq 2^{-\lambda}$$

from a state

$$|\text{Eval}'(1)\rangle \propto |1\rangle \otimes \sum_{(\mathbf{x}, \mathbf{e}')} \sqrt{p(\mathbf{e}' + \mathbf{e})} |\langle \mathbf{x}, \mathbf{e}' \rangle\rangle \otimes |\mathbf{Ax} + (\mathbf{As} + \mathbf{e}) + \mathbf{e}'\rangle^{\otimes 2}$$

Setting $\mathbf{x}' = \mathbf{x} - \mathbf{s}$ and $\mathbf{e}'' = \mathbf{e} + \mathbf{e}'$, this state can be rewritten as

$$|\text{Eval}'(1)\rangle \propto |1\rangle \otimes \sum_{(\mathbf{x}', \mathbf{e}'')} \sqrt{p(\mathbf{e}'')} |\langle \mathbf{x}' + \mathbf{s}, \mathbf{e}'' - \mathbf{e} \rangle\rangle \otimes |\mathbf{Ax}' + \mathbf{e}''\rangle^{\otimes 2}$$

Now consider the result of $(I \otimes \text{Recover}) \circ (I \otimes \text{Exp}')$ on an arbitrary state $\alpha_0 |\phi_0\rangle \otimes |0\rangle + \alpha_1 |\phi_1\rangle \otimes |1\rangle$. By the trace distance bound above, the result of running $I \otimes \text{Exp}'$ on this state has $\leq 2^{-\lambda}$ trace distance to

$$\begin{aligned} &\alpha_0 |\phi_0\rangle \otimes |\text{Eval}(0)\rangle + \alpha_1 |\phi_1\rangle \otimes |\text{Eval}'(1)\rangle \\ &= \sum_{(\mathbf{x}, \mathbf{e}')} \sqrt{p(\mathbf{e}')} \left(\sum_{b \in \{0,1\}} \alpha_b |\phi_b\rangle \otimes |b\rangle \otimes |\mathbf{x} - b\mathbf{s}, \mathbf{e}' - b\mathbf{e}\rangle \right) \otimes |\mathbf{Ax} + \mathbf{e}'\rangle^{\otimes 2} \end{aligned}$$

Since Recover successfully inverts $(\mathbf{x}, \mathbf{e}')$ from any $\mathbf{Ax} + \mathbf{e}'$ with overwhelming probability over $(\mathbf{A}, \mathbf{t}_\mathbf{A})$ (Theorem 9), the result of running $I \otimes \text{Recover}$ on this state is

$$\left(\sum_{b \in \{0,1\}} \alpha_b |\phi_b\rangle \otimes |b\rangle \right) \sum_{(\mathbf{x}, \mathbf{e}')} \sqrt{p(\mathbf{e}')} \otimes |0\rangle \otimes |\mathbf{Ax} + \mathbf{e}'\rangle^{\otimes 2}$$

To complete the experiment, the register containing the extra copy of $\mathbf{A}\mathbf{x} + \mathbf{e}'$ is measured in the computational basis to obtain y and both registers containing y are traced out, which cannot increase the trace distance. $\square$

6 State-Preserving Arguments for NP

In this section, we construct an interactive argument for NP with the property that if the (honest) prover starts with a state $|\psi\rangle = \sum_w \alpha_w |w\rangle$ which is a superposition over valid witnesses, then after running the protocol it is left with a state ρ that is negligibly far from its original state $|\psi\rangle$. We call such an argument system a *state preserving argument*.

Definition 12. A state-preserving argument for an NP language $\mathcal{L}$ is an interactive protocol between a QPT prover P and a PPT verifier V that satisfies the following guarantees.

- **Completeness and State Preservation.** For every $x \in \mathcal{L}$ and every superposition $|\psi\rangle = \sum_w \alpha_w |w\rangle_{\mathcal{W}} \otimes |\phi_w\rangle_{\mathcal{B}}$ over witnesses for x (encoded in the computational basis and potentially entangled with an external state $|\phi_w\rangle$), there exist negligible functions μ_1 and μ_2 such that for every $\lambda \in \mathbb{N}$,

$$\Pr \left[\left(\frac{1}{2} \| |\psi\rangle\langle\psi| - \rho \|_1 \leq \mu_1(\lambda) \right) \wedge (b = \text{Accept}) : (\rho, b) \leftarrow \langle P(|\psi\rangle), V \rangle(1^\lambda, x) \right] \geq 1 - \mu_2(\lambda),$$

where P acts only on register $\mathcal{W}$ and where the notation $(\rho, b) \leftarrow \langle P(|\psi\rangle), V \rangle(1^\lambda, x)$ means that V outputs $b \in \{\text{Accept}, \text{Reject}\}$ and P 's residual state is ρ .

When we wish to be more precise about the repair guarantees, we say that the argument is μ_1 -state-preserving.

- **Computational Soundness.** For every polynomial $\ell = \ell(\lambda)$, there exists a negligible function μ such that for every $\lambda \in \mathbb{N}$, every $x \notin \mathcal{L}$ of size $\leq \ell(\lambda)$ and every QPT cheating prover P^* with auxiliary input $|\psi\rangle$ consisting of at most $\text{poly}(\ell(\lambda))$ qubits

$$\Pr[\langle P^*(|\psi\rangle), V \rangle(1^\lambda, x) = \text{Accept}] \leq \mu(\lambda)$$

We say a state-preserving argument for NP is an δ -argument of knowledge if there exists a QPT extractor Ext such that for every QPT adversarial prover P^* and every statement x ,

$$\Pr[\text{Accept} \leftarrow \langle P^*, V \rangle(1^\lambda, x)] \leq \Pr[(x, w) \in \mathcal{R}_{\mathcal{L}} : w \leftarrow \text{Ext}(P^*)] + \delta$$

where $\mathcal{R}$ is the NP relation for $\mathcal{L}$. If $\delta = \text{negl}(\lambda)$, we simply say that it is an argument of knowledge.

State preservation composes in parallel since it also preserves entanglement with an external register. As a result, state-preserving arguments can be parallelly amplified whenever parallel repetition amplifies soundness: coherently classical-copy the witness to produce $\sum_w \alpha_w |w\rangle^{\otimes n}$ and independently run a state-preserving argument on each witness copy.

Lemma 9. *The n -fold parallel repetition of a $(1 - \delta)$ -state-preserving argument is $(1 - n\delta)$ -state preserving.*

Proof. Since each repetition acts on disjoint registers from the others, the repetitions commute (for the purposes of determining the leftover state). Therefore we may consider running the repetitions one at a time. Let ρ_i be the leftover state after repetition i , with $\rho_0 = |\psi\rangle\langle\psi|$ being the initial witness superposition. By state-preservation and union bound, $\frac{1}{2}\|\rho_i - \rho_{i+1}\|_1 \leq \delta$ for every i with overwhelming probability. By triangle inequality the leftover state has distance $\leq n\delta$ from the prover's initial input $|\psi\rangle\langle\psi|$ with overwhelming probability. $\square$

6.1 Construction

We construct witness preserving arguments for **NP** by building on top of the well-known witness indistinguishable argument for the **NP**-complete 3-coloring language. Instances for 3-coloring are graphs $x = (\mathcal{V}, \mathcal{E})$. Witnesses are a 3-coloring of the graph, i.e., a function $w : \mathcal{V} \rightarrow [3]$, such that for every edge $(i, j) \in \mathcal{E}$, the vertices i and j have different colors: $w(i) \neq w(j)$. We consider w to be written as a truth-table and denote the color it assigns to vertex i by w_i .

Our construction makes use of a dual-mode trapdoor function family $\mathcal{F}$, with a corresponding setup algorithm, denoted by $\text{Setup}_{\mathcal{F}}$ (see Definition 11). We describe a version with $1/\text{poly}(\lambda)$ soundness. Soundness can be amplified by running the protocol sequentially, without losing state preservation.

Construction 2 (State-Preserving Arguments for **NP**). Let $\mathcal{L}$ be the 3-coloring language and consider any statement $x \in \mathcal{L}$. The prover starts with a superposition $\sum_w \alpha_w |w\rangle$ over witnesses w for $x \in \mathcal{L}$.

During the protocol, the prover will maintain three registers $\mathcal{W}$, $\mathcal{P}$, and $\mathcal{R}$.

- $\mathcal{W} = (\mathcal{W}_1, \dots, \mathcal{W}_{|\mathcal{V}|})$ will contain the witness and is initialized to $\sum_w \alpha_w |w\rangle$.
- $\mathcal{P}$ will contain a permutation $\pi : [3] \rightarrow [3]$ and is initialized to $\propto \sum_{\pi \in \text{Perm}([3])} |\pi\rangle$.
- $\mathcal{R} = (\mathcal{R}_1, \dots, \mathcal{R}_{|\mathcal{V}|})$ will contain randomness r_i for evaluating a trapdoor function and is initialized to $\bigotimes_{i \in \mathcal{V}} |\text{rand}_{\text{pp}}\rangle$.

1. **Verifier:** Sample $(\text{pp}, \text{td}) \leftarrow \text{Setup}_{\mathcal{F}}(1^\lambda, \text{recovery})$.

Send pp to the prover.

2. **Prover:**

(a) For every $i \in \mathcal{V}$ apply an isometry

$$|w_i, \pi, r_i\rangle_{\mathcal{W}_i, \mathcal{P}, \mathcal{R}_i} \mapsto |w_i, \pi, r_i\rangle_{\mathcal{W}_i, \mathcal{P}, \mathcal{R}_i} \otimes |f_{\text{pp}}(\pi(w_i); r_i)\rangle$$

and measure the last register to obtain y_i .

- (b) Send $y_{\mathcal{V}} := (y_1, \dots, y_{|\mathcal{V}|})$ to the verifier.
- 3. **Verifier:** Sample a random edge $(i, j) \leftarrow \mathcal{E}$ and send (i, j) to the prover.
- 4. **Prover:** Measure $(\pi(w_i), \pi(w_j), r_i, r_j)$ using registers $\mathcal{P}$, $\mathcal{W}_i$, $\mathcal{W}_j$, $\mathcal{R}_i$, and $\mathcal{R}_j$, and send it to the verifier.
- 5. **Verifier:** Upon receiving $c = (c_i, c_j, r_i, r_j)$ from the prover, check that $c_i \neq c_j$, that $y_i = f_{\text{pp}}(c_i; r_i)$, and that $y_j = f_{\text{pp}}(c_j; r_j)$. If these are all true, output **Accept** and send **td** to the prover. Otherwise, output **Reject**.
- 6. **Prover:** For each $k \in \mathcal{V} \setminus \{i, j\}$, run $\text{Recover}_{\mathcal{F}}(\text{td}, \cdot, y_k)$ on registers $(\mathcal{W}_k, \mathcal{P}_k)$.

Next, observe that for every witness w , there is a unique permutation $\pi_{c,w,(i,j)}$ such that $c_i = \pi_{c,w}(w_i)$ and $c_j = \pi_{c,w}(w_j)$; this holds because $w_i \neq w_j$ and fixing two outputs of a permutation on $[3]$ fully determines the permutation. Compute the unitary swapping

$$|w, \pi_{c,w,(i,j)}\rangle_{\mathcal{W},\mathcal{P}} \leftrightarrow |w, \text{id}\rangle_{\mathcal{W},\mathcal{P}}$$

where **id** is the identity permutation, and acting as the identity elsewhere.

Output register $\mathcal{W}$.

Theorem 10. *Construction 2 is a $(1 - 2^{-\lambda})$ -state-preserving argument for NP with soundness $1 - \frac{1}{|\mathcal{V}|}$, assuming the existence of dual-mode trapdoor function family with $(1 - 2^{-\lambda}/|\mathcal{V}|)$ -state recovery (which in turn can be constructed from the post-quantum hardness of LWE [BCM⁺18]). Furthermore, its parallel repetition is an argument of knowledge.*

Remark To improve the soundness error to be negligible, one can repeat the protocol in Construction 2 sequentially $\text{poly}(\lambda)$ times. As shown in Lemma 9, repeating k times results in a $(1 - k2^{-\lambda})$ -state-preserving argument. This can be modified to $(1 - 2^{-\lambda})$ by adjusting the security parameter.

Proof. We need to prove that the completeness, soundness and state preservation properties hold.

Completeness. Completeness follows from the observation that $\pi(w_i) \neq \pi(w_j)$ for all valid witnesses w .

Soundness and Argument of Knowledge. It suffices to prove soundness in the hybrid experiment where in the first round the verifier sends **pp** which are generated in injective mode: $(\text{pp}, \text{td}) \leftarrow \text{Setup}_{\mathcal{F}}(1^\lambda, \text{injective})$, as opposed to lossy mode. The dual-mode indistinguishability property implies that this experiment is indistinguishable from the real execution. Therefore, for any QPT prover P^* , its success probability changes only by a negligible factor.

In injective mode, each y_i corresponds to a unique preimage color c_i . If $x \notin \mathcal{L}$, then there must exist an edge $(i, j) \in \mathcal{E}$ such that $c_i = c_j$ or such that $c_i \notin [3]$. Therefore, P^* 's success probability is at most $1 - 1/|\mathcal{E}|$ in the hybrid experiment, and thus at most $1 - 1/|\mathcal{E}| - \text{negl}(\lambda)$ in a real execution.

The argument of knowledge extractor **Ext** interacts with P^* using an injective mode **pp** until it receives y_V for each instance. Then, it inverts each y_V using the trapdoor to obtain a coloring c_V for each instance. It checks which coloring is valid and outputs that coloring. As before, P^* 's success probability is unmodified by this extractor. However, if the extractor does not succeed, every instance's preimage has an edge with the same color; therefore P^* can only succeed with $(1 - 1/|\mathcal{E}|)^\lambda$ probability, which is negligible for $\lambda = \omega(|\mathcal{E}| \log(|\mathcal{E}|))$.

State preservation. At the start of the protocol, the prover initializes their state to

$$\propto \sum_w \alpha_w |w\rangle \otimes \sum_{\pi \in \text{Perm}([3])} |\pi\rangle \otimes \bigotimes_{i \in V} \sum_{r_i \in \mathcal{R}} \beta_{r_i} |r_i\rangle$$

where β_{r_i} is an amplitude determined by $\mathcal{F}$ and **pp**. Suppose that the prover (with the verifier's aid) ran all computations on $k \in \mathcal{V} \setminus \{i, j\}$ before doing any computation on registers corresponding to i and j . Explicitly, they apply the isometry from step 2, measure y_k , and run $\text{Recover}_{\mathcal{F}}(\text{td}, \cdot, y_k)$ before doing any operations to the registers corresponding to i and j . Since these computations are done on disjoint registers, reordering them produces an identical state.

The trapdoor property of $\mathcal{F}$ in **recovery** mode implies that the state immediately after operating on all $k \in \mathcal{V} \setminus \{i, j\}$, before operating on i and j , has trace distance $(|\mathcal{V}| - 2)2^{-\lambda}/|\mathcal{V}| < 2^{-\lambda}$ from

$$\propto \sum_w \alpha_w |w\rangle \otimes \sum_{\pi \in \text{Perm}([3])} |\pi\rangle \otimes \sum_{r_i, r_j \in \mathcal{R}} \beta_{r_i} \beta_{r_j} |r_i, r_j\rangle$$

We will now rewrite this state in terms of the prover's possible last messages – specifically, the revealed colors $c = (c_i, c_j)$. Fix any c and any witness w . The possible settings of π are constrained by $\pi(w_i) = c_i$ and $\pi(w_j) = c_j$. Since $w_i \neq w_j$ for valid witnesses, this leaves only one possibility for π to map the last color. In other words, there is a unique $\pi_{c,w,(i,j)}$ which is consistent with c , w , and (i, j) . Thus, we can rewrite the state as

$$\propto \sum_w \alpha_w |w\rangle \otimes \sum_c |\pi_{c,w,(i,j)}\rangle \otimes \sum_{r_i, r_j \in \mathcal{R}} \beta_{r_i} \beta_{r_j} |r_i, r_j\rangle$$

Measuring c , r_i , r_j , and the evaluations of f_{pp} on indices i and j collapses the state to

$$\sum_w \alpha_w |w\rangle \otimes |\pi_{c,w,(i,j)}\rangle$$

Finally, applying a unitary swapping

$$|w, \pi_{c,w,(i,j)}\rangle \leftrightarrow |w, \text{id}\rangle$$

and discarding the register containing $|\text{id}\rangle$ recovers the original state $\sum_w \alpha_w |w\rangle$, up to $2^{-\lambda}$ trace distance. $\square$

7 Non-Destructive CVQC for QMA

In this section we construct a CVQC for any QMA language with nearly perfect completeness and soundness, while using only a *single* copy of the QMA witness, and moreover the protocol is witness preserving (as formally defined below).

Clearly, we cannot hope to obtain negligible soundness with *any* witness, since the witness can be a superposition of a “good” witness and “junk.” Indeed, we define the set of “repairable” witnesses for every $x \in \mathcal{L}_{\text{yes}}$, and argue that the honest prover only needs a single copy of a “repairable” witness, to convince the verifier that $x \in \mathcal{L}_{\text{yes}}$ with overwhelming probability.

The set of repairable witnesses includes every witness $|w\rangle$ that is accepted by $V_{\text{QMA},x}$ with overwhelming probability, as well as all of the Marriot-Watrous witnesses. So, as a special case, we obtain the following theorem.

Theorem 11. *Fix any $\mathcal{L} \in \text{QMA}_{a,b}$ with a QMA verifier $V_{\text{QMA}} = \{V_{\text{QMA},x}\}_{x \in \{0,1\}^*}$. Then there exists a CVQC protocol (P, V) for $\mathcal{L}$ w.r.t. V_{QMA} with the following properties:*

- **Computational soundness:** *It has $\text{negl}(\lambda)$ computational soundness error (as defined in Definition 6), assuming the post-quantum hardness of LWE.*
- **Completeness:** *For every negligible function μ_1 there exists a negligible function μ_2 s.t. for every $x \in \mathcal{L}_{\text{yes}}$ and every (single) witness $|w\rangle$ s.t.*

$$\Pr[V_{\text{QMA},x}(1^\lambda, |w\rangle) = 1] = 1 - \mu_1(\lambda) \quad (6)$$

it holds that

$$\Pr[(P(|w\rangle), V)(1^\lambda, x) = 1] = 1 - \mu_2(\lambda).$$

- **Witness preserving:** *There exist negligible functions μ_1 and μ_2 such that for every $x \in \mathcal{L}$ and every entangled witness superposition $|\tilde{w}\rangle_{\mathcal{A},\mathcal{B}} = \sum_i \alpha_i |w_i\rangle_{\mathcal{A}} \otimes |\phi_i\rangle_{\mathcal{B}}$ such that each $|w_i\rangle$ satisfies Equation (6),*

$$\Pr_{(|\psi\rangle, b) \leftarrow (P(|\tilde{w}\rangle), V)(1^\lambda, x)} \left[\frac{1}{2} \left\| |\psi\rangle\langle\psi| - |\tilde{w}\rangle\langle\tilde{w}| \right\|_1 \leq \mu_1(\lambda) \right] \geq 1 - \mu_2(\lambda),$$

where P operates only on register $\mathcal{A}$.

- **Complexity:** *The number of rounds, the communication complexity and the computational complexity are $\text{poly}(\lambda, |x|)$.*
- **η -Argument of Knowledge:** *It is an η -classical argument of quantum knowledge (Definition 7) with $\eta = \frac{1}{p(\lambda)}$ for some polynomial $p(\cdot)$.*

All of the above holds as long as $|x| > \lambda$, which can be ensured w.l.o.g. via padding.

More generally, we present a generic compiler that converts any non-adaptive CVQC protocol (P, V) for $\mathcal{L}$ w.r.t. $V_{\text{QMA}} = \{V_{\text{QMA},x}\}_{x \in \{0,1\}^*}$, with arbitrary completeness c and soundness s , into a new CVQC protocol with negligible soundness and nearly perfect completeness, while using only a *single copy* of a “repairable” witness, and moreover the protocol preserves the witness.

We defer the precise definition of a “repairable” witness to Definition 15 in Section 7.1, but mention some brief intuition here. Generally, the set of repairable witnesses (corresponding to an instance $x \in \mathcal{L}_{\text{yes}}$) is associated with two parameters:

- **Quality:** A parameter $p \in [0, 1]$ corresponding to the probability that the witness is accepted by $V_{\text{QMA},x}$.
- **Repair Decay:** A parameter $\epsilon \in (0, 1]$ bounding how much worse the witness gets after running the protocol: an ϵ -non-destructive protocol which takes in a witness in $\text{Repairable}_{p,\epsilon}$ should output one in $\text{Repairable}_{p-\epsilon,\epsilon}$.

We denote by $\text{Repairable}_{p,\epsilon}(x)$ the set of repairable witnesses corresponding to $x \in \mathcal{L}_{\text{yes}}$, associated with parameters p, ϵ . We mention that every witness $|w\rangle \in \mathcal{ES}_{\geq p}(V_{\text{QMA},x})$ is in $\text{Repairable}_{p,\epsilon}(x)$ for every $\epsilon \geq 0$. Recall from Definition 1 that $\mathcal{ES}_{\geq p}$ is the span of eigenvectors of $V_{\text{QMA},x}$ with eigenvalue $\geq p$, which is precisely the set of Marriot-Watrous witnesses when $p \geq a$, and a is the QMA threshold.

We next provide the definition of witness preserving.

Definition 13 (Witness-Preserving CVQC). *Let $\mathcal{L} = (\mathcal{L}_{\text{yes}}, \mathcal{L}_{\text{no}})$ be a language in $\text{QMA}_{a,b}$ with a QMA verifier V_{QMA} . A CVQC protocol (P, V) for $\mathcal{L}$ w.r.t. V_{QMA} is witness-preserving with respect to a family of witnesses $\{W_x\}_{x \in \mathcal{L}_{\text{yes}}}$, where for every $x \in \mathcal{L}_{\text{yes}}$,*

$$W_x \subseteq \{|w\rangle : \Pr[V_{\text{QMA},x}(|w\rangle) = 1] \geq a\},$$

if there exist negligible functions μ_1 and μ_2 such that for every $x \in \mathcal{L}_{\text{yes}}$, every entangled witness state $|\tilde{w}\rangle_{\mathcal{A},\mathcal{B}} = \sum_i \alpha_i |w_i\rangle_{\mathcal{A}} \otimes |\phi_i\rangle_{\mathcal{B}}$ where $|w_i\rangle \in W_x$, and every $\lambda \in \mathbb{N}$,

$$\Pr_{(|\psi\rangle, b) \leftarrow (P(|\tilde{w}\rangle), V)(1^\lambda, x)} \left[\frac{1}{2} \left\| |\psi\rangle\langle\psi| - |\tilde{w}\rangle\langle\tilde{w}| \right\|_1 \leq \mu_1(\lambda) \right] \geq 1 - \mu_2(\lambda),$$

where P operates only on register $\mathcal{A}$.

Our generic compiler is formalized below.

Theorem 12. *Fix any language $\mathcal{L} \in \text{QMA}_{a,b}$ and a corresponding QMA verifier V_{QMA} . Assume the existence of the following two cryptographic primitives (both which can be instantiated assuming the post-quantum hardness of LWE):*

1. *A dual-mode trapdoor function family with $(1 - 2^{-\lambda})$ -state recovery (Definition 11).*
2. *A $(1 - 2^{-\lambda})$ -state-preserving argument for NP (Definition 12).*

There exists an efficient compiler, associated with a parameter $\epsilon = \epsilon(\lambda)$, that converts any non-adaptive CVQC protocol (P, V) for $\mathcal{L}$ w.r.t. V_{QMA} , with completeness c and (testable) soundness s s.t. $c - s \geq 1/\text{poly}(\lambda)$ into a new (non-adaptive) CVQC protocol (P', V') for $\mathcal{L}$ w.r.t. V_{QMA} , such that (P', V') has the following properties, whenever $|x| > \lambda, N \geq \frac{\lambda^2}{(c-s)^2}$:

- **Computational soundness:** It has $\text{negl}(\lambda)$ computational soundness error (as defined in Definition 6).
- **Completeness:** For every $x \in \mathcal{L}_{\text{yes}}$ and every $|w\rangle \in \text{Repairable}_{a+\epsilon \cdot 4N, \epsilon}(x)$,

$$\Pr[(P'(|w\rangle), V')(1^\lambda, x) = 1] \geq 1 - \text{negl}(\lambda).$$

- **Witness preserving:** For every $x \in \mathcal{L}_{\text{yes}}$, the protocol is witness preserving w.r.t. all witnesses in $\text{Repairable}_{a+\epsilon \cdot 4N, \epsilon}(x)$.
- **Complexity:** The number of rounds, the communication complexity and the computational complexity grow by a multiplicative factor of $\text{poly}(\lambda, |x|, (c-s)^{-1})$. The prover's computational complexity additionally grows by an additive $\text{poly}(\lambda, |x|, (c-s)^{-1}, \epsilon^{-1})$ factor.
- **Argument of Knowledge:** If the underlying CVQC is a (d, s, δ) -testable argument of knowledge, then whenever $N \geq \frac{\lambda^{d+3}|x|^d}{(c-s)^2}$, the compiled CVQC is a δ/N -argument of quantum knowledge (Definition 7).

To use Theorem 12, we need an underlying CVQC protocol with completeness and soundness c and s , respectively, such that $c - s \geq 1/\text{poly}(\lambda)$, and such that the (honest) prover uses a *single* copy of a (repairable) QMA witness.

One way to obtain such a CVQC protocol is to use Mahadev's protocol [Mah18] as the underlying CVQC protocol. This requires some care since Mahadev's CVQC consists of many repetitions of an underlying CVQC (which [Mah18] constructs), and hence the prover needs many copies of the witness, one copy per each repetition. Since we want a protocol where the prover uses a single copy of the witness, it is tempting to use only one execution of Mahadev's protocol. However, a single copy of the protocol has worse completeness than soundness! To demystify this, we note that while the soundness is worse than the completeness, the *testable* soundness is better than the completeness (see Definition 8).

Specifically, in Mahadev's protocol, the verifier V is defined via two verification algorithms $(V_{\text{T}}, V_{\text{C}})$; it implements V_{T} with probability $1/2$, and implements V_{C} with probability $1/2$. V_{T} has completeness 1 while V_{C} has (low) completeness c .²⁴ The guarantee is that for every $x \in \mathcal{L}_{\text{no}}$ and every cheating prover P^* that convinces V_{T} to accept with probability close to 1 (say probability 0.99), it holds that P^* can convince V_{C} to accept with probability at most s which is smaller than c . In addition, a single execution of Mahadev's protocol has an (s, δ) testable argument of knowledge guarantee (Definition 9) with $\delta = 1 - \frac{1}{\lambda}$.

²⁴This follows from the fact that the QMA witness is converted into a Morimae-Fitzsimons [MF16] QMA witness, which has low completeness.

Due to this, we need to ensure that the compiler in Theorem 12 also compiles CVQC protocols with *testable* soundness and *testable* argument of knowledge properties, which it indeed does (as we show below).

We prove Theorem 12 in three stages.

Stage 1 (Sections 7.1 to 7.4): We first present a compiler that converts any non-adaptive CVQC protocol (possibly with only testable soundness) into one with the same completeness and (testable) soundness up to a negligible loss, and which is “ ϵ -non-destructive” (defined below) assuming the original witness is in $\text{Repairable}_{p,\epsilon}$ for $p \geq a + \epsilon$ (where ϵ is a tunable parameter).

Definition 14 (ϵ -Non-Destructive). *We say that a CVQC protocol (P, V) for $\mathcal{L} \in \text{QMA}_{a,b}$ is ϵ -non-destructive if there exists a negligible function μ such that for every $\lambda \in \mathbb{N}$, every $x \in \mathcal{L}_{\text{yes}}$ and every witness in $|w\rangle \in \text{Repairable}_{p,\epsilon}(x)$, for $p \geq a + \epsilon$,*

$$\Pr_{(|\psi\rangle, b) \leftarrow (P(|w\rangle), V)(1^\lambda, x)} [|\psi\rangle \in \text{Repairable}_{p-\epsilon, \epsilon}(x)] = 1 - \mu(\lambda).$$

Theorem 13. *Let $\mathcal{L} = (\mathcal{L}_{\text{yes}}, \mathcal{L}_{\text{no}}) \in \text{QMA}_{a,b}$ with a QMA verifier V_{QMA} . Assume the existence of the following primitives:*

1. *A dual-mode trapdoor function family with $(1 - 2^{-\lambda})$ -state recovery.*
2. *A $(1 - 2^{-\lambda})$ -state-preserving argument for NP.*

Then there exists a generic compiler, associated with a parameter $\epsilon = \epsilon(\lambda)$, that converts any non-adaptive CVQC protocol (P, V) w.r.t. V_{QMA} , with completeness c and (testable) soundness s , into a new CVQC protocol (P', V') for $\mathcal{L}$ (described in Construction 3), with the following guarantees:

1. **Computational soundness:** (P', V') has (testable) soundness $s + \text{negl}(\lambda)$.
2. **Completeness:** For every $x \in \mathcal{L}_{\text{yes}}$, the protocol (P', V') has completeness $c - \text{negl}(\lambda)$, using any single witness for $V_{\text{QMA}, x}$.
3. **Witness preserving if $c = 1 - \text{negl}(\lambda)$:** If $c = 1 - \text{negl}(\lambda)$ then Construction 3, with the witness recovery algorithm in Construction 4, is witness-preserving w.r.t. the set of all witnesses that are accepted in the underlying CVQC with probability $1 - \text{negl}(\lambda)$.
4. **4 ϵ -Non-Destructive:** There exists a negligible function μ such that for every $\lambda \in \mathbb{N}$, every $x \in \mathcal{L}_{\text{yes}}$ and every witness in $|w\rangle \in \text{Repairable}_{p,\epsilon}(x)$, for $p \geq a + 4\epsilon$,

$$\Pr_{(|\psi\rangle, b) \leftarrow (P'(|w\rangle), V')(1^\lambda, x)} [|\psi\rangle \in \text{Repairable}_{p-4\epsilon, \epsilon}(x)] = 1 - \mu(\lambda).$$

5. **Complexity:** The communication complexity and computational complexity grow by a multiplicative factor of $\text{poly}(\lambda)$.

Additionally, in the general case, the prover's computational complexity grows by an additive factor of $\text{poly}(\lambda, 1/\epsilon)$.

6. **Argument of Knowledge:** If (P, V) is a (d, s, δ) (testable) argument of knowledge, then (P', V') is a $(d + 1, s + \text{negl}(\lambda), \delta - \text{negl}(\lambda))$ (testable) argument of knowledge.

Stage 2 (Section 7.5): We show a general compiler that converts any CVQC protocol that is ϵ -non-destructive (Theorem 13), into one with nearly perfect completeness and soundness and is also ϵ -non-destructive.

Theorem 14. Let $\mathcal{L} = (\mathcal{L}_{\text{yes}}, \mathcal{L}_{\text{no}}) \in \text{QMA}_{a,b}$ with a QMA verifier V_{QMA} . There exists an efficient compiler that converts any ϵ -non-destructive CVQC protocol (P, V) for $\mathcal{L}$ w.r.t. V_{QMA} (as defined in Theorem 13), with completeness c and (testable) soundness s , into a new CVQC protocol (P', V') for $\mathcal{L}$ w.r.t. V_{QMA} , that has the following properties, whenever $|x| > \lambda, N \geq \frac{\lambda^2 |x|^2}{(c-s)^2}$:

1. **Computational soundness:** It has soundness $\text{negl}(\lambda)$.
2. **Completeness:** It has completeness $1 - \text{negl}(\lambda)$ using a single witness in $\text{Repairable}_{p,\epsilon}$, for any $\epsilon \geq 0$ and $p \geq a + \epsilon N$.
3. **ϵN -Non-Destructive:** There exists a negligible function μ such that for every $\lambda \in \mathbb{N}$, every $x \in \mathcal{L}_{\text{yes}}$ and every witness $|w\rangle \in \text{Repairable}_{p,\epsilon}(x)$, where $p \geq a + \epsilon N$,
$$\Pr_{(|\psi\rangle, b) \leftarrow (P'(|w\rangle), V')(1^\lambda, x)} [|\psi\rangle \in \text{Repairable}_{p-\epsilon N, \epsilon}(x)] = 1 - \mu(\lambda).$$
4. **Complexity:** The number of rounds, the communication complexity and the computational complexity grow by a multiplicative factor of $\text{poly}(\lambda, |x|, (c-s)^{-1})$.
5. **Argument of Knowledge:** If the underlying CVQC is a (d, s, δ) -testable argument of knowledge, then whenever $N \geq \frac{\lambda^{d+2} |x|^d}{(c-s)^2}$, the compiled CVQC is a δ/N -argument of knowledge according to Definition 7.²⁵

Loosely speaking, the new CVQC protocol (P', V') simply repeats the underlying protocol (P, V) sequentially $N = \text{poly}(\lambda, |x|, (c-s)^{-1})$ times. The verifier V' will accept if and only if at least $(\frac{c+s}{2}) \cdot N$ of the executions are accepting. If we start with a witness in $\text{Repairable}_{p,\epsilon}$ then after N executions, with overwhelming probability we are left with a witness in $\text{Repairable}_{p-\epsilon N, \epsilon}$, as desired. The end result is an ϵN -non-destructive CVQC protocol with completeness $1 - \text{negl}(\lambda)$ and soundness $\text{negl}(\lambda)$. Thus, after running the entire protocol, we are left with a witness that remains in $\text{Repairable}_{p-\epsilon N, \epsilon}(x)$ for some $p \geq a + \epsilon N$.

Note that if we started with a protocol that is non-destructive (i.e., ϵ -non-destructive for $\epsilon = \text{negl}(\lambda)$) then the resulting protocol is also non-destructive. For the general completeness case, Theorem 13 is restricted to $\epsilon = 1/\text{poly}(\lambda)$ due to $\text{poly}(1/\epsilon)$ runtime increase.

²⁵See Remark 4 for an explanation of the factor N loss.

Stage 3: Finally we show how to compile any ϵ -non-destructive CVQC protocol with completeness $1 - \text{negl}(\lambda)$ and soundness $\text{negl}(\lambda)$, into a state preserving CVQC protocol satisfying the properties of Theorem 12.

This is done by running the compiler from Theorem 13 (again), and using the fact that if the underlying CVQC protocol has completeness $c = 1 - \text{negl}(\lambda)$ then the resulting CVQC is witness preserving.

We next formally use Theorems 13 and 14 to prove Theorem 12.

Proof of Theorem 12 Fix any language $\mathcal{L} \in \text{QMA}_{a,b}$ and a corresponding QMA verifier V_{QMA} . We next describe the compiler, associated with parameter $\epsilon = \epsilon(\lambda)$. Given a CVQC protocol (P, V) for $\mathcal{L}$ w.r.t. V_{QMA} , with completeness c , (testable) soundness s , and (testable) (d, s, δ) argument of knowledge with some corresponding constant $d > 1$, the compiler does the following.

1. Let $N := \frac{|x|^{d\lambda^{d+3}}}{(c-s)^2}$.
2. Let (P_1, V_1) denote the CVQC protocol obtained by applying the compiler given in Theorem 13, with parameter ϵ , to (P, V) .
3. Let (P_2, V_2) denote the CVQC protocol obtained by applying the compiler given in Theorem 14.
4. Let (P_3, V_3) denote the CVQC protocol obtained by applying the compiler given in Theorem 13 to (P_2, V_2) .²⁶
5. Output (P_3, V_3) .

By Theorem 13, (P_1, V_1) has the same properties as (P, V) up to negligible factors, but is 4ϵ -non-destructive, which means that if the prover starts with a witness $|w\rangle \in \text{Repairable}_{p,\epsilon}(x)$, where $p \geq a + 4\epsilon$, then at the end of the protocol, with overwhelming probability, the prover has a state $|\psi\rangle \in \text{Repairable}_{p-4\epsilon,\epsilon}(x)$. Therefore, by Theorem 14, (P_2, V_2) has nearly perfect completeness and soundness and if the prover starts with a witness $|w\rangle \in \text{Repairable}_{p,\epsilon}(x)$ for $p \geq a + \epsilon 4N$ then at the end of the protocol, with overwhelming probability, the prover has a state $|w\rangle \in \text{Repairable}_{p-\epsilon 4N,\epsilon}(x)$. Therefore, by Theorem 13, the final protocol (P_3, V_3) also has nearly perfect completeness and soundness, but is also witness preserving for any witness $|w\rangle \in \text{Repairable}_{a+\epsilon 4N,\epsilon}(x)$, as desired. □

Thus, it remains to prove Theorems 13 and 14, which we do next.

²⁶Here the the parameter ϵ associated with the compiler is not important since (P_2, V_2) has completeness $1 - \text{negl}(\lambda)$, and the parameter ϵ is only used by the compiler in the case where the verifier rejects.

7.1 Proof of Theorem 13 (ϵ -Non-Destructive CVQC)

In this section, we prove Theorem 13. To this end, fix the following ingredients:

- A dual-mode trapdoor function family with $(1 - 2^{-\lambda})$ -state recovery $\mathcal{F}$ (Definition 11).
- A $(1 - 2^{-\lambda})$ -state-preserving argument SPA_{NP} for NP (Definition 12), which can be constructed from any dual-mode trapdoor function family with state recovery (Theorem 10).

We prove completeness and soundness in Section 7.2 (Claims 3 and 4). We prove witness preservation for the special case of $c = 1 - \text{negl}(\lambda)$ in Section 7.3 (Claim 5) and ϵ -nondestructiveness for the general case in Section 7.4 (Claim 6).

For the general case, we will also need the notion of a repairable witness.

Definition 15 ($\text{Repairable}_{p,\epsilon}$). *Fix any language $\mathcal{L} = (\mathcal{L}_{\text{yes}}, \mathcal{L}_{\text{no}}) \in \text{QMA}_{a,b}$ and any QMA verifier V_{QMA} . Then for any $x \in \mathcal{L}_{\text{yes}}$, $|w\rangle \in \text{Repairable}_{p,\epsilon}(x)$ if and only if for every $\lambda \in \mathbb{N}$, and for $\delta = 2^{-\lambda}$*

$$\Pr \left[p' < p - \epsilon : (p', \rho) \leftarrow \text{ValEst}_{V_{\text{QMA},x}(1^\lambda, \cdot)}(|w\rangle, \epsilon, \delta) \right] \leq \sqrt{2\delta}$$

where ValEst is the almost projective measurement defined in Lemma 4.

By Lemma 4, every $|w\rangle \in \text{Repairable}_{p,\epsilon}(x)$ satisfies

$$\Pr[\text{Accept} \leftarrow V_{\text{QMA},x}(|w\rangle)] \geq p - \epsilon - \sqrt{2\delta} - \delta$$

Conversely, by Corollary 1, every $|w\rangle$ which is accepted with probability $1 - \text{negl}(\lambda)$ belongs to $\text{Repairable}_{1-1/\text{poly}(\lambda),\epsilon}$ for any $1/\text{poly}(\lambda)$, for example $1/\text{poly}(\lambda) = 0.1$. Lemma 6 gives a more precise characterization of the Repairable set.

Notation for the underlying non-adaptive CVQC protocol. Before giving our construction, we define some additional notation for the underlying CVQC, similar to the notation used in Section 3.6.

- $\ell = \ell(|x|, \lambda)$ denotes the number of rounds in the protocol.
- The verifier is denoted by $V = (V_1, V_2)$, and its messages by $(q_1, \dots, q_\ell) \leftarrow V_1(x, 1^\lambda)$.
- The prover's messages are denoted by $(a_1, \dots, a_\ell)$.
- For every $i \in [\ell]$ we denote by

$$q_{[i]} = (q_1, \dots, q_i) \quad \text{and} \quad a_{[i]} = (a_1, \dots, a_i)$$

- We denote the partial transcript in the first i rounds by

$$\tau_{[i]} = (q_1, a_1, \dots, q_i, a_i).$$

- We denote the state of the prover after the first i rounds as $|\psi_{\tau[i]}\rangle$, and denote its initial state by $|\psi_\emptyset\rangle$.
- For every round $i \in [\ell]$, and for every message q_i that the verifier sends in the i 'th round, denote by $P_i = P_i(q_i)$ the unitary that the prover applies to its state upon receiving q_i from the verifier. Namely,

$$P_i : |\psi_{\tau[i-1]}\rangle \mapsto \sum_{a_i} \left(\beta_{a[i]} |a[i]\rangle \otimes |\psi_{\tau[i]}\rangle \right)$$

- For every $i \in [\ell]$ and every query sequence $q[i] = (q_1, \dots, q_i)$, we denote the coherent state of the prover after the i 'th round, which is the state obtained by applying $\prod_{j=1}^i P_j$ to its initial state denoted by $|\psi_\emptyset\rangle$ (without doing any measurements), by

$$\sum_{a[i]} \beta_{a[i]} |a[i]\rangle \otimes |\psi_{\tau[i]}\rangle$$

After measuring the first i answers its state becomes $|\psi_{\tau[i]}\rangle$.

In what follows we construct our ϵ -non-destructive CVQC protocol that preserves the completeness and soundness of the underlying CVQC protocol, up to negligible factors.

Construction 3 (ϵ -Non-Destructive CVQC). Fix a QMA language $\mathcal{L} = (\mathcal{L}_{\text{yes}}, \mathcal{L}_{\text{no}}) \in \text{QMA}_{a,b}$ with verifier V_{QMA} . To prove to the verifier that $x \in \mathcal{L}_{\text{yes}}$, the prover begins with a witness state $|w\rangle \in \text{Repairable}_{p,\epsilon}(x)$, where $p \geq a + 4\epsilon$, and the protocol proceeds as follows.

1. Prover:

- (a) **This step is skipped for the special case of $c = 1 - \text{negl}(\lambda)$. Otherwise:**

Compute $(p^*, \rho) \leftarrow \text{ValEst}_{V_{\text{QMA}},x}(|w\rangle, \epsilon, \delta)$, with $\delta = 2^{-\lambda}$.

This value p^* is needed by the witness recovery protocol (in Construction 5 for the general case).

Note that if $|w\rangle \in \text{Repairable}_{p,\epsilon}(x)$ then by Lemma 4 and Lemma 1, with overwhelming probability ρ has overwhelming probability mass on pure states $|\psi\rangle$ such that

$$|\psi\rangle \in \text{Repairable}_{p-\epsilon,\epsilon}(x).$$

Therefore, for the sake of simplicity (and without loss of generality), we abuse notation, and denote the resulting state by $|w\rangle \in \text{Repairable}_{p-\epsilon,\epsilon}(x)$.

- (b) Initialize three registers $(\mathcal{A}, \mathcal{P}, \mathcal{R})$.

- $\mathcal{A} = (\mathcal{A}_1, \dots, \mathcal{A}_\ell)$ is initialized to $|0\rangle$ and will contain a superposition over messages $a[i]$ that the prover could have sent in the base CVQC protocol.
- $\mathcal{P}$ is initialized to $|\psi_\emptyset\rangle = |w\rangle \otimes |0\rangle$ and will contain the prover's internal state $|\psi_{\tau[i]}\rangle$ for the base CVQC protocol.

- $\mathcal{R} = (\mathcal{R}_1, \dots, \mathcal{R}_\ell)$ is initialized to $|0\rangle$ and will contain randomness used for evaluating the trapdoor family $\mathcal{F}$.

2. **Verifier:**

- Compute the verifier's internal state and queries $(\mathbf{st}, q_1, \dots, q_\ell) \leftarrow V_1(x, 1^\lambda)$ for the base CVQC.
- Sample $(\mathbf{pp}, \mathbf{td}) \leftarrow \text{Setup}_{\mathcal{F}}(1^\lambda, \text{recovery})$.
- Send $\mathbf{pp}$ to the prover.

3. For each round $i \in [\ell]$ of the base CVQC protocol:

- Verifier:** Send q_i to the prover.
- Prover:**

- Apply $P_i(q_i)$ to registers $(\mathcal{A}_i, \mathcal{P})$, after which the prover's state is

$$\sum_{a_{[i]}} \alpha_{a_{[i]}} |a_{[i]}\rangle_{\mathcal{A}} \otimes |\psi_{\tau_{[i]}}\rangle_{\mathcal{P}}$$

- Coherently evaluate $f_{\mathbf{pp}}$ on $\mathcal{A}_i$ and measure the result. Specifically, apply the isometry

$$|a_i\rangle_{\mathcal{A}_i} \otimes |0\rangle_{\mathcal{R}_i} \mapsto \frac{1}{\sqrt{|\mathcal{R}|}} \sum_{r_i \in \mathcal{R}} |a_i\rangle_{\mathcal{A}_i} \otimes |r_i\rangle_{\mathcal{R}_i} \otimes |f_{\mathbf{pp}}(a_i; r_i)\rangle$$

to registers $(\mathcal{A}_i, \mathcal{R}_i)$ and measure the last register to obtain y_i .
At this step, the state of the prover is

$$\propto \left(I_{\mathcal{A}, \mathcal{P}, \mathcal{R}} \otimes \Pi_{y_{[i]}} \right) \left(\sum_{a_{[i]}} \alpha_{a_{[i]}} |a_{[i]}\rangle_{\mathcal{A}} \otimes |\psi_{\tau_{[i]}}\rangle_{\mathcal{P}} \otimes \sum_{r_{[i]} \in \mathcal{R}^i} |r_{[i]}\rangle_{\mathcal{R}} \otimes |f_{\mathbf{pp}}(a_{[i]}; r_{[i]})\rangle \right) \quad (7)$$

where $\Pi_{y_{[i]}}$ is a projector onto $|y_{[i]}\rangle = |(y_1, \dots, y_i)\rangle$ and where

$$f_{\mathbf{pp}}(a_{[i]}; r_{[i]}) = (f_{\mathbf{pp}}(a_1; r_1), \dots, f_{\mathbf{pp}}(a_i; r_i)).$$

- Send y_i to the verifier.

4. **Verifier.** Send $\mathbf{st}$ to the prover.

5. **Prover.** Project the state to answers that are accepted or rejected by V_2 . Namely,

- Denote by U_{V_2} the unitary that computes the verdict $V_2(\mathbf{st}, a_{[\ell]})$.
- denote by M the measurement of this verdict.

(c) Apply $U_{V_2}^\dagger MU_{V_2}$ to the state, to obtain a verdict bit $b \in \{0, 1\}$.

The residual state becomes:

$$\propto \left(I_{\mathcal{A}, \mathcal{P}, \mathcal{R}} \otimes \Pi_{y_{[\ell]}} \right) \left(\sum_{a_{[\ell]}: V_2(\text{st}, a_{[\ell]})=b} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle_{\mathcal{A}} \otimes |\psi_{\tau_{[\ell]}}\rangle_{\mathcal{P}} \otimes \sum_{r_{[i]} \in \mathcal{R}^\ell} |r_{[i]}\rangle_{\mathcal{R}} \otimes |f_{\text{pp}}(a_{[\ell]}; r_{[i]})\rangle \right)$$

6. If $b = 0$ then abort. Otherwise, continue.

7. **Both:** The prover proves to the verifier that the statement $(\text{st}, \text{pp}, y_{[\ell]})$ is in the NP language

$$\mathcal{L}_{\text{NDCVQC}} := \left\{ (\text{st}, \text{pp}, y_{[\ell]}) : \begin{array}{l} \exists (a_{[\ell]}, r_{[\ell]}) \text{ s.t.} \\ V_2(\text{st}, a_{[\ell]}) = \text{Accept} \\ \wedge y_i = f_{\text{pp}}(a_i; r_i) \quad \forall i \in [\ell] \end{array} \right\} \quad (8)$$

using SPA_{NP} , the state-preserving argument for NP. The prover uses the contents of registers $(\mathcal{A}, \mathcal{R})$ as the witness.

8. **Verifier:** Output the result of SPA_{NP} . Send td to the prover.

9. **Prover:** Run the witness recovery algorithm (Construction 4 for the case where the completeness or the underlying protocol is $1 - \text{negl}(\lambda)$, and Construction 5 for the general case) using td and all three internal registers $(\mathcal{A}, \mathcal{P}, \mathcal{R})$.

7.2 Completeness and Soundness

Claim 3. Let (P, V) be a CVQC and let (P', V') be result of compiling (P, V) using Construction 3. If (P, V) has completeness c , then (P', V') has completeness $c - \text{negl}(\lambda)$.

Proof. Let $|w'\rangle$ be the state after running ValEst in step 1a. As noted previously, if $|w\rangle \in \text{Repairable}_{p, \epsilon}(x)$, then with overwhelming probability $|w'\rangle \in \text{Repairable}_{p-\epsilon, \epsilon}$. Therefore

$$\Pr[\text{Accept} \leftarrow V_{\text{QMA}, x}(|w'\rangle)] \geq p - \epsilon - \sqrt{2\delta} - \delta \geq a + 2\epsilon$$

By the completeness of (P, V) , the prover's measurement in step 5 outputs a verdict bit $b = 1$ with probability $\geq c$. Whenever this occurs, the prover's state is supported on witnesses for $\mathcal{L}_{\text{NDCVQC}}$, so completeness of SPA_{NP} implies that the verifier accepts with overall probability $c - \text{negl}(\lambda)$. $\square$

Claim 4. Let (P, V) be a CVQC and let (P', V') be result of compiling (P, V) using Construction 3.

If (P, V) has s -testable soundness and is a (d, s, δ) -testable proof of quantum knowledge for the relation decided by the QMA verifier $V_{\text{QMA}}(1^\lambda, \cdot, \cdot)$, then (P', V') has $s + \text{negl}(\lambda)$ -testable soundness and is a $(d + 1, s + \text{negl}(\lambda), \delta + \text{negl}(\lambda))$ -testable proof of quantum knowledge.

Furthermore, if (P, V) is an (s, δ) -testable proof of quantum knowledge for the relation decided by the QMA verifier $V_{\text{QMA}}(1^\lambda, \cdot, \cdot)$, then (P', V') is an $(s + \text{negl}(\lambda), \delta + \text{negl}(\lambda))$ -testable proof of quantum knowledge.

Proof. Let $x \in \mathcal{L}_{\text{yes}} \cup \mathcal{L}_{\text{no}}$. First, note that when V is composed of two possible verifier algorithms V_{T} and V_{C} , then V' can also be decomposed into analogous verification algorithms V'_{T} and V'_{C} . We consider V_{T} to be the “test” verifier and V_{C} to be the “honest” verifier from Definition 8 (or Definition 9 for argument of knowledge). We show that for every QPT prover P^* which passes $V'_{\text{T}}(x)$ (resp., $V'_{\text{C}}(x)$) with probability p in an execution of (P', V') , there exists a QPT prover $\tilde{P}^*$ that convinces $V_{\text{T}}(x)$ (resp., $V_{\text{C}}(x)$) in an execution of the underlying (P, V) protocol with probability $\geq p - \text{negl}(\lambda)$. Thus, if (P, V) is sound, then so is (P', V') , and if it is a proof of quantum knowledge, then so is (P', V') .

$\tilde{P}^*$ begins by sampling $(\text{pp}, \text{td}) \leftarrow \text{Setup}_{\mathcal{F}}(1^\lambda, \text{injective})$. Then, for each round i of the underlying (P, V) protocol, it receives the external verifier’s query q_i and forwards it internally to P^* , who sends back a message y_i . $\tilde{P}^*$ runs the trapdoor function extractor $a'_i \leftarrow \text{Ext}_{\mathcal{F}}(\text{td}, y_i)$, then sends a'_i to the external verifier.

We now analyze the success probability of $\tilde{P}^*$. Let st be the verifier’s final internal state. Suppose that $(\text{st}, \text{pp}, y_{[\ell]}) \in \mathcal{L}_{\text{NDCVQC}}$. Then there exist $(a_{[\ell]}, r_{[\ell]})$ such that the following two conditions hold:

$$\begin{aligned} V_2(\text{st}, a_{[\ell]}) &= \text{Accept} \\ y_i &= f_{\text{pp}}(a_i; r_i) \quad \forall i \in [\ell] \end{aligned}$$

where $a_{[\ell]} = (a_1, \dots, a_\ell)$, where $r_{[\ell]} = (r_1, \dots, r_\ell)$.

The trapdoor property of $\mathcal{F}$ in injective mode (Definition 11) implies that the extractor outputs $a'_i = a_i$ with overwhelming probability. Furthermore, the first condition says that sending each a_i to the verifier constitutes an accepting transcript. Thus, it suffices to show that

$$\Pr_{\tilde{P}^*}[(\text{st}, \text{pp}, y_{[\ell]}) \in \mathcal{L}_{\text{NDCVQC}}] = p - \text{negl}(\lambda)$$

Consider the following hybrid experiments.

- **Hyb₀** is an execution of $(P^*, V'(x))$ up to the stage where V decides its output. The experiment ends *before* V' sends td to P^* for witness recovery. Let b be the decision bit of V' , which is its decision in the execution of $\text{SPANP}\langle P^*, V \rangle(1^\lambda, (\text{st}, \text{pp}, y_{[\ell]}))$.
- **Hyb₁** replaces the trapdoor function specified by pp with an *injective-mode* function $(\text{pp}, \text{td}) \leftarrow \text{Setup}_{\mathcal{F}}(1^\lambda, \text{injective})$.

Hyb₁ can be run using an externally generated pp and generating everything else internally. Therefore dual-mode indistinguishability of $\mathcal{F}$ implies that

$$\left| \Pr_{\text{Hyb}_1}[b = \text{Accept}] - \Pr_{\text{Hyb}_0}[b = \text{Accept}] \right| = \text{negl}(\lambda)$$

The completeness and soundness of SPANP imply that

$$\Pr_{\text{Hyb}_1}[b = \text{Accept}] = \Pr_{\text{Hyb}_1}[(\text{st}, q_{[\ell]}, \text{pp}, y_{[\ell]}) \in \mathcal{L}_{\text{NDCVQC}}] - \text{negl}(\lambda)$$

Observe that Hyb_1 is distributed precisely as in the internal evaluation of $\tilde{P}^*$. Combining this with the observations from before, $\tilde{P}^*$ convinces the verifier in an execution of (P, V) with probability $p - \text{negl}(\lambda)$.

Thus, if V'_T accepts P^* with probability p , then V_T accepts $\tilde{P}^*$ with probability at least $p - \text{negl}(\lambda)$, which by the s -testable soundness of (P, V) implies that V_C accepts $\tilde{P}^*$ with probability at most s . In turn this implies that V'_C accepts P^* with probability at most $s + \text{negl}(\lambda)$, as desired. Furthermore, if (P, V) is a (s, δ) -testable proof of quantum knowledge, then running its extractor $\mathcal{E}$ on $\tilde{P}^*$ produces a witness ρ with $\geq \delta$ probability, that is,

$$\Pr[\text{Accept} \leftarrow \mathcal{R}(1^\lambda, x, \rho) : \rho \leftarrow \mathcal{E}^{\tilde{P}^*}(\rho, 1^\lambda, x)] \geq \delta(\lambda)$$

□

7.3 Witness Recovery: High Completeness

We first show the witness recovery algorithm in the case where the underlying CVQC has completeness $\geq 1 - \text{negl}(\lambda)$. We will later generalize it to any completeness parameter.

Construction 4 (Witness Recovery for $c \geq 1 - \text{negl}(\lambda)$). As we argue below, at the end of the CVQC protocol the prover holds a state that is negligibly close to

$$\propto \left(I_{\mathcal{A}, \mathcal{P}, \mathcal{R}} \otimes \Pi_{y_{[\ell]}} \right) \left(\sum_{a_{[\ell]}} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle_{\mathcal{A}} \otimes |\psi_{\tau_{[\ell]}}\rangle_{\mathcal{P}} \otimes \sum_{r_{[\ell]} \in \mathcal{R}^\ell} |r_{[\ell]}\rangle_{\mathcal{R}} \otimes |f_{\text{pp}}(a_{[\ell]}; r_{[\ell]})\rangle \right)$$

Thus the witness recovery algorithm proceeds as follows:

1. For every $i \in [\ell]$ run the trapdoor function recovery algorithm $\text{Recover}_{\mathcal{F}}(\text{td}, \cdot, y_i)$ on registers $(\mathcal{A}_i, \mathcal{R}_i)$, to obtain the state

$$\sum_{a_{[\ell]}} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle_{\mathcal{A}} \otimes |\psi_{\tau_{[\ell]}}\rangle_{\mathcal{P}}$$

2. For $i \in [\ell]$ apply $P_i^\dagger$ to registers $(\mathcal{A}_i, \mathcal{P})$, to obtain the initial state $|\psi_\emptyset\rangle = |w\rangle \otimes |0\rangle$.

Claim 5. *Let $|w\rangle$ be a witness for $x \in \mathcal{L}_{\text{yes}}$. Suppose the underlying CVQC protocol has completeness $c \geq 1 - \text{negl}(\lambda)$. Let NDCVQC be the protocol from Construction 3 together with witness recovery from Construction 4. Then NDCVQC is witness-preserving.*

Proof. Note that step 1a, where ValEst is run, is skipped in the case of $c \geq 1 - \text{negl}(\lambda)$. NDCVQC goes directly to computing the verifier's queries $q_{[\ell]}$ and the prover's messages $y_{[\ell]}$. By Equation (7), at the end of the ℓ 'th round of the CVQC protocol, the prover holds the state

$$\propto \left(I_{\mathcal{A}, \mathcal{P}, \mathcal{R}} \otimes \Pi_{y_{[\ell]}} \right) \left(\sum_{i, a_{[\ell]}} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle_{\mathcal{A}} \otimes |\psi_{\tau_{[\ell]}}\rangle_{\mathcal{P}} \otimes \sum_{r_{[\ell]} \in \mathcal{R}^\ell} |r_{[\ell]}\rangle_{\mathcal{R}} \otimes |f_{\text{pp}}(a_{[\ell]}; r_{[\ell]})\rangle \otimes |\phi_i\rangle \right)$$

where $\sum_{a_{[\ell]}} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle_{\mathcal{A}}$ is the result of coherently running the base CVQC prover on one of the input witnesses $|w_i\rangle$ in the support of $|\tilde{w}\rangle$.

By the trapdoor recovery property of $\mathcal{F}$ (Definition 11), it suffices to argue that the state of the prover, after running the state-preserving argument for NP (SPANP), is negligibly close to the state above. To see that this is the case, first recall that the underlying CVQC has completeness $1 - \text{negl}(\lambda)$. Thus, measuring whether $a_{[\ell]}$ is an accepting transcript is almost deterministic on any mixture of valid witnesses $|w_i\rangle$, so this measurement is gentle on the above state. Conditioned on the measurement accepting (which occurs with overwhelming probability), the prover holds a superposition over NP witnesses for $\mathcal{L}_{\text{NDCVQC}}$, so the SPANP protocol further disturbs the state by a negligible amount. $\square$

7.4 Witness Recovery: Arbitrary Completeness

Next, we present a witness recovery algorithm for the case that the underlying CVQC protocol has arbitrary completeness c . This witness recovery protocol is associated with a tunable parameter $\epsilon = \epsilon(\lambda)$ and has the guarantee that if the initial witness $|w\rangle$ is in $\text{Repairable}_{p,\epsilon}(x)$ for $p \geq a + 4\epsilon$, then the witness recovery algorithm will output a witness that is in $\text{Repairable}_{p-4\epsilon,\epsilon}(x)$.

Construction 5 (Witness Recovery for any completeness parameter c). This witness recovery algorithm takes as input a parameter ϵ and an estimate $p^* \in [0, 1]$ of the quality of the witness, as computed by P' in the first step of the protocol (Construction 3).

At the end of the CVQC protocol given in Construction 3, the prover either has the state

$$|\varphi_{\text{Reject}}\rangle = \left(I_{\mathcal{A}, \mathcal{P}, \mathcal{R}} \otimes \Pi_{y_{[\ell]}} \right) \left(\sum_{a_{[\ell]}: V_2(\text{st}, a_{[\ell]})=0} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle_{\mathcal{A}} \otimes |\psi_{\tau_{[\ell]}}\rangle_{\mathcal{P}} \otimes \sum_{r_{[\ell]} \in \mathcal{R}^\ell} |r_{[\ell]}\rangle_{\mathcal{R}} \otimes |f_{\text{pp}}(a_{[\ell]}; r_{[\ell]})\rangle \right)$$

or the state that is negligibly far from the state

$$|\varphi_{\text{Accept}}\rangle = \left(I_{\mathcal{A}, \mathcal{P}, \mathcal{R}} \otimes \Pi_{y_{[\ell]}} \right) \left(\sum_{a_{[\ell]}: V_2(\text{st}, a_{[\ell]})=1} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle_{\mathcal{A}} \otimes |\psi_{\tau_{[\ell]}}\rangle_{\mathcal{P}} \otimes \sum_{r_{[\ell]} \in \mathcal{R}^\ell} |r_{[\ell]}\rangle_{\mathcal{R}} \otimes |f_{\text{pp}}(a_{[\ell]}; r_{[\ell]})\rangle \right).$$

In what follows, we use the [CMSZ22] repair procedure (Lemma 5) to repair the state back to a state $\sigma^* \in \text{Repairable}_{p-4\epsilon,\epsilon}$. This is done as follows:

1. For every $i \in [\ell]$, run the trapdoor function recovery algorithm $\text{Recover}_{\mathcal{F}}(\text{td}, \cdot, y_i)$ on registers $(\mathcal{A}_i, \mathcal{R}_i)$, to obtain the state

$$|\psi^*\rangle = \sum_{a_{[\ell]}: V_2(\text{st}, a_{[\ell]})=b} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle_{\mathcal{A}} \otimes |\psi_{\tau_{[\ell]}}\rangle_{\mathcal{P}}$$

where b is the verdict bit computed by the prover in Item 5 in the CVQC protocol.

2. Denote by $U_{q_{[\ell]}}$ the unitary that converts the state $|\psi_\emptyset\rangle = |w\rangle \otimes |\mathbf{0}\rangle$ to the state

$$U_{q_{[\ell]}} |\psi_\emptyset\rangle = \sum_{a_{[\ell]}} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle_{\mathcal{A}} \otimes |\psi_{\tau_{[\ell]}}\rangle_{\mathcal{P}}$$

by emulating the CVQC protocol, with the honest prover given the state $|\psi_\emptyset\rangle$ and the honest verifier using queries $q_{[\ell]} = (q_1, \dots, q_\ell)$.

3. Denote by Π'_b the projective measurement that projects the state $U_{q_{[\ell]}} |\psi_\emptyset\rangle$ onto the state

$$|\psi^*\rangle = \sum_{a_{[\ell]}: V_2(\mathbf{st}, a_{[\ell]})=b} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle_{\mathcal{A}} \otimes |\psi_{\tau_{[\ell]}}\rangle_{\mathcal{P}}$$

4. Let $\Pi = U_{q_{[\ell]}}^\dagger \cdot \Pi' \cdot U_{q_{[\ell]}}$.

5. Let $|\psi^{**}\rangle = U_{q_{[\ell]}}^\dagger |\psi^*\rangle = U_{q_{[\ell]}}^\dagger \Pi' U_{q_{[\ell]}} |\psi_\emptyset\rangle$.

6. Let $\mathbf{M} = \text{ValEst}_{\text{VQMA},x}(\cdot, \epsilon, \delta)$ be the almost projective measurement, where $\delta = 2^{-\lambda}$, and where ValEst is from Lemma 4.

7. Compute $\sigma^* \leftarrow \text{Repair}_{\mathbf{M},\Pi}(|\psi^{**}\rangle, b, p^*, T)$, where $T = \sqrt{2\delta}$, where p^* is the quality estimate of the initial witness, and where Repair is from Lemma 5.

8. Estimate $(p^{**}, \sigma^{**}) \leftarrow \text{ValEst}_{\text{VQMA},x}(\sigma^*, \epsilon, \delta)$ and output σ^{**} along with the new quality estimate p^{**} .²⁷

Claim 6. *Let NDCVQC be the protocol from Construction 3 together with witness recovery from Construction 5. It is 4ϵ -non-destructive with respect to all witnesses in $\text{Repairable}_{p,\epsilon}$, for any $p \geq a + 4\epsilon$ (as defined in Definition 15). Furthermore, it runs in expected quantum polynomial time.*

Proof. Fix any $x \in \mathcal{L}_{\text{yes}}$ and let $|w\rangle \in \text{Repairable}_{p,\epsilon}(x)$ for any $p \geq a + 4\epsilon$.

We need to prove that running NDCVQC on such a $|w\rangle$ results in a witness in $\text{Repairable}_{p-4\epsilon,\epsilon}(x)$ with overwhelming probability, and that the runtime is expected quantum polynomial. Consider the following hybrid experiments.

- **Hyb₀** runs NDCVQC and outputs the (honest) prover's output state along with the number of times $\text{Repair}_{\mathbf{M},\Pi}$ makes oracle calls to M or Π .
- **Hyb₁** modifies **Hyb₀** by computing the transcript

$$\sum_{a_{[\ell]}: V_2(\mathbf{st}, a_{[\ell]})=b} \alpha_{a_{[\ell]}} |a_{[\ell]}\rangle_{\mathcal{A}} \otimes |\psi_{\tau_{[\ell]}}\rangle_{\mathcal{P}}$$

before evaluating f_{pp} on it. The overall description of this hybrid can be broken into the following steps:

²⁷As a small optimization when sequentially repeating the protocol, this estimate p^{**} can be used in place of making a fresh estimate in step 1(a) of Construction 3.

1. The prover estimates $(p^*, \rho) \leftarrow \text{ValEst}_{\text{V}_{\text{QMA},x}}(|w\rangle, \epsilon, \delta)$.
 2. The prover coherently computes the *entire* transcript $a_{[\ell]}$, where the verifier's queries $q_{[\ell]}$ are pre-sampled.
 3. **Evaluate f_{pp}** : The prover evaluates f_{pp} on $a_{[\ell]}$ and measures the result.
 4. **Measure Acceptance**: The prover measures whether the base CVQC verifier would accept $a_{[\ell]}$ (after receiving st from the verifier).
 5. **Prove in SPA_{NP}** : If the base CVQC verifier would accept $a_{[\ell]}$, prove this fact using SPA_{NP} .
 6. **Recover from f_{pp}** : The prover runs $\text{Recover}_{\mathcal{F}}$ on the registers containing the transcript $a_{[\ell]}$ and f_{pp} evaluation randomness, using the td_{pp} from the verifier.
 7. The prover uncomputes the base CVQC for $q_{[\ell]}$, then runs $\text{Repair}_{M,\Pi}$ on the transcript to obtain ρ .
 8. The prover runs $(p^{**}, \sigma^{**}) \leftarrow \text{ValEst}_{\text{V}_{\text{QMA},x}}(\rho, \epsilon, \delta)$ and outputs the residual state ρ' .
- Hyb_2 removes step 5.
 - Hyb_3 removes steps 3 and 6.

The output of Hyb_0 is equivalent to the output of Hyb_1 because the prover is honest.

$$\text{TraceDist}[\text{Hyb}_1, \text{Hyb}_2] \leq 2^{-\lambda}$$

by the $(1 - 2^{-\lambda})$ -state preservation of SPA_{NP} (Definition 12) because SPA_{NP} is only run when the prover's state is successfully projected onto witnesses for $(\text{st}, \text{pp}, y_{[\ell]}) \in \mathcal{L}_{\text{NDCVQC}}$.

$$\text{TraceDist}[\text{Hyb}_2, \text{Hyb}_3] \leq 2^{-\lambda}$$

by the $(1 - 2^{-\lambda})$ -state recovery of $\mathcal{F}$ and in particular Lemma 8, which states that measuring a predicate of the computational basis on a state $|\psi\rangle$ (step 4) is close to evaluating f_{pp} on $|\psi\rangle$, then measuring the predicate, and finally running $\text{Recover}_{\mathcal{F}}$ on the residual state (steps 3, 4, 6). Therefore

$$\text{TraceDist}[\text{Hyb}_0, \text{Hyb}_3] \leq 2^{-\lambda+1}$$

Observe that Hyb_3 is exactly the repair experiment from Lemma 5, with a binary outcome damaging measurement. Therefore the state ρ prior to running ValEst in Hyb_3 satisfies

$$\Pr_{(p^{**}, \sigma') \leftarrow \text{ValEst}(\rho, \epsilon, \delta)} [|p^{**} - p^*| \geq 2\epsilon] \leq 2 \left(\delta + \sqrt{\delta} \right) + 4\sqrt{\delta}$$

Furthermore, $\text{Repair}_{M,\Pi}$ makes 5 oracle calls to M and Π in expectation in Hyb_3 .

Using the trace distance of Hyb_0 to Hyb_3 ,

$$\Pr_{\text{Hyb}_0} [p^{**} \leq p^* - 2\epsilon] \leq 2^{-\lambda+1} + 2 \left(\delta + \sqrt{\delta} \right) + 4\sqrt{\delta} = 2^{-\lambda+2} + 5 \cdot 2^{-\lambda/2}$$

since $\delta = 2^{-\lambda}$. Furthermore in Hyb_0 , $\text{Repair}_{M,\Pi}$ makes $5 + 2^{-\lambda+1}T < 6$ oracle calls to M and Π in expectation since $\text{Repair}_{M,\Pi}$ makes at most $T = 1/\sqrt{\delta} = 2^{\lambda/2}$ oracle calls.²⁸

Since $|w\rangle \in \text{Repairable}_{p,\epsilon}$, by union bound we have

$$\begin{aligned} \Pr_{\substack{(\sigma^*, p^*) \leftarrow \text{ValEst}(|w\rangle, \epsilon, \delta) \\ (\sigma^{**}, p^{**}) \leftarrow \text{ValEst}(\sigma^*, \epsilon, \delta)}} [p^{**} \leq p - 3\epsilon] &\leq \Pr_{\substack{(\sigma^*, p^*) \leftarrow \text{ValEst}(|w\rangle, \epsilon, \delta) \\ (\sigma^{**}, p^{**}) \leftarrow \text{ValEst}(\sigma^*, \epsilon, \delta)}} [p^* \leq p - \epsilon \text{ or } p^{**} \leq p^* - 2\epsilon] \\ &\leq \sqrt{2\delta} + 2^{-\lambda+2} + 5 \cdot 2^{-\lambda/2} \\ &< 2^{-\lambda+2} + 7 \cdot 2^{-\lambda/2} \end{aligned}$$

Now consider estimating again the probability of $V_{\text{QMA},x}(\sigma^{**})$ accepting. The (ϵ, δ) -almost projectivity of $\text{ValEst}(\cdot, \epsilon, \delta)$ implies that the new estimate $(\sigma^{***}, p^{***}) \leftarrow \text{ValEst}_{V_{\text{QMA},x}}(\sigma^{**}, \epsilon, \delta)$ satisfies $|p^{***} - p^{**}| < \epsilon$ except with probability δ . Thus, we can bound the conditional probability

$$\begin{aligned} \Pr_{\substack{(\sigma^*, p^*) \leftarrow \text{ValEst}(|w\rangle, \epsilon, \delta) \\ (\sigma^{**}, p^{**}) \leftarrow \text{ValEst}(\sigma^*, \epsilon, \delta) \\ (\sigma^{***}, p^{***}) \leftarrow \text{ValEst}(\sigma^{**}, \epsilon, \delta)}} \left[p^{***} \geq p - 4\epsilon \mid p^{**} \geq p - 3\epsilon \right] \\ &\geq \frac{1 - \Pr_{p^{**}, p^{***}} [|p^{***} - p^{**}| \geq \epsilon \text{ or } p^{**} < p - 3\epsilon]}{\Pr_{p^{**}} [p^{**} \geq p - 3\epsilon]} \\ &\leq \frac{1 - (\delta + 2^{-\lambda+2} + 7 \cdot 2^{-\lambda/2})}{1 - (2^{-\lambda+2} + 7 \cdot 2^{-\lambda/2})} \\ &= 1 - \frac{\delta}{1 - 2^{-\lambda+2} + 7 \cdot 2^{-\lambda/2}} \\ &\leq 1 - 2\delta \end{aligned}$$

since $2^{-\lambda+2} + 7 \cdot 2^{-\lambda/2} < 1/2$. By Lemma 1, conditioned on $p^{**} \geq p - 3\epsilon$, the residual state σ^{**} after the *second* ValEst (which occurs in the actual repair procedure) has at least $1 - \sqrt{2\delta} = 1 - \text{negl}(\lambda)$ probability mass on pure states $|\psi\rangle$ such that

$$\Pr_{(\sigma^{***}, p^{***}) \leftarrow \text{ValEst}(|\psi\rangle, \epsilon, \delta)} [p^{***} \geq p - 4\epsilon] \geq 1 - \sqrt{2\delta}$$

Since $p^{**} \geq p - 3\epsilon$ with $1 - \text{negl}(\lambda)$ probability as well, for every initial witness $|w\rangle \in \text{Repairable}_{p,\epsilon}(x)$,

$$\Pr_{(|\psi\rangle, b) \leftarrow \text{NDCVQC}(P(|w\rangle), V)(1^\lambda, x)} [(|\psi\rangle \in \text{Repairable}_{p-4\epsilon, \epsilon}(x))] = 1 - \text{negl}(\lambda).$$

□

²⁸Note that $M = \text{ValEst}_{V_{\text{QMA},x}}(\cdot, \epsilon, \delta)$ takes $\text{poly}(1/\epsilon, \log(\delta))$ time, so the overall expected runtime of $\text{Repair}_{M,\Pi}$ is $\text{poly}(1/\epsilon, \log(\delta))$.

7.5 Proof of Theorem 14 (Sequential Repetition)

In this section we prove Theorem 14. Namely, we construct a compiler that converts any ϵ -non-destructive CVQC into one that has negligible soundness and almost perfect completeness, while still being ϵ -non-destructive (as defined in Theorem 14). The compiler consists of many sequential repetitions of the underlying CVQC protocol, and the verifier accepts if the number of accepting executions exceeds a threshold.

Construction 6 (Sequential Amplification of ϵ -Non-Destructive CVQC). Fix a language $\mathcal{L} = (\mathcal{L}_{\text{yes}}, \mathcal{L}_{\text{no}}) \in \text{QMA}_{a,b}$ with a QMA verifier $V_{\text{QMA}} = \{V_{\text{QMA},x}\}_{x \in \mathcal{L}_{\text{yes}} \cup \mathcal{L}_{\text{no}}}$. Let (P, V) be an ϵ -non-destructive CVQC for $\mathcal{L}$ w.r.t. V_{QMA} , with completeness c and testable soundness s , where $c - s \geq 1/\text{poly}(\lambda)$, and where $V = (V_{\text{T}}, V_{\text{check}})$.

For N large enough (that we set below), we construct an $N\epsilon$ -non-destructive CVQC protocol (P', V') for $\mathcal{L}$ w.r.t. V_{QMA} with negligible soundness and overwhelming completeness. P' and V' take as input $(1^\lambda, x)$ where $x \in \mathcal{L}_{\text{yes}}$, and in P' takes as an additional input a witness $|w_0\rangle \in \text{Repairable}_{p,\epsilon}(x)$. The protocol $(P'(|w_0\rangle), V)(1^\lambda, x)$ proceeds as follows:

1. Set $p_0 = 1 - 2^{-\lambda}$.
2. Set $N = \frac{|x|^d \lambda^{d+2}}{(c-s)^2}$.
3. For $i = 1, \dots, N$, do the following:
 Emulate the protocol $\left(P(|w_{i-1}\rangle), V\right)(1^\lambda, x)$, where V implements V_{T} with probability $1/2$ and implements V_{check} with probability $1/2$.
 - (a) If V implemented V_{T} and the test failed (i.e., V_{T} rejects), then V' aborts and outputs **Reject**.
 - (b) If V implemented V_{check} then denote by b_i the output bit of V (1 denotes accept, 0 denotes reject).
 - (c) Denote by $|w_i\rangle$ the residual state of P .
4. Denote by $B \subseteq [N]$ all the repetitions where V executed V_{check} .
5. V' outputs **Accept** if $\sum_{i \in B} b_i \geq \left(\frac{c+s}{2}\right) |B|$. Otherwise, V' outputs **Reject**.
6. P' outputs $|w_N\rangle$.

Proof. We prove that (P', V') is ϵN -non-destructive, and that it has $1 - \text{negl}(\lambda)$ completeness and $\text{negl}(\lambda)$ computational soundness error. Below, we assume that $|x| > \lambda$, which can be achieved without loss of generality by padding the instance.

ϵN -Non-Destructiveness.

$$\begin{aligned}
& \Pr[|w_N\rangle \in \text{Repairable}_{p-N\epsilon,\epsilon}(x)] \geq \\
& \Pr[|w_N\rangle \in \text{Repairable}_{p-N\epsilon,\epsilon}(x) \mid |w_{N-1}\rangle \in \text{Repairable}_{p-(N-1)\epsilon,\epsilon}(x)] \cdot \\
& \Pr[|w_{N-1}\rangle \in \text{Repairable}_{p-(N-1)\epsilon,\epsilon}(x)] \geq \\
& (1 - \mu_N) \cdot \Pr[|w_{N-1}\rangle \in \text{Repairable}_{p-(N-1)\epsilon,\epsilon}(x)] \geq \\
& (1 - \mu_N)(1 - \mu_{N-1}) \cdot \Pr[|w_{N-2}\rangle \in \text{Repairable}_{p-(N-2)\epsilon,\epsilon}(x)] \geq \\
& \dots \\
& (1 - \mu_N)(1 - \mu_{N-1}) \cdot \dots \cdot (1 - \mu_1) \Pr[|w_0\rangle \in \text{Repairable}_{p,\epsilon}(x)] = \\
& (1 - \mu_N)(1 - \mu_{N-1}) \cdot \dots \cdot (1 - \mu_1) = \\
& 1 - \text{negl}(\lambda).
\end{aligned}$$

where all the equations except the last two follow from the fact that (P, V) is ϵ -non-destructive, the second to last equality follows by assumption, and the last equation follows from the fact that $N = \text{poly}(|x|, \lambda)$ and that each $\mu_i = \text{negl}(\lambda)$.

Completeness: Fix any $x \in \mathcal{L}_{\text{yes}}$ and fix any $|w_0\rangle \in \text{Repairable}_{p,\epsilon}(x)$ for $p \geq a + \epsilon N$. By the ϵN -non-destructiveness, with overwhelming probability each intermediate witness $|w_i\rangle \in \text{Repairable}_{p-N\epsilon,\epsilon}$. When this occurs, each $|w_i\rangle$ is accepted by V_{QMA} with probability $\geq a$. Therefore, the c -completeness of the underlying CVQC implies that for every $i \in B$,

$$\Pr[b_i = 1] \geq c$$

Therefore the random variable $\sum_{i=1}^N b_i$ is bounded below by $\text{Binomial}(N, c)$. Applying Hoeffding's inequality shows that

$$\Pr \left[\sum_{i=1}^N b_i \leq Nc - N \frac{c-s}{2} \right] \leq \exp \left(\frac{-2 \left(N \frac{c-s}{2} \right)^2}{N} \right) = \text{negl}(\lambda)$$

since $N > \frac{\lambda^{d+3}|x|^d}{(c-s)^2}$, $d > 1$ and $c - s < 1$.

Argument of Knowledge. Fix any $x \in \mathcal{L}_{\text{yes}} \cup \mathcal{L}_{\text{no}}$, and any efficient prover strategy P^* with arbitrary initial quantum advice ρ^* such that the verifier in the sequentially repeated protocol accepts with noticeable probability $\frac{1}{p_{\text{acc}}(\lambda)}$.

Our goal is to build a proof-of-knowledge extractor for the repeated protocol. Our extractor is inspired by [VZ21]. Looking ahead, our extractor will simply pick a random execution $j \leftarrow [N]$ and run the extractor of the underlying CVQC on this execution. We show that such an extractor succeeds with probability $\frac{\delta}{p_{\text{acc}}N}(\lambda) - \text{negl}(\lambda)$, where $\delta(\lambda)$ is the success probability of the underlying CVQC extractor.

Formally, we will first prove that for accepting transcripts, nearly all prefixes of the transcript satisfy the following: the prover, conditioned on this prefix, will pass V_T with high probability in the upcoming round. We call such a transcript a **Good** transcript.

Definition 16 (Good Transcripts). We say that a transcript $\tau = \tau_\lambda \in \text{Good}_\lambda$ if (1) τ is accepted by the verifier and (2) there exists a large subset of repetitions $S_\tau \subseteq [N]$, $|S_\tau| = (N - \lambda^{d+1})$ such that for all $i \in S_\tau$,

$$\Pr [P_i^*(\rho_{i-1}), V_{i,T}(1^\lambda, x) = 1 | \tau_{i-1}] > \left(1 - \frac{1}{\lambda^d}\right)$$

where ρ_{i-1} denotes the state on the prover's registers at the end of $i - 1$ repetitions with $\rho_0 = \rho^*$, $V_{i,T}$ denotes the output of the verification algorithm of the underlying CVQC in round i , and the probability is over the randomness of the protocol upto round i .

The next claim proves that an overwhelming fraction of accepting transcripts are **Good**.

Claim 7. There exists a negligible function $\mu(\cdot)$ such that

$$\Pr[\tau_\lambda \in \text{Good}_\lambda | \tau_\lambda \text{ is accepted}] \geq (1 - \mu(\lambda))$$

where the probability is over the randomness of the experiment.

Proof. Suppose the claim is not true, then there is a polynomial $\text{poly}(\cdot)$ such that for infinitely many $\lambda \in \mathbb{N}$,

$$\Pr[\tau_\lambda \notin \text{Good}_\lambda | \tau_\lambda \text{ is accepted}] \geq \frac{1}{\text{poly}(\lambda)}$$

Because P^* convinces the verifier with noticeable probability $\frac{1}{p_{\text{acc}}(\lambda)}$, this implies that with probability at least $\frac{1}{p_{\text{acc}}(\lambda)\text{poly}(\lambda)}$, the sampled transcript is (1) accepted and (2) has a set T of size at least λ^{d+1} such that for all $i \in T$,

$$\Pr [P_i^*(\rho_{i-1}), V_{i,T,\text{NDCVQC}}(1^\lambda, x) = (\cdot, 1) | \tau_{i-1}] < \left(1 - \frac{1}{\lambda^d}\right)$$

Since the verifier picks “test” with constant probability, such accepting transcripts are only generated with probability at most $(1 - \frac{1}{\lambda^d})^{\lambda^{d+1}} = \text{negl}(\lambda)$, a contradiction. $\square$

We will now prove that there exists a polynomial $p(\cdot)$ such that for an overwhelming fraction of accepting transcripts τ , there exists $i \in S_\lambda$ such that P_i^* passes V_C with probability greater than $s + \frac{1}{p(\lambda)}$ conditioned on τ .

To do this, we will first prove the following claim, which gives tail bounds on the number of accepts in rounds where the verifier picks V_C .

Claim 8. Define a sequence of variables $X_1 \dots X_N$ as follows. For each $i \in [N]$, denote by $X_i = 1$ the event that (1) the verifier picked V_C in repetition i , and (2) the prover passed V_C in repetition i (and otherwise, $X_i = 0$). Then for every $\gamma \in (0, 1)$,

$$\Pr \left[\sum_{i \in [N]} X_i - \sum_{i \in [N]} \mathbb{E}[X_i | \tau_{i-1}] \geq \gamma n_C \right] \leq \exp(-2\gamma^2 n_C) \quad (9)$$

Proof. For any i and transcript τ , define $Y_i = X_i - \mathbb{E}[X_i | \tau_{i-1}]$, where τ_{i-1} denotes a truncation of the transcript upto round $i - 1$. We have that for every $i \in S$ and every τ_{i-1} ,

$$\mathbb{E}[Y_i | \tau_{i-1}] = \mathbb{E}[X_i - \mathbb{E}[X_i | \tau_{i-1}] | \tau_{i-1}] = 0$$

Moreover, for every $i \leq N$ and transcript τ , $Y_i \in [-1, 1]$.

Then $S_0 = 0$, $S_k = \sum_{i=1}^k Y_k$ is a martingale because $\mathbb{E}[S_{k+1} | S_1, \dots, S_k] = \mathbb{E}[S_{k+1} | Y_1, \dots, Y_k] = S_k$, and it satisfies $|S_k - S_{k-1}| \leq 1$.

Then by Azuma-Hoeffding (Theorem 6) applied to Martingales with distance $c_k = 1$, we have

$$\Pr[S_m - S_0 \geq t] \leq \exp(-2t^2/m).$$

Let n_C denote the total number of repetitions in S where the verifier picks V_C . Let $t = \gamma n_C$. Then,

$$\Pr[S_{n_C} \geq \gamma n_C] \leq \exp(-2(\gamma n_C)^2/n_C) = \exp(-2\gamma^2 n_C)$$

which is the same as saying

$$\Pr \left[\sum_{i \in [N]} X_i - \sum_{i \in [N]} \mathbb{E}[X_i | \tau_{i-1}] \geq \gamma n_C \right] \leq \exp(-2\gamma^2 n_C)$$

This completes the proof. $\square$

Equation (9) above proves that for nearly all transcripts, the total number of accepts in rounds where the verifier picks V_C deviates from its expected value by only a small amount. We will now use the definition of **Good** to bound the expected value of the total number of accepts in check rounds, and use this bound to argue (in the proof of the following claim) that there must exist a round where both V_T accepts with probability $> \zeta$ and V_C accepts with probability $> s$. Therefore on this round, the underlying extractor will succeed.

Claim 9. *With probability $(1 - \text{negl})$ over $\tau = \tau_\lambda \in \text{Good}_\lambda$, there exists $i \in S_\tau$ such that*

$$\Pr [P_i^*(\rho_{i-1}), V_{i,C,\text{NDCVQC}}(1^\lambda, x) = (\cdot, 1) | \tau_{i-1}] > s + \frac{1}{p(\lambda)} \quad (10)$$

Proof. Suppose towards a contradiction that with noticeable probability over $\tau = \tau_\lambda \in \text{Good}_\lambda$, for all $i \in S_\tau$,

$$\Pr [P_i^*(\rho_{i-1}), V_{i,C,\text{NDCVQC}}(1^\lambda, x) = (\cdot, 1) | \tau_{i-1}] \leq s + \text{negl}(\lambda) \quad (11)$$

which (by the definition of X_i) is the same as saying that for $i \in S_\tau$, $\mathbb{E}[X_i | \tau_{i-1}] \leq s_H + \text{negl}(\lambda)$. This implies that with noticeable probability over $\tau = \tau_\lambda \in \text{Good}_\lambda$,

$$\sum_{i \in [n_C]} \mathbb{E}[X_i | \tau_{i-1}] \leq s n_C + \lambda^{d+1} + \text{negl}(\lambda)$$

Equation (9) proves that with all but an inverse-exponential probability over transcripts, the actual sum of X_i does not exceed its expected value except by an inverse-exponentially small amount. Thus, upon combining with equation (9) we have that with noticeable probability over $\tau_\lambda \in \text{Good}_\lambda$

$$\sum_{i \in S} X_i > \left(s + \frac{\lambda^{d+1}}{n_C} + \gamma \right) n_C + \text{negl}(\lambda)$$

Setting $\gamma = \frac{\lambda}{\sqrt{n_C}}$, we have that with noticeable probability over $\tau_\lambda \in \text{Good}_\lambda$,

$$\sum_{i \in S} X_i > \left(s + \frac{\lambda^{d+1}}{n_C} + \frac{\lambda}{\sqrt{n_C}} \right) n_C + \text{negl}(\lambda)$$

Fix any transcript that satisfies the above equation. By a Chernoff bound, with overwhelming probability, $n_C \geq \frac{N}{3}$, and therefore for all but a negligible fraction of such transcripts,

$$\frac{\sum_{i \in n_C} X_i}{n_C} \geq \left(s + 3 \frac{\lambda^{d+1}}{N} + 3 \frac{\lambda}{\sqrt{N}} \right)$$

Recall that $N = \left(\frac{\lambda^{d+2}}{(c-s)^2} \right)$, thus the fraction $\frac{\sum_{i \in n_C} X_i}{n_C}$ is at most

$$\left(s + 3 \frac{\lambda^{d+1}(c-s)^2}{\lambda^{d+2}} + 3 \frac{\lambda(c-s)}{\lambda^{3/2}} \right)$$

which, for large enough λ , is at most

$$\left(s + \frac{c-s}{3} \right) \leq \left(\frac{c+s}{2} \right)$$

and therefore this noticeable fraction of **Good** transcripts τ are not accepted, a contradiction. $\square$

Combining Claim 7 and Claim 9, we have that with probability $(1 - \text{negl}(\lambda))$ over $\tau_\lambda \in \text{Good}_\lambda$, there exists $j \in [N]$ such that:

$$\Pr [P_j^*(\rho_{j-1}), V_{j,T,\text{NDCVQC}}(1^\lambda, x) = (\cdot, 1) | \tau_{j-1}] > \left(1 - \frac{1}{\lambda^d} \right)$$

and

$$\Pr [P_j^*(\rho_{j-1}), V_{j,C,\text{NDCVQC}}(1^\lambda, x) = (\cdot, 1) | \tau_{j-1}] > s + \frac{1}{p(\lambda)}$$

and furthermore, the event $\tau_\lambda \in \text{Good}_\lambda$ happens with probability $\frac{1}{p(\lambda)} - \text{negl}(\lambda)$.

Thus, an extractor that randomly picks a sequential round $j \leftarrow [N]$ and runs the extraction procedure of the underlying CVQC on round j will, with probability $\frac{1}{pN}(\lambda) - \text{negl}(\lambda)$, sample j satisfying the two conditions above, and therefore will output a witness with probability $\frac{\delta}{pN}(\lambda) - \text{negl}(\lambda)$, where $\delta(\lambda)$ is the (non-negligible) probability that the underlying extractor succeeds.

Remark 4. *The factor N loss is difficult to eliminate due to the following case: the adversary participates in $\frac{c+s}{2}N$ executions with an extremely good witness that the CVQC accepts with probability 1, and uses a junk state in the remaining executions. In this case, the extractor would need to identify an execution where the adversary uses a good witness.*

Soundness. Fix any $x \in \mathcal{L}_{\text{no}}$ and any prover P^* that generates accepting proofs for x with probability $\frac{1}{p_{\text{acc}}(\lambda)}$, for some polynomial p_{acc} . Towards a contradiction, for such a prover, note that by Claim 7 and Claim 9 above, there exists $j \in [N]$, transcript prefix τ_{j-1} and corresponding prover state ρ_{j-1} such that:

$$\Pr [P_j^*(\rho_{j-1}), V_{j,T,\text{NDCVQC}}(1^\lambda, x) = (\cdot, 1) | \tau_{j-1}] > \left(1 - \frac{1}{\lambda^d}\right)$$

and

$$\Pr [P_j^*(\rho_{j-1}), V_{j,C,\text{NDCVQC}}(1^\lambda, x) = (\cdot, 1) | \tau_{j-1}] > s + \frac{1}{p(\lambda)}$$

This circuit P_j^* with input state ρ_{j-1} contradicts testable s -soundness of the underlying CVQC, as desired. $\square$

8 Acknowledgments

DK was supported in part by AFOSR, NSF CAREER CNS-2238718, NSF 2112890, NSF CNS-2247727 and a Google Research Scholar award. This material is based upon work supported by the Air Force Office of Scientific Research under award number FA9550-23-1-0543. This work was partially done while the authors were visiting the Simons Institute for the Theory of Computing, Berkeley.

References

- [ACGH20] Gorjan Alagic, Andrew M. Childs, Alex B. Grilo, and Shih-Han Hung. Non-interactive classical verification of quantum computation. In Rafael Pass and Krzysztof Pietrzak, editors, *TCC 2020, Part III*, volume 12552 of *LNCS*, pages 153–180. Springer, Cham, November 2020.
- [AMR22] Navid Alamati, Giulio Malavolta, and Ahmadreza Rahimi. Candidate trapdoor claw-free functions from group actions with applications to quantum protocols. In Eike Kiltz and Vinod Vaikuntanathan, editors, *TCC 2022, Part I*, volume 13747 of *LNCS*, pages 266–293. Springer, Cham, November 2022.
- [BCM⁺18] Zvika Brakerski, Paul F. Christiano, Urmila Mahadev, Umesh V. Vazirani, and Thomas Vidick. A cryptographic test of quantumness and certifiable randomness from a single quantum device. In Mikkel Thorup, editor, *59th IEEE Annual*

Symposium on Foundations of Computer Science, FOCS 2018, Paris, France, October 7-9, 2018, pages 320–331. IEEE Computer Society, 2018.

- [BGK⁺24] James Bartusek, Vipul Goyal, Dakshita Khurana, Giulio Malavolta, Justin Raizes, and Bhaskar Roberts. Software with certified deletion. In Marc Joye and Gregor Leander, editors, *EUROCRYPT 2024, Part IV*, volume 14654 of *LNCS*, pages 85–111. Springer, Cham, May 2024.
- [BK25] James Bartusek and Dakshita Khurana. On the power of oblivious state preparation. In *CRYPTO 2025, Part II*, *LNCS*, pages 575–607. Springer, Cham, August 2025.
- [BKL⁺22] James Bartusek, Yael Tauman Kalai, Alex Lombardi, Fermi Ma, Giulio Malavolta, Vinod Vaikuntanathan, Thomas Vidick, and Lisa Yang. Succinct classical verification of quantum computation. In Yevgeniy Dodis and Thomas Shrimpton, editors, *CRYPTO 2022, Part II*, volume 13508 of *LNCS*, pages 195–211. Springer, Cham, August 2022.
- [BKL⁺25] Matilde Baroni, Igor Klep, Dominik Leichtle, Marc-Olivier Renou, Ivan Šupić, Lucas Tendick, and Xiangling Xu. Quantitative quantum soundness for all multipartite compiled nonlocal games, 2025.
- [BKM⁺23] James Bartusek, Dakshita Khurana, Giulio Malavolta, Alexander Poremba, and Michael Walter. Weakening assumptions for publicly-verifiable deletion. In Guy N. Rothblum and Hoeteck Wee, editors, *TCC 2023, Part IV*, volume 14372 of *LNCS*, pages 183–197. Springer, Cham, November / December 2023.
- [BKM⁺25] Kaniuar Bacho, Alexander Kulpe, Giulio Malavolta, Simon Schmidt, and Michael Walter. Compiled nonlocal games from any trapdoor claw-free function. In *CRYPTO 2025, Part II*, *LNCS*, pages 642–673. Springer, Cham, August 2025.
- [BKP23] James Bartusek, Dakshita Khurana, and Alexander Poremba. Publicly-verifiable deletion via target-collapsing functions. In Helena Handschuh and Anna Lysyanskaya, editors, *CRYPTO 2023, Part V*, volume 14085 of *LNCS*, pages 99–128. Springer, Cham, August 2023.
- [BLJŠ25] Matilde Baroni, Dominik Leichtle, Siniša Janković, and Ivan Šupić. Bounding the asymptotic quantum value of all multipartite compiled non-local games, 2025. Asymptotic quantum soundness for all multipartite compiled games.
- [BM22] James Bartusek and Giulio Malavolta. Indistinguishability obfuscation of null quantum circuits and applications. In Mark Braverman, editor, *ITCS 2022*, volume 215, pages 15:1–15:13. LIPIcs, January / February 2022.
- [CCY20] Nai-Hui Chia, Kai-Min Chung, and Takashi Yamakawa. Classical verification of quantum computations with efficient verifier. In Rafael Pass and Krzysztof

- Pietrzak, editors, *TCC 2020, Part III*, volume 12552 of *LNCS*, pages 181–206. Springer, Cham, November 2020.
- [CKP25] Rohit Chatterjee, Srijita Kundu, and Supartha Podder. Uncloneable quantum states are necessary as proofs and advice. In Michal Koucký and Nikhil Bansal, editors, *Proceedings of the 57th Annual ACM Symposium on Theory of Computing, STOC 2025, Prague, Czechia, June 23–27, 2025*, pages 1817–1827. ACM, 2025.
- [CLLZ21] Andrea Coladangelo, Jiahui Liu, Qipeng Liu, and Mark Zhandry. Hidden cosets and applications to unclonable cryptography. In Tal Malkin and Chris Peikert, editors, *CRYPTO 2021, Part I*, volume 12825 of *LNCS*, pages 556–584, Virtual Event, August 2021. Springer, Cham.
- [CMSZ22] Alessandro Chiesa, Fermi Ma, Nicholas Spooner, and Mark Zhandry. Post-quantum succinct arguments: breaking the quantum rewinding barrier. In *2021 IEEE 62nd Annual Symposium on Foundations of Computer Science (FOCS)*, pages 49–58. IEEE, 2022.
- [GKNV25] Sam Gunn, Yael Tauman Kalai, Anand Natarajan, and Agi Villanyi. Classical commitments to quantum states. In *Proceedings of the 57th Annual ACM Symposium on Theory of Computing (STOC 2025)*. ACM, 2025.
- [GR25] Vipul Goyal and Justin Raizes. Proofs of no intrusion. *IACR Cryptol. ePrint Arch.*, page 1826, 2025.
- [Gup] Aparna Gupte. Personal Communication.
- [GV24] Aparna Gupte and Vinod Vaikuntanathan. How to construct quantum FHE, generically. In Leonid Reyzin and Douglas Stebila, editors, *CRYPTO 2024, Part III*, volume 14922 of *LNCS*, pages 246–279. Springer, Cham, August 2024.
- [Ji15] Zhengfeng Ji. Classical verification of quantum proofs, 2015.
- [KLKY23] Yael Tauman Kalai, Alex Lombardi, Vinod Vaikuntanathan, and Lisa Yang. Quantum advantage from any non-local game. In *Proceedings of the 55th Annual ACM Symposium on Theory of Computing (STOC 2023)*, pages 1617–1628. ACM, 2023.
- [KMP⁺25] Alexander Kulpe, Giulio Malavolta, Connor Paddock, Simon Schmidt, and Michael Walter. A bound on the quantum value of all compiled nonlocal games. In *Proceedings of the 57th Annual ACM Symposium on Theory of Computing (STOC ’25)*, pages 222–233. ACM, 2025.
- [KNY23] Fuyuki Kitagawa, Ryo Nishimaki, and Takashi Yamakawa. Publicly verifiable deletion from minimal assumptions. In Guy N. Rothblum and Hoeteck Wee,

- editors, *TCC 2023, Part IV*, volume 14372 of *LNCS*, pages 228–245. Springer, Cham, November / December 2023.
- [LMS22a] Russell WF Lai, Giulio Malavolta, and Nicholas Spooner. Quantum rewinding for many-round protocols. In *Theory of Cryptography Conference*, pages 80–109. Springer, 2022.
 - [LMS22b] Alex Lombardi, Fermi Ma, and Nicholas Spooner. Post-quantum zero knowledge, revisited or: How to do quantum rewinding undetectably. In *63rd FOCS*, pages 851–859. IEEE Computer Society Press, October / November 2022.
 - [Mah18] Urmila Mahadev. Classical verification of quantum computations. In *2018 IEEE 59th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 259–267, Paris, France, 2018. IEEE.
 - [MF16] Tomoyuki Morimae and Joseph F Fitzsimons. Post hoc verification with a single prover. *arXiv preprint arXiv:1603.06046*, 2016.
 - [MNZ24] Tony Metger, Anand Natarajan, and Tina Zhang. Succinct arguments for QMA from standard assumptions via compiled nonlocal games. In *65th FOCS*, pages 1193–1201. IEEE Computer Society Press, October 2024.
 - [MP11] Daniele Micciancio and Chris Peikert. Trapdoors for lattices: Simpler, tighter, faster, smaller. *IACR Cryptol. ePrint Arch.*, page 501, 2011.
 - [MW05] Chris Marriott and John Watrous. Quantum arthur-merlin games. *Comput. Complex.*, 14(2):122–152, 2005.
 - [NZ23] Anand Natarajan and Tina (Xiaodi) Zhang. Bounding the quantum value of compiled nonlocal games: From CHSH to BQP verification. In *2023 IEEE 64th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 1342–1348. IEEE, 2023.
 - [VZ21] Thomas Vidick and Tina Zhang. Classical proofs of quantum knowledge. In Anne Canteaut and François-Xavier Standaert, editors, *Advances in Cryptology – EUROCRYPT 2021*, volume 12697 of *Lecture Notes in Computer Science*, pages 630–660, Cham, 2021. Springer.
 - [Zha20] Mark Zhandry. Schrödinger’s pirate: How to trace a quantum decoder. In Rafael Pass and Krzysztof Pietrzak, editors, *TCC 2020, Part III*, volume 12552 of *LNCS*, pages 61–91. Springer, Cham, November 2020.
 - [Zha22] Jiayu Zhang. Classical verification of quantum computations in linear time. In *63rd FOCS*, pages 46–57. IEEE Computer Society Press, October / November 2022.

A Proofs of No-Intrusion

As an application of state-preserving arguments for NP, we show how to construct public-key encryption (PKE) with *proofs of no-intrusion* (PoNIs). PKE with PoNIs was introduced by [GR25]. Roughly, it allows a classical verifier to test that a quantum ciphertext has not been stolen from a server, in the sense that nobody besides the server can decrypt it even given the key, without destroying the server's copy of the ciphertext.

Our construction achieves a stronger notion of verifiability than [GR25]. In their construction, only the original encryptor can verify their own ciphertext because the PoNI verifier requires secret information about it. In our construction, the encryptor may publish part of their verification key to allow *anyone* to act as the PoNI verifier.

A.1 Definition

Definition 17 (Encryption with Proofs of No-Intrusion). *A (public-key) **encryption scheme with a proof of no-intrusion** consists of the following QPT algorithms.*

- $(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda)$ takes as input the security parameter 1^λ then outputs a public key pk and a secret key sk .
- $\text{vk} \leftarrow \text{VKGen}(1^\lambda)$ takes as input the security parameter 1^λ then outputs a verification key vk .
- $\text{ct} \leftarrow \text{Enc}(\text{pk}, m, \text{vk})$ takes as input a public key pk , a message m , and a verification key vk , then outputs a ciphertext ct .
- $m \leftarrow \text{Dec}(\text{sk}, \text{ct})$ takes as input a secret key sk and a ciphertext ct , then outputs a message m .

Additionally, it is equipped with a **proof of no-intrusion**, which is an interactive protocol with classical communication between a QPT prover P holding a state $|\psi\rangle$ and a PPT verifier holding a verification key vk :

$$(|\psi'\rangle, b) \leftarrow \text{PoNI}(P(|\psi\rangle), V(\text{vk}))$$

At the end of the protocol, P outputs a state $|\psi'\rangle$ and V outputs a decision bit $b \in \{\text{Accept}, \text{Reject}\}$.

The scheme must satisfy the following properties:

- **Correctness.** For every message m , every key pair (pk, sk) in the support of $\text{KeyGen}(1^\lambda)$, and every verification key vk in the support of $\text{VKGen}(1^\lambda)$,

$$\Pr[m = \text{Dec}(\text{sk}, \text{Enc}(\text{pk}, m, \text{vk}))] = 1 - \text{negl}(\lambda)$$

- **Semantic Security.** For every pair of messages (m_0, m_1) ,

$$\left\{ \begin{array}{l} (\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda) \\ (\text{ct}_0, \text{vk}) : \quad \text{vk} \leftarrow \text{VKGen}(1^\lambda) \\ \quad \text{ct}_0 \leftarrow \text{Enc}(\text{pk}, m_0, \text{vk}) \end{array} \right\} \approx_c \left\{ \begin{array}{l} (\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda) \\ (\text{ct}_1, \text{vk}) : \quad \text{vk} \leftarrow \text{VKGen}(1^\lambda) \\ \quad \text{ct}_1 \leftarrow \text{Enc}(\text{pk}, m_1, \text{vk}) \end{array} \right\}$$

- **PoNI Correctness and State Preservation:** Let (pk, sk) be in the support of $\text{KeyGen}(1^\lambda)$, let m be an arbitrary message, let vk be in the support of VKGen , and let $|ct\rangle$ be a ciphertext which decrypts to m with certainty.

$$(|\psi'\rangle, b) \leftarrow \text{PoNI}\langle P(|ct\rangle), V(vk) \rangle$$

$b = \text{Accept}$ and $|\psi'\rangle$ belongs to the space of ciphertexts which decrypt to m .

- **PoNI Security:** The construction satisfies either search PoNI security (Definition 18) or decisional PoNI security (Definition 19).

For security, we consider a general scenario where the server receives a ciphertext, then splits it into two pieces after interacting in many PoNIs. After the split, the server simultaneously gives a PoNI using one piece while the “hacker” receives the secret key and attempts to decrypt the ciphertext using the other piece.

Definition 18 (Proofs of No-Intrusion for Encryption: Search Security). *Consider the following security game $\text{PoNI-Enc-S}_n(\mathcal{A})$, played by an adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_P, \mathcal{A}_H)$ consisting of three QPT algorithms (with auxiliary quantum input) and parameterized by a non-negative integer n .*

1. Sample a message $m \leftarrow \{0, 1\}^\lambda$, a key pair $(pk, sk) \leftarrow \text{KeyGen}(1^\lambda)$, a verification key $vk \leftarrow \text{VKGen}(1^\lambda)$, and a ciphertext $ct \leftarrow \text{Enc}(pk, m, vk)$.
2. Initialize $\mathcal{A}_1$ with (pk, ct) .
3. Perform $\text{PoNI}\langle \mathcal{A}_1, V(vk) \rangle$ a total of n times. If the verifier outputs **Reject** in any of these executions, the adversary immediately loses (output 0).
4. $\mathcal{A}_1$ outputs two registers $(\mathcal{H}, \mathcal{P})$.
5. Perform $b \leftarrow \text{PoNI}\langle \mathcal{A}_P(\mathcal{P}), V(vk) \rangle$, where b is the verifier’s decision bit.
6. Run $m' \leftarrow \mathcal{A}_H(\mathcal{H}, sk, vk)$.
7. Output 1 (the adversary wins) if $b = \text{Accept}$ and $m' = m$. Otherwise output 0 (the adversary loses).

We say the PoNI has n -time search security if for all QPT adversaries $\mathcal{A}$,

$$\Pr[1 \leftarrow \text{PoNI-Enc-S}_n(\mathcal{A})] = \text{negl}(\lambda)$$

If this holds for all $n = \text{poly}(\lambda)$, then we simply say that the PoNI has search security.

If this holds for QPT $\mathcal{A}_1$ and $\mathcal{A}_P$, but unbounded $\mathcal{A}_H$, we say that the PoNI has (n -time) everlasting search security.

Definition 19 (Proofs of No-Intrusion for Encryption: Decisional Security). *Consider the following security game $\text{PoNI-Sec}_n(m)$, played by an adversary $\mathcal{A}(|\psi\rangle)$ and parameterized by a message bit $m \in \{0, 1\}$ along with a non-negative integer n .*

1. Sample a key pair $(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda)$, a verification key $\text{vk} \leftarrow \text{VKGen}(1^\lambda)$ and a ciphertext $\text{ct} \leftarrow \text{Enc}(\text{pk}, m, \text{vk})$.
2. Initialize $\mathcal{A}(|\psi\rangle)$ with (pk, ct) .
3. Perform $\text{PoNI}(\mathcal{A}(\dots), V(\text{vk}))$ a total of n times. If the verifier outputs **Reject** in any of these executions, immediately output $\perp$.
4. $\mathcal{A}$ outputs a register $\mathcal{R}_{\text{Dec}}$.
5. Perform $\text{PoNI}(\mathcal{A}(\dots), V(\text{vk}))$. If the prover rejects, output $\perp$. Otherwise output $(\mathcal{R}_{\text{Dec}}, \text{sk}, \text{vk})$.

We say the PoNI has n -time security if for all QPT adversaries $\mathcal{A}(|\psi\rangle)$,

$$\{\text{PoNI-Sec}_n(0)\} \approx \{\text{PoNI-Sec}_n(1)\}$$

If this holds for all $n = \text{poly}(\lambda)$, then we simply say that the PoNI is secure.

If these two distributions are instead statistically close, we say that the PoNI is (n -time) everlasting secure.

Goyal and Raizes show that search PoNI security can be generically compiled to decision PoNI security in the quantum random oracle model (QROM).

Theorem 15. *Any PKE with search PoNI security can be generically compiled into a PKE with decisional PoNI security in the quantum random oracle model.*

A.2 Tools

PKE with Publicly Verifiable Certified Deletion. The construction uses a PKE with publicly-verifiable certified deletion, which are known from a variety of assumptions including post-quantum PKE [BKP23, BGK⁺24, BKM⁺23, KNY23]. Roughly, this primitive allows generation of a quantum ciphertext that can be destructively measured to produce a certificate **cert**. The certificate can be checked by anyone using a public verification key; if the certificate is valid, then the ciphertext can no longer be decrypted, even using the secret decryption key.

Definition 20 (PKE with Publicly Verifiable Certified Deletion). *A PKE scheme with publicly verifiable certified deletion has the following syntax.*

- $(\text{sk}, \text{pk}) \leftarrow \text{KeyGen}(1^\lambda)$ takes as input the security parameter then outputs a secret decryption key **sk** and public encryption key **pk**.
- $(|\text{ct}\rangle, \text{vk}) \leftarrow \text{Enc}(\text{pk}, m)$ takes as input a public key and a message bit b then outputs a ciphertext and an associated verification key.
- $m \leftarrow \text{Dec}(\text{sk}, |\text{ct}\rangle)$ takes as input a secret key and a ciphertext then outputs a message.

- $\text{cert} \leftarrow \text{Del}(|\text{ct}\rangle)$ takes as input a ciphertext then outputs a (classical) certificate.
- $\text{Accept/Reject} \leftarrow \text{Ver}(\text{vk}, \text{cert})$ takes as input a verification key and a certificate, then outputs accept or reject.

It must satisfy the standard decryption correctness and semantic security properties for PKE.

Additionally, it must satisfy **correctness of deletion**: for any message m ,

$$\Pr \left[\begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{KeyGen}(1^\lambda) \\ (\text{ct}, \text{vk}) \leftarrow \text{Enc}(\text{pk}, m) \\ \text{cert} \leftarrow \text{Dec}(\text{sk}, \text{ct}) \end{array} : \text{Accept} \leftarrow \text{Ver}(\text{vk}, \text{cert}) \right] = 1 - \text{negl}(\lambda)$$

Finally, it must also satisfy **publicly verifiable certified deletion security**: for every QPT adversary $\mathcal{A}$,

$$\frac{1}{2} \|\text{PVCD-Exp}(0) - \text{PVCD-Exp}(1)\|_1 = \text{negl}(\lambda)$$

where the experiment $\text{PVCD-Exp}(b)$ is defined as follows:

1. Sample keys $(\text{sk}, \text{pk}) \leftarrow \text{KeyGen}(1^\lambda)$ and encrypt $(|\text{ct}\rangle, \text{vk}) \leftarrow \text{Enc}(\text{pk}, m)$.
2. Run $\mathcal{A}(|\text{ct}\rangle, \text{pk}, \text{vk})$ and parse their output as a certificate cert and a state on register $\mathcal{R}_{\mathcal{A}}$.
3. If $\text{Ver}(\text{vk}, \text{cert}) = \text{Accept}$, output $\mathcal{R}_{\mathcal{A}}$. Otherwise output $\perp$.

Simultaneous Extraction. Additionally, the analysis requires the following lemma from [GR25]. Informally, it states that if two adversaries B and C simultaneously succeed in a search task with noticeable probability without communicating, then they also simultaneously succeed with noticeable probability even when C 's challenge is modified in a way which is indistinguishable to C , but which may be distinguishable to B .

Lemma 10. Let $\mathcal{D}$ be a distribution outputting three registers $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ along with a classical string k_C . Let $\mathcal{D}'$ be a distribution taking as input k_C then outputting $\mathcal{C}'$ such that

$$\{(\mathcal{A}, \mathcal{C}, k_C) : (\mathcal{A}, \mathcal{B}, \mathcal{C}, k_C) \leftarrow \mathcal{D}\} \approx_c \left\{ (\mathcal{A}, \mathcal{C}', k_C) : \begin{array}{l} (\mathcal{A}, \mathcal{B}, \mathcal{C}, k_C) \leftarrow \mathcal{D} \\ \mathcal{C}' \leftarrow \mathcal{D}'(k_C) \end{array} \right\}$$

For all QPT algorithms (A, C) and all (unbounded) quantum algorithms B , if

$$\Pr \left[1 \leftarrow B(\mathcal{B}, \mathcal{A}_B) \wedge 1 \leftarrow C(\mathcal{C}, k_C, \mathcal{A}_C) : \begin{array}{l} (\mathcal{A}, \mathcal{B}, \mathcal{C}, k_C) \leftarrow \mathcal{D} \\ (\mathcal{A}_B, \mathcal{A}_C) \leftarrow A(\mathcal{A}) \end{array} \right] \geq 1/p$$

for some $p = \text{poly}(\lambda)$, then there exists $q = \text{poly}(\lambda)$ such that

$$\Pr \left[1 \leftarrow B(\mathcal{B}, \mathcal{A}_B) \wedge 1 \leftarrow C(\mathcal{C}', k_C, \mathcal{A}_C) : \begin{array}{l} (\mathcal{A}, \mathcal{B}, \mathcal{C}, k_C) \leftarrow \mathcal{D} \\ (\mathcal{A}_B, \mathcal{A}_C) \leftarrow A(\mathcal{A}) \\ \mathcal{C}' \leftarrow \mathcal{D}'(k_C) \end{array} \right] \geq 1/q$$

A.3 Construction

We construct a PKE with *search* PoNI security. As mentioned previously, any such scheme can be generically transformed into one with decisional PoNI security in the quantum random oracle model. We make use of our specific construction of state-preserving arguments for NP to enable compatibility with Lemma 10.²⁹ The construction uses the following tools:

- A PKE scheme with certified deletion $\text{PKE}(\text{KeyGen}, \text{Enc}, \text{Dec}, \text{Del}, \text{Ver})$.
- A signature scheme $\text{Sig}(\text{KeyGen}, \text{Sign}, \text{Ver})$.
- The state-preserving argument for NP SPA_{NP} from Construction 2.

Construction 7. The core algorithms are as follows:

- $\text{KeyGen}(1^\lambda)$ samples and outputs $(\text{sk}, \text{pk}) \leftarrow \text{PKE.KeyGen}(1^\lambda)$.
- $\text{VKGen}(1^\lambda)$ samples a signing key pair $(\text{sk}_{\text{Sig}}, \text{pk}_{\text{Sig}}) \leftarrow \text{Sig.KeyGen}(1^\lambda)$. It outputs $\text{vk} = (\text{sk}_{\text{Sig}}, \text{pk}_{\text{Sig}})$.
- $\text{Enc}(\text{pk}, m; \text{vk})$ parses $\text{vk} = (\text{sk}_{\text{Sig}}, \text{pk}_{\text{Sig}})$. Then, it samples $x \leftarrow \{0, 1\}^\lambda$ and encrypts $(|\text{ct}\rangle, \text{vk}') \leftarrow \text{PKE.Enc}(\text{pk}, x)$. Finally, it signs $\sigma \leftarrow \text{Sig.Sign}(\text{sk}_{\text{Sig}}, \text{vk}')$ and outputs

$$|\text{ct}'\rangle = (|\text{ct}\rangle, \text{vk}', \sigma)$$

- $\text{Dec}(\text{sk}, |\text{ct}'\rangle)$ parses $|\text{ct}'\rangle = (|\text{ct}\rangle, \text{vk}', \sigma)$, then computes $m \leftarrow \text{PKE.Dec}(\text{sk}, |\text{ct}\rangle)$. It outputs m .

The **proof of no-intrusion** is as follows.

1. **Prover:** Parse $|\text{ct}'\rangle = (|\text{ct}\rangle, \text{vk}', \sigma)$ and send (vk', σ) to the verifier. Coherently compute $\text{Del}(|\text{ct}\rangle)$ to obtain $\sum_{\text{cert}} \alpha_{\text{cert}} |\text{cert}, \text{garbage}_{\text{cert}}\rangle$.
2. **Verifier:** Parse $\text{vk} = (\text{sk}_{\text{Sig}}, \text{pk}_{\text{Sig}})$. Then, compute $\text{Sig.Ver}(\text{pk}_{\text{Sig}}, \sigma, \text{vk}')$. If it rejects, immediately output reject.
3. **Both.** Interact in an execution of SPA_{NP} for the language

$$\mathcal{L} = \{\text{vk}' : \exists \text{cert s.t. } \text{PKE.Ver}(\text{vk}', \text{cert}) = \text{Accept}\}$$

and the statement vk' .

4. **Prover:** Uncompute the coherent implementation of Del .

²⁹Roughly, Lemma 10 allows reasoning about the simultaneous success rate of two searchers B and C when C 's input is changed in a way that is indistinguishable to C (but may be distinguishable to B). It does *not* allow wholesale replacement of C by some other algorithm $\text{Ext}(C)$, which might in general behave very differently. In our construction of state-preserving arguments for NP, Ext is a straightline extractor which works by sending indistinguishable messages for the first part of the protocol.

Theorem 16. *Construction 7 is a PKE with everlasting search proofs of no-intrusion, assuming the existence of PKE with publicly verifiable certified deletion (which can be based on post-quantum PKE [BKM⁺23, KNY23]), post-quantum signature schemes, and that Construction 2 is a state-preserving argument of knowledge for NP (all of which can be based on LWE).*

Proof. Decryption correctness follows from inspection and the correctness of PKE. PoNI correctness follows from the deletion correctness of PKE, the correctness of Sig, and the correctness of SPA_{NP}. State preservation follows from the state preservation of SPA_{NP}, the deletion correctness of PKE, and the fact that no other measurements are made. Semantic security follows from the semantic security of PKE.

To see PoNI security, first observe that the adversary may simulate any number of PoNIs locally using vk_{Sig} . Furthermore, in the final PoNI, the NP statement proved in the SPA_{NP} execution must be for the original vk' generated at encryption time, or else the unforgeability of Sig is broken.

We show PoNI-Enc-S search security by reducing to the publicly verifiable certified deletion property of PKE. Specifically, given a PoNI-Enc-S search adversary $(\mathcal{A}_1, \mathcal{A}_P, \mathcal{A}_H)$, the reduction plays the certified deletion security game as follows.

1. Sample random $m_0 \leftarrow \{0, 1\}^\lambda$ and $m_1 \leftarrow \{0, 1\}^\lambda$ to be the challenge plaintexts. Receive $(|ct_b\rangle, \text{vk}')$, an encryption of a random m_b along with the corresponding verification key, from the certified deletion challenger.
2. Sample $\text{vk} = (\text{sk}_{\text{Sig}}, \text{pk}_{\text{Sig}})$ and compute $\sigma = \text{Sig.Sign}(\text{sk}_{\text{Sig}}, \text{vk}')$ to complete the ciphertext $|ct'\rangle = (|ct_b\rangle, \text{vk}', \sigma)$, then give it to $\mathcal{A}_1$.
3. Run the initial PoNI executions locally with $\mathcal{A}_1$.
4. Receive the adversary's split registers $(\mathcal{H}, \mathcal{P})$.
5. Execute the final PoNI with $\mathcal{A}_P(\mathcal{P})$ using the argument of knowledge extractor $\text{Ext}(\mathcal{A})$ for SPA_{NP}. If it succeeds in extraction, it outputs cert such that $\text{PKE.Ver}(\text{vk}', \text{cert}) = \text{Accept}$ (since vk' is the statement used as argued previously). If extraction fails, set $\text{cert} = \perp$.
6. Send cert and $\mathcal{H}$ to the certified deletion challenger.
7. Receive either $\mathcal{H}$ or $\perp$ from the certified deletion challenger, depending on whether cert is valid.
8. If the latter, output a random bit. If the former, run $m' \leftarrow \mathcal{A}_H(\mathcal{H})$. Then, output b if $m' = m_b$ and otherwise guess randomly.

We now analyze the reduction's advantage in the certified deletion security game, supposing that the adversary violates PoNI-Enc-S search security for the sake of contradiction. Observe that $\mathcal{A}_H$'s view is independent of m_{1-b} , so $m' = m_{1-b}$ with negligible probability.

Since the reduction guesses randomly in this case unless $m' = m_b$ or if **cert** is invalid, it suffices to show that with noticeable probability $m' = m_b$ and simultaneously **cert** is valid.

Consider the hybrid experiment where steps 5 and 7 are replaced by running the honest execution of SPA_{NP} (i.e. the honest PoNI), and running $m' \leftarrow \mathcal{A}_H(\mathcal{H})$ if the PoNI accepts (instead of checking an extracted **cert**). This matches the PoNI-Enc-S game for random message m_b , so by assumption $\mathcal{A}_H$ outputs $m' = m_b$ with noticeable probability.³⁰

Now consider the following simultaneous search task: $\mathcal{A}_1$ acts as the splitter A ; $\mathcal{A}_H(\mathcal{H})$ acts as (unbounded) B and searches for m_b ; C takes as input a view from $\mathcal{A}_P$ just before the opening phase in Construction 2 and searches for an opening that would lead to an accepting response. This simultaneous search task matches the hybrid experiment described above, so B and C succeed with noticeable probability. Now consider replacing C 's input by a view from $\mathcal{A}_P$ in an execution of SPA_{NP} where the public parameters are generated in *injective* mode (this matches the extractor from the proof of Theorem 10). This view is computationally indistinguishable to C because the trapdoor has not yet been revealed in the SPA_{NP} execution. Therefore Lemma 10 implies that B and C simultaneously succeed in their search tasks with noticeable probability even when C receives the modified view. Finally, we argue that if C succeeds in finding an accepting opening, then with overwhelming probability **Ext** successfully extracts a witness. This follows from the fact that if **Ext** would fail, then for the overwhelming majority of challenges in SPA_{NP} , there does not exist an accepting opening.

Therefore with noticeable probability, **Ext** extracts a valid **cert** and $\mathcal{A}_H$ outputs $m' = m_b$. As argued previously, this contradicts the publicly verifiable certified deletion security of PKE. $\square$

³⁰It is tempting to assume that this immediately implies the above reduction also finds $m' = m_b$ with noticeable probability. However, because we consider unbounded $\mathcal{A}_H$ for everlasting security, running **Ext** is distinguishable to them, which may affect the simultaneous success probability of $\mathcal{A}_H$ and $\mathcal{A}_P$. We rely on Lemma 10 to avoid this issue.